



Verified Carbon Standard

WASTEWATER TO BIOGAS GROUPED PROJECT



Document Prepared By Green Lagoon Technology Sdn Bhd And Eco-Ideal Consulting Sdn Bhd

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Prepared by	Green Lagoon Technology Sdn Bhd and Eco-Ideal Consulting Sdn Bhd

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1 PROJECT DETAILS

1.1 Summary Description of the Project

Grouped Project Description

Project description and location: Wastewater to Biogas Grouped Project (hereinafter referred to as the project) is a grouped project which consists of biogas recovery projects located in Malaysia. The project is owned by Green Lagoon Technology Sdn Bhd. (hereinafter referred to as the project owner), and implementation of the projects are by the individual plant operators of each project activity instance (PAI). The purpose of this Grouped Project is to recover biogas generated from wastewater treatment process which would have otherwise, been emitted into the atmosphere resulting in GHG emissions. The recovered biogas is expected to be utilized for various purposes:

- a. To upload to grid, and/or
- b. Excess biogas will be flared in open/enclosed flare.

Project Activity Instance 1 (PAI1) Description

The first PAI (PAI1) is located at Tian Siang Oil Mill (Triang) Sdn Bhd ("Tian Siang POM") in the State of Pahang, Malaysia, about 6 KM Off Jalan Triang Ke Kemayang, Triang, Pahang and 180km north of Kuala Lumpur. GLT Maju Sdn Bhd is the owner of the biogas plant. Green Lagoon Technology Sdn Bhd has signed an agreement with GLT Maju Sdn Bhd, who has the legal rights to operate PAI1. The mill operator, Tian Siang POM has signed an agreement with GLT Maju Sdn Bhd. for ongoing access to the palm oil mill effluent (POME), in addition to permitting the construction and operation of PAI1. GLT Maju Sdn Bhd is owned by Green Lagoon Technology Sdn Bhd.

Baseline Scenario: Prior to the implementation of PAI1, the wastewater at Tian Siang POM is distributed into open anaerobic lagoons for anaerobic treatment. (a series of eight (8) treatment ponds or lagoons, i.e. four (4) anaerobic, two (2) aerobic, one (1) settling basin and one (1) maturation pond). This treatment method is one of the most common and conventional methods for treating the wastewater whereby approximately 85% of palm oil mills in Malaysia employ this method¹. The anaerobic environment within the anaerobic lagoons with an average mean surface air temperature of 26.98°C² in the region result in baseline methane generation from the organic content of the wastewater (estimated about 35,282 tCO₂e/year) for a period of seven (7) years. After the anaerobic treatment, the treated wastewater is discharged to the river. The sludge from the lagoons is disposed of via land application within the vicinity of the project area.

¹ Yap et. al. (2020). Comparison of different industrial scale palm oil mill effluent anaerobic systems in degradation of organic contaminants and kinetic performance. Journal of Cleaner Production 262 (2020) 121361 (<https://www.sciencedirect.com/science/article/abs/pii/S0959652620314086>)

² As for Pahang, 1991-2020, available from [World Bank Group – Climate Change Knowledge Portal at https://climateknowledgeportal.worldbank.org/country/malaysia/climate-data-historical](https://climateknowledgeportal.worldbank.org/country/malaysia/climate-data-historical)

Project Scenario: The project involves the improvement of wastewater treatment by introducing an anaerobic digester, to be installed in a new lagoon whereby a membrane is installed over the lagoon to capture the methane emitted. After the digester, the residue will be further treated through a series of aerobic processes before finally being discharged to river. The biogas generated from the digester will be captured and utilized for power generation to be uploaded to the grid. This action leads to the reduction of greenhouse gas emissions whereby the estimated annual average and total GHG emission reductions during the first seven (7) year crediting period of the PAI1 is tabulated below:

No	Project activity instance	Annual Average GHG Emission (tCO ₂ e)	Total GHG emission reductions for first crediting period (tCO ₂ e)
PAI1	Tian Siang POM	35,217	246,518

1.2 Audit History

Audit type	Period	Program	Validation/verification body name	Number of years
Validation/verification	24-04-2024 to 23-04-2031	VCS	Control Union Certifications Germany GmbH	7 years

1.3 Sectoral Scope and Project Type

The project activity under consideration is a grouped project activity. The project activity instances as part of the grouped project are defined as follow:

Sectoral scope³	13 - Waste handling and disposal
Project activity type	III - Other project activities AMS-III.H (Version 19.0) - Methane recovery in wastewater treatment
Sectoral scope	1 - Energy industries (renewable/non-renewable sources)
Project activity type	I - Renewable energy projects AMS-I.D (Version 18.0) - Grid-connected renewable electricity generation

³ Projects, activities, or methodologies may be developed under any of the 16 VCS sectoral scopes: <https://verra.org/programs/verified-carbon-standard/vcs-program-details/#sectoral-scopes>

1.4 Project Eligibility

1.4.1 General eligibility

As per section 2.1.1 of VCS Standard (version 4.7), the scope of the VCS Program includes:

1. The seven Kyoto Protocol greenhouse gases: The proposed grouped project is designed to treat the wastewater by the installation of a new biogas digester and avoid methane emission from open anaerobic lagoons during the baseline scenario. The recovered biogas is expected to be utilized for generation of electricity to be uploaded to the grid. Methane and carbon dioxide are the greenhouse gases under Kyoto Protocol, therefore the project is applicable to this scope.
2. Ozone-depleting substances (ODS): Not applicable.
3. Project activities supported by a methodology approved under the VCS Program through the methodology development and review process: Not applicable.
4. Project activities supported by a methodology approved under an approved GHG program, unless explicitly excluded (see the Verra website for exclusions): The applied methodologies AMS-III.H (Version 19.0) and/or AMS-I.D (Version 18.0) of the grouped project are methodologies approved under the CDM Program, which is a VCS approved GHG program.
5. Jurisdictional REDD+ programs and nested REDD+ projects as set out in the VCS Program document Jurisdictional and Nested REDD+ (JNR) Requirements: Not applicable.

As per section 2.1.2 of VCS Standard (version 4.7), the scope of the VCS Program excludes projects that can reasonably be assumed to have generated GHG emissions primarily for the purpose of their subsequent reduction, removal or destruction: The proposed grouped project involves the utilization and destruction of biogas generated from the biogas digester to replace the existing open anaerobic lagoons, which does not generate GHG emissions primarily for the purpose of their subsequent reduction, removal or destruction.

As per section 2.1.3 of VCS Standard (version 4.7), the VCS Program also excludes the following project activities under the circumstances indicated in Table 1: Excluded Project Activities. The proposed grouped project involves the utilization and destruction of biogas generated from the biogas digester to replace the existing open anaerobic lagoons, which does not fall under the exclusion listed in table 1.

The project started operation on 24-Apr-2024, it shall complete validation before 23-Mar-2026.

According to VCS, the projects seeking registration in the VCS Program that comply with all VCS Program rules may use methodologies from other approved GHG programs, including Clean Development Mechanism (CDM). The methodologies to be applied for the grouped project are adopted from CDM methodologies which are AMS-III.H, and AMS-I.D. For small-scale projects,

the measures are limited to those that result in aggregate emission reductions of less than or equal to 60 ktCO₂ equivalent annually from all Type III components of the project activity. The PAIs under the grouped project activity will result in less than or equal to 60ktCO₂ equivalent annually from all Type III, as demonstrated in section 3.2 of PD

1.4.2 AFOLU project eligibility

Not applicable as the grouped project activity is not AFOLU project.

1.4.3 Transfer project eligibility

Not applicable as the grouped project activity is not transferred from other GHG Programs.

1.5 Project Design

- ☐ Single location or installation
- ☐ Multiple locations or project activity instances (but not a grouped project)
- ☒ Grouped project

1.5.1 Grouped project design

The project is a grouped project and the following are the eligibility criteria for the inclusion of new project activity instances into the grouped project activity:

- **Applicability Conditions:** The project activity instances shall meet applicability conditions for applicable methodologies (AMS III.H version 19.0 and/or AMS I.D version 18.0) as defined in section 3.2. There are three (3) potential individual methodologies or combination methodologies for new PAIs, which are listed below:
 - AMS-III.H.
 - AMS-I.D.
 - AMS-III.H. and AMS-I.D.
- **Geographical Area:** The project activity instance is located in the geographic boundary for the grouped project, i.e., Malaysia (Peninsular Malaysia, Sabah, and Sarawak), and owned by the project owner or the subsidiary of the project owner. The geographical area should be defined as per requirements under section 3.6.10 and 3.11.1 of the VCS Standard (version 4.7) and presented in KML file.
- **Baseline scenario:** All Project Activity Instances shall have the following existing conditions at their respective PAI level, following the combination of methodology(ies) outlined above. Future PAIs will only be eligible for inclusion if their existing conditions at the PAI level are consistent with the baseline scenario and applicability conditions that defined in the project description section 3.4, in accordance with Sections 3.6.11 of the VCS Standard

- **AMS-III.H.:** For wastewater related to palm oil mill or other type of mill, it would have been treated in anaerobic open lagoons; and
- **AMS-I.D.:** For electricity generation, it would have been generated by operation of grid connected power plants and by addition of new generation sources into the grid
- **Technology type:** The project activity instances to be included in the grouped project activity is the implementation of biogas recovery, flaring and/or utilization systems in palm oil mills using either an anaerobic digester or covered anaerobic lagoon, and gas engine for power generation.
- **Additionality:** The project activity instances to be added as part of the grouped project will meet additionality criteria as set out in the methodology at respective PAI level, following the combination of methodology(ies) above.
 - For any combination involves wastewater related methodologies (AMS-III.H.), the respective PAIs will apply the methodological tool Positive lists of technologies - Version 04.0, A positive list of technology and project activity types that are defined as automatically additional if the PAIs apply such technologies and meet specified conditions. The positive list of technology including the methane recovery in wastewater treatment where the specific conditions to be met is listed below:
 - The existing treatment system is an anaerobic lagoon and the wastewater discharged meets the host country regulation. The PP can submit the engineering drawings of the open anaerobic lagoon, as well as documentation related to wastewater discharge permits and compliance
 - There is no regulation in the host country that requires the management of biogas from domestic, industrial and agricultural sites. In all three (3) regions (Peninsular Malaysia, Sabah, and Sarawak), the industry is regulated at the federal level by the Department of Environment (DOE), which oversees environmental compliance and standards. In addition, the licensing and operational requirements of palm oil mills are specifically regulated by the Malaysian Palm Oil Board (MPOB) under the Malaysian Palm Oil Board Licensing Regulations. The requirement to install methane capture facilities under the Malaysian Palm Oil Board Licensing Regulations is determined based on the mill's start date and any subsequent capacity additions. To demonstrate that this project is not subject to a mandatory methane capture requirement, the PP can submit documentation confirming the start date of the mill (e.g., trade license/operating permit).
 - There is no capacity increase in the wastewater treatment system. The requirement to install methane capture facilities under the Malaysian Palm Oil Board Licensing Regulations is determined based on the mill's start date and any subsequent capacity additions. The PP can submit documentation confirming there is no capacity addition to the original approved mill capacity

- No other alternative economic activity is expected to be undertaken on the land of the existing lagoon. The PP can submit relevant documents to the VVB.
 - The biogas is used to generate electricity in one or more power plants, and the total nameplate capacity is below 5 MW. The PP can submit the engine specification to the VVB.
- For project activity instances that solely apply AMS-I.D., the additionality criteria will apply the tool “Demonstration of additionality of small-scale project activities” Version 13.1.
- The additionality will be demonstrated for each designated geographic area (i.e., Peninsular Malaysia, Sabah, and Sarawak) for each of the PAIs in accordance with Section 3.6.14 of the VCS Standard
- **Start Date:** The start date of each project activity instance under the grouped project should not be prior to the start date of the grouped project. The start date of each project activity instance will be determined through documentary evidence.

Future project activity instances will only be included at verification if they are located within the defined geographic area, are subject to the established baseline scenario, and meet the additionality demonstration provided at registration, in accordance with the VCS Standard v4.7 (Sections 3.6.10 and 3.6.11)

The following table demonstrates how the initial project activity instance (PAI1) being included in the grouped project activity fulfil above mentioned eligibility conditions:

Table 1: PAI1 Eligibility Criteria

No	Eligibility Criteria	PAI1 Eligibility
1	Applicability Conditions: The project activity instances shall meet applicability conditions for applicable methodology (AMS III.H version 19.0 and/or AMS I.D version 18.0). There are three (3) potential individual methodologies or combination methodologies for new PAIs, which are listed below: • AMS-III.H. • AMS-I.D. • AMS-III.H. and AMS-I.D.	The initial project activity instance (PAI1) has the combination methodologies of AMS-III.H. and AMS-I.D., therefore PAI1 is subject to meet respective applicability conditions of methodology AMS-III.H version 19.0 and AMS-I.D version 18.0., as well as related tools. Hence this condition is fulfilled. Please refer section 3.2 of this document for applicability conditions.
2	Geographical Area: The project activity instance locates in the	The initial project activity instance being included in the grouped project are located

No	Eligibility Criteria	PAI1 Eligibility
	geographic boundary for the grouped project, i.e., Malaysia, and owned by the project owner or the subsidiary of the project owner. The geographical area should be defined as per requirements under section 3.6.10 and 3.11.1 of the VCS Standard (version 4.7) and presented in KML file.	within geographic boundaries of Malaysia, with the area is defined and KML file is submitted as per requirements under section 3.6.10 and 3.11.1 of the VCS Standard (version 4.7). Hence this condition is fulfilled. Please refer to KML file for the Geographical area of project activity
3	Baseline scenario: All Project Activity Instances shall have the existing conditions at their respective PAI level, following the combination of methodology(ies) outlined above,	The initial project activity instance has the existing conditions at its PAI level following its combination methodologies of AMS-III.H, and AMS-I.D., which are: - Wasterwater related to palm oil mill would be treated in anaerobic open lagoons - Electricity generation would have been generated by the operation of grid connected power plants and by addition of new generation sources into the grid. The same baseline as per methodology is detailed in subsequent sections. Hence this eligibility criteria are fulfilled. Please refer section 3.4 of this document for baseline scenario.
4	Technology type: The project activity instances to be included in the grouped project activity is the implementation of biogas recovery, flaring and/or utilization systems in palm oil mill using either anaerobic digester or covered anaerobic lagoon, and gas engine for power generation.	The initial project activity instance being included in the grouped project activity is the implementation of biogas recovery, flaring and/or utilization systems in the palm oil mill. The recovered biogas is utilized for power generation and uploaded to the grid. Hence this condition is fulfilled.
5	Additionality: The project activity instances to be added as part of the grouped project will meet additionality criteria as set out in the methodology at respective PAI level,	The initial project activity instance being included in grouped project activity is a combination of AMS-III.H. and AMS-I.D., which will apply the methodological tool Positive lists of technologies – Version 04.0. Their

No	Eligibility Criteria	PAI1 Eligibility
	following the combination of methodology(ies) above.	additionality is demonstrated as discussed in section 3.5 of this document. Hence this condition is fulfilled. Please refer section 3.5 of this document for additionality.
6	Start Date: The start date of each project activity instance under the grouped project should not be prior to the start date of the grouped project. The start date of each project activity instance will be determined through documentary evidence	The start date of grouped project is considered on 24-Apr-2024 which is the date of the biogas plant in operation for PAI1. Hence this condition is fulfilled. Please refer section 1.8 of this document or commissioning certificate for start date of PAIs.

1.6 Project Proponent

Organization name	Green Lagoon Technology Sdn Bhd
Contact person	Chan Sow Keong
Title	CEO
Address	H-3A-01, Pusat Perdagangan Bandar Bukit Jalil, Persiaran Jalil 1, 57000 Kuala Lumpur, Malaysia
Telephone	+603 2733 2960
Email	chansk@gl.t.my

1.7 Other Entities Involved in the Project

Organization name	GLT Maju Sdn Bhd
Role in the project	Project Operator of PAI1
Contact person	Chan Sow Keong
Title	CEO
Address	H-3A-01, Pusat Perdagangan Bandar Bukit Jalil, Persiaran Jalil 1, 57000 Kuala Lumpur, Malaysia
Telephone	+603 2733 2960
Email	chansk@gl.t.my

1.8 Ownership

Grouped Project

Green Lagoon Technology Sdn Bhd is the sole project owner of the project. Every additional PAI to be added to this Grouped Project is required to provide evidence of project ownership.

PAI1

GLT Maju Sdn Bhd is the owner of the biogas plant. Green Lagoon Technology Sdn Bhd has signed an agreement with GLT Maju Sdn Bhd, who has the legal rights to operate PAI1. The mill operator, Tian Siang POM has signed an agreement with GLT Maju Sdn Bhd. for ongoing access to the palm oil mill effluent (POME), in addition to permitting the construction and operation of PAI1. GLT Maju Sdn Bhd is owned by Green Lagoon Technology Sdn Bhd

1.9 Project Start Date

Project start date	24-Apr-2024
Justification	According to VCS Standard, version 4.7, the project start date of a non-AFOLU project is the date on which the project began generating GHG emission reductions or carbon dioxide removals. The Grouped Project start date is 24-April-2024, which is the commencement of emissions reductions at PAI1 when the biogas plant is in operation.

1.10 Project Crediting Period

Crediting period	☒ Seven years, twice renewable ☐ Ten years, fixed ☐ Other (state the selected crediting period and justify how it conforms with the VCS Program requirements)
Start and end date of first or fixed crediting period	24-Apr-2024 to 23-Apr-2031

1.11 Project Scale and Estimated GHG Emission Reductions or Removals

☒ < 300,000 tCO₂e/year (project)

☐ ≥ 300,000 tCO₂e/year (large project)

Calendar year of crediting period	Estimated GHG emission reductions or removals (tCO ₂ e)
24-April-2024 – 31-December-2024	23,335
01-January-2025 – 31-December-2025	35,372
01-January-2026 – 31-December-2026	35,372
01-January-2027 – 31-December-2027	35,372
01-January-2028 – 31-December-2028	35,372
01-January-2029 – 31-December-2029	35,372
01-January-2030 – 31-December-2030	35,372
01-January-2031 – 23- April-2031	10,951
Total estimated ERRs during the first or fixed crediting period	246,518
Total number of years	7
Average annual ERRs	35,217

1.12 Description of the Project Activity

Grouped Project

The Grouped Project will consist of multiple PAIs for the implementation of biogas recovery, flaring and/or utilization systems from palm oil mill wastewater in Malaysia. The Grouped Project will reduce greenhouse gas (GHG) emissions from sources of palm oil wastewater by capturing the biogas that would otherwise have been generated and released to the atmosphere from the baseline wastewater treatment systems.

The grouped project activity will encompass multiple individual project activity instances that apply a set of technologies for methane avoidance and biogas utilization. These instances are characterized by two (2) stages:

Stage 1: Biogas Generation and Recovery (Methane Avoidance)

This stage involves the anaerobic digestion of organic waste streams to capture methane (CH₄) that would otherwise be released into the atmosphere. The feedstock sources and recovery technologies for each instance will include:

- **Feedstock Sources:**
 - **Palm Oil Mill Effluent (POME):** Wastewater generated from the palm oil milling process (AMS-III.H.).
- **Methane Recovery Technologies:**
 - **Anaerobic Digesters:** Enclosed tanks designed for optimal anaerobic digestion conditions. These offer controlled temperature and mixing for efficient methane production.
 - **Covered Anaerobic Lagoons:** The In-Ground Bio Reactor System with Biogas Storage (IGBR) System. The reactor uses an in-ground pond and covered High-Density Polyethylene Geomembrane (HDPE) membrane. The reactor is also integrated with hydraulic sludge mixers and a sludge removal system inside the digester.

Stage 2: Biogas Utilization

The raw biogas from each project activity instance will be managed and utilized through one or a combination of the following systems:

- **Energy Utilization Systems:**
 - **Gas Engines for Power Generation:** Biogas will be directly combusted in gas engines for power generation and exported to the national grid (AMS-I.D).
- **Flaring:** Excess biogas will be safely combusted in an open flare

The generic process flow for the grouped project is illustrated below:

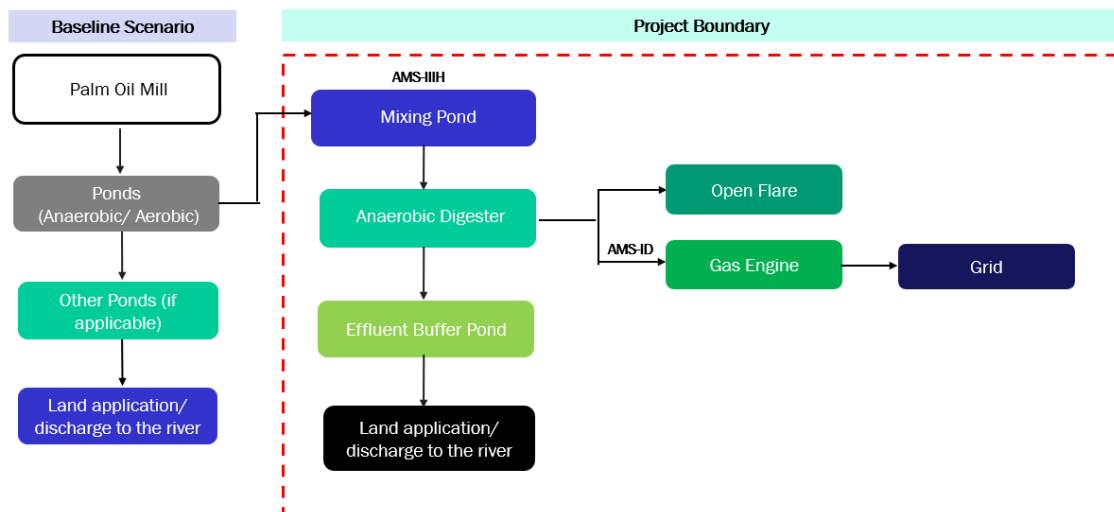


Figure 1: Grouped Project Generic Process Flow

As the Project Proponent is exploring opportunities with diverse suppliers, name of the manufacturer, and the average lifespan of the equipment are not available. However, in the Malaysian context, the operational longevity of biogas equipment, particularly gas engines and digester components, is demonstrably extended through diligent adherence to rigorous

preventive maintenance schedules, often exceeding typical estimated lifespans. The specific information on the technologies, including a mass flow diagram, will be provided under each PAI level during inclusion into the grouped project.

PAI1

The PAI1 is named the Tian Siang Biogas Plant. It is implemented at Tian Siang Palm Oil Mill ("Tian Siang POM") in the State of Pahang, Malaysia, about 6km from the Temerloh and 160km north of Kuala Lumpur. Tian Siang POM is an established palm oil mill that is processing approximately 198,891 MT/yr of FFB. The processing of FFB creates a significant volume of wastewater that requires treatment, due to a high level of organic matter, as measured by the wastewater's Chemical Oxygen Demand (COD). The existing wastewater treatment system includes a series of ponds (anaerobic and aerobic). The breakdown of COD in an anaerobic environment result in the emission of methane into the atmosphere.

In PAI1, the biogas digester is installed in a new lagoon, and a membrane is installed over the lagoon to capture the methane emitted. By capturing methane emissions and combusting the biogas, the PAI leads to reduced emissions of greenhouse gases. This methane is converted into biogas before processing for use in the generation of electricity.

PAI1 will include the following technologies:

1. An anaerobic digester – The In-Ground Bio Reactor System with Biogas Storage (IGBR) System is approximately 104m × 44m × 8m depth, with an effective volume of around 30,000m³. The reactor uses an in-ground pond and covered High-Density Polyethylene Geomembrane (HDPE) membrane, which also acts concurrently as a biogas storage facility, capable of storing up to 21,000m³ of biogas. The hydraulic retention time is 35 days. The reactor is also integrated with hydraulic sludge mixers and a sludge removal system inside the digester. Based on technology supplier, the digester offers strong resistance to chemical, UV and mechanical degradation with a technical service life of up to 50 years.
2. Biogas treatment system - This includes 1 unit of scrubber and a dehumidifier and chiller. The biological scrubbers with a capacity of 650m³/hr are used to oxidize hydrogen sulphide (H₂S) to sulphate. The result of the scrubbing process is clean biogas with H₂S content reduced to less than 200 ppm. The biogas is passed from the scrubber to a dehumidifier/chiller system to reduce the water content in the biogas, from 75-90% to 40%, and to cool the biogas to the point it can be used in the biogas engine.
3. Biogas engine – A single Jenbacher biogas engine model JGS 416 GS-B.L, which has a gross electrical output of 1,202kW is installed at GLT Maju Biogas Plant. Based on technology supplier, the biogas engine has a technical lifespan exceeding 20 years, with major maintenance intervals typically between 60,000 to 80,000 hours of operation. An open flare is also installed, although it is not expected to be needed due to the storage capacity of the biogas membrane.
4. Control and monitoring system – All parameters necessary for the operation and management of the biogas plant are integrated to a Supervisory Control and Data Acquisition (SCADA) system.

The process flow for the PAI1 is illustrated below:

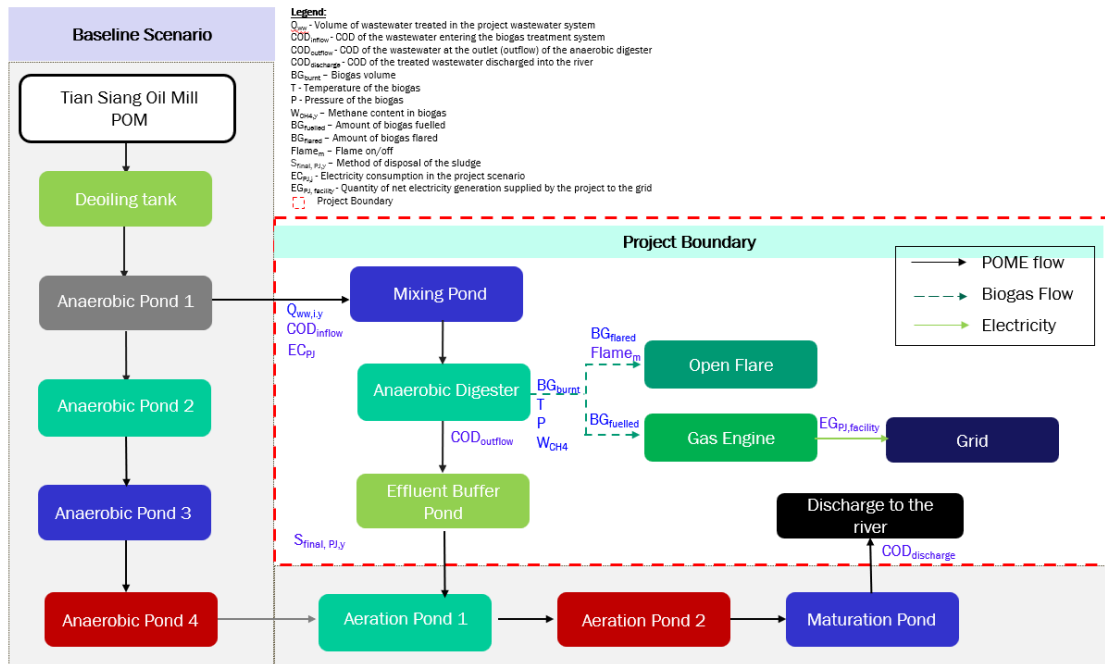


Figure 2: PAI1 Mass Flow Diagram

1.13 Project Location

Grouped Project

Project Activity Instances within the border of Malaysia, consisting of Peninsular Malaysia, Sabah and Sarawak, are allowed to be added to and included in this Grouped Project. Geodetic coordinates are provided as a KML file.



Figure 3: Grouped Project Location

PAI1

PAI1 is situated at the site of Tian Siang Oil Mill (Triang) Sdn Bhd in the State of Pahang, Malaysia, about 6 KM Off Jalan Triang Ke Kemayang, Triang, Pahang and 160km north of Kuala Lumpur. The geographical coordinates for the site are N 3°10'58.6" and E 102°27'27.3". The exact location of the site is shown in Figure below:

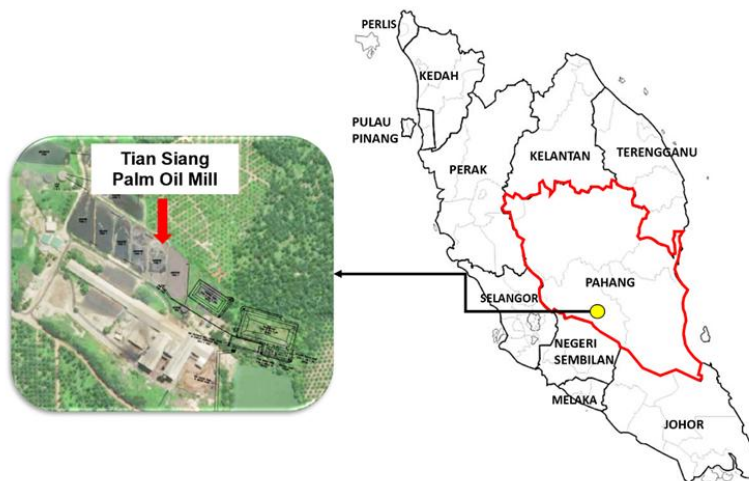


Figure 4: PAI1 - Project Location

1.14 Conditions Prior to Project Initiation

The existing scenario prior to the start of the implementation of each project activity instance is: The wastewater was left at uncovered open lagoon and methane is emitted to the atmosphere directly without any methane recovery and destruction facility. The conditions existing prior to the project initiation are the same as the baseline scenario,

According to AMS-III.H version 19.0, the baseline scenario involves a baseline system with anaerobic wastewater treatment system without biogas recovery, which is the case the baseline scenario is determined to be the continuation of the existing open lagoon-based treatment of wastewater without methane recovery as commonly practiced because it only requires small investment and maintenance cost where the treatment system only consume less energy for pumping and mixing.

According to AMS-I.D version 18.0, the baseline scenario for greenfield power plant, electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources into the grid.

Therefore, as a result of the combination of baseline analysis for all the above methodologies, it is concluded that the baseline scenario of the proposed Grouped Project is, in the absence of the project activity:

- For wastewater related to palm oil mill would have been treated in anaerobic open lagoons; and

- For electricity generation, it would have been generated by operation of grid connected power plants and by addition of new generation sources into the grid

1.15 Compliance with Laws, Statutes and Other Regulatory Frameworks

Grouped Project

Malaysia consists of three (3) regions which are Peninsular Malaysia, Sabah and Sarawak, where each regions has the right to enact its legislative framework. However, in all three (3) regions (Peninsular Malaysia, Sabah, and Sarawak), the industry is regulated at the federal level by the Department of Environment (DOE), which oversees environmental compliance and standards. In addition, the licensing and operational requirements of palm oil mills are specifically regulated by the Malaysian Palm Oil Board (MPOB) under the Malaysian Palm Oil Board Licensing Regulations. Every additional PAI to be added to this Grouped Project is required to demonstrate compliance with all and any relevant local, regional and national laws, statutes and regulatory frameworks.

PAI1

PAI1 is in compliance with the conditions of local laws and regulations of Malaysia, including the following of particular relevance:

- Environmental Quality Regulations (Prescribed Premises) (Crude Palm Oil) 1977 – P.U.(A) 342/77 Regulation 6 - Making Changes that change the quality of the effluent ⁴
- Written Declaration on the Design and Construction of Air Pollution Control Systems [Regulation 7(5)] DOE ASPUB⁵
- Construction Site Registration - DOSH JKJ 103⁶
- Environmental Quality (Clean Air) Regulations 2014
- Environmental Quality (Scheduled Wastes) Regulations 2005
- Act A1648 OSH Act 2022
- Minimum Wage Order 2022
- Employment Act 1955
- Employees Provident Fund Act 1991 (Act 452)
- ACT 177, Industrial Relations Act 1967
- Act 840 Anti-Sexual Harassment Act 2022
- Act 800 Employment Insurance System Act 2017
- Act 4 Employees' Social Security Act 1969 (Amended 2003)

⁴ Peraturan Kualiti Alam Sekeliling (Premis Yang Ditetapkan) (Minyak Kelapa Sawit Mentah) 1977 – P.U.(A) 342/77 Peraturan 6 - Membuat Perubahan-perubahan yang mengubah kualiti effluen

⁵ Perisytiharan Bertulis Mengenai Reka Bentuk Dan Pembinaan Sistem Kawalan Pencemaran Udara [Peraturan 7(5)] DOE ASPUB

⁶ DOSH JKJ 103

The other licenses and approvals specific to PAI 1 is listed below:

- Company registration with Companies Commission of Malaysia (SSM)
- Licensing from Energy Commission (ST)
- Registration of construction site with CIDB
- Feed-In Approval (FiAH)
- Feed-in Tariff (FiT)
- Approved plan for POME discharge under Section 18(1), Act 127, Regulation 14, P.U (A) 342/1977 (JP/KKS/2018/2019/003238) dated 12.06.2018 signed by DOE, Pahang
- Grid connection feasibility study and power purchase agreement from TNB

There are no regulations in Malaysia that requires management of biogas from PAI1. Condition in the criteria and guidelines⁷ for an application for licensing by the Malaysian Palm Oil Board (MPOB) require new palm-oil mills constructed after 01-Jan-2014 to install wastewater treatment systems with methane capture. The requirement was also placed on existing mills who applied for capacity extensions after 01-Jan-2014, although an exemption effective 22-May-2021 was given to those existing mills who applied for capacity extensions below 270,000 MT FFB per year, noting this threshold has been revised several times.

Tian Siang POM was established in 2000 and was not a mill constructed after 01-Jan-2014 and has not undergone a capacity expansion since 01-Jan-2014. Therefore, the new criteria do not apply and there is no obligation to recover the methane and/or use the biogas.

1.16 Double Counting and Participation under Other GHG Programs

1.16.1 No Double Issuance

Is the project receiving or seeking credit for reductions and removals from a project activity under another GHG program?

☐ Yes ☒ No

1.16.2 Registration in Other GHG Programs

Has the project registered under any other GHG programs?

☐ Yes ☒ No

Is the project active under the other program?

☐ Yes ☒ No

⁷ Kriteria dan Panduan Permohonan Lesen MPOB, available from: https://e-lesen.mpob.gov.my/document/CRITERIA%20AND%20GUIDELINES%20ON%20MPOB%20LICENSE%20APPLICATION_EnglishVersion_latest.pdf

1.16.3 Projects Rejected by Other GHG Programs

Has the project been rejected by any other GHG programs?

☐ Yes ☒ No

1.17 Double Claiming, Other Forms of Credit, and Scope 3 Emissions

1.17.1 No Double Claiming with Emissions Trading Programs or Binding Emission Limits

Are project reductions and removals or project activities also included in an emissions trading program or binding emission limit? See the *VCS Program Definitions* for definitions of emissions trading program and binding emission limit.

☐ Yes ☒ No

1.17.2 No Double Claiming with Other Forms of Environmental Credit

Has the project activity sought, received, or is planning to receive credit from another GHG-related environmental credit system? See the *VCS Program Definitions* for definition of GHG-related environmental credit system.

☐ Yes ☒ No

1.17.3 Supply Chain (Scope 3) Emissions

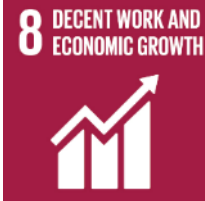
Do the project activities specified in Section 1.12 affect the emissions footprint of any product(s) (goods or services) that are part of a supply chain?

☐ Yes ☒ No

1.18 Sustainable Development Contributions

The project captures the methane release to produce biogas for electricity generation. The project achieves significant GHG emission reductions by destroying methane gas that would have been vented into the atmosphere directly in the absence of the project activity. The project also provides long term job opportunities to local people and contributes to the economic prosperity.

The Project will contribute to sustainable development in the following ways:

SDG	Project Activity Contribution
 SDG 7 “Ensure access to affordable, reliable, sustainable and modern energy for all”	The biogas generated from the project activity will be used for electricity generation for the project activity and upload to the grid. The biogas plant will capture the methane to convert into renewable energy and turn in producing valuable climate and clean air energy solution
 SDG 8 “Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all”	During the construction and operation of the project will provide new job opportunities which will increase the income of local residences and accelerate economy development in the areas.
 SDG 13 “Take urgent action to combat climate change and its impacts	The project will achieve a GHG emission reduction during the crediting period. The project will recover and destroy methane gas that would otherwise be released directly into the atmosphere, The expected total emission reductions are 246,518 tCO₂e during the first 7-year crediting period, with an average annual emission reduction 35,217 tCO₂e

During this monitoring period, the Project will contribute to sustainable development in the following ways.

SDG	SDG Target	SDG Indicator	Net Impact on SDG Indicator	Current Project Activity Contribution	Contributions over project lifetime
7	7.2	Amount of electricity generation	Implemented activities to increase	An average of 7,884 MWh per year of power is uploaded to the grid.	A total of 55,188 MWh of power will be uploaded to the grid during the first crediting period
8	8.5	Number of job opportunities provided	Implemented activities to increase	A total number of 8 job opportunity will be provided to the local	At least total number of 8 job opportunity will be provided to the local

SDG	SDG Target	SDG Indicator	Net Impact on SDG Indicator	Current Project Activity Contribution	Contributions over project lifetime
13	13.0	Tons of greenhouse gas emissions avoided	Implemented activities to increase	An annual average emission reduction of 35,217 tCO ₂ e was achieved from the project activity	A total of 246,518 tCO ₂ e of emission reduction will be achieved during the first crediting period

1.19 Additional Information Relevant to the Project

1.19.1 Leakage Management

In accordance with AMS III.H, Ver. 19, no leakage emissions are considered as the technology is not using equipment transferred from other activity.

In accordance with AMS I.D, Ver. 18, “General guidance on leakage in biomass project activities” shall be followed. As grouped project and PAIs are not demonstrating competing use of biomass, there will be no leakage to be considered

1.19.2 Commercially Sensitive Information

Grouped Project

Where applicable, PAIs to be added to this Grouped Project shall briefly describe the items to which commercially sensitive information has been excluded from the public version of the Project Description if this is relevant.

PAI1

This is not applicable to PAI1 as there is no commercially sensitive information that is or needs to be excluded from the public of the project description.

1.19.3 Further Information

Not applicable. There is no additional relevant legislative, technical, economic, sectoral, social, environmental, geographic, site-specific and/or temporal information that may have a bearing on the eligibility of the project, the GHG emission reductions or carbon dioxide removals, or the quantification of the project’s reductions or removals.

2 SAFEGUARDS AND STAKEHOLDER ENGAGEMENT

2.1 Stakeholder Engagement and Consultation

2.1.1 Stakeholder Identification

Stakeholder Identification

The procedure for the stakeholder identification is outlined and will be used as a guideline for future PAIs. The description of the process to identify stakeholders for the grouped project is as follows:

Step 1: Identify the nearest sensitive receptors (settlements, institutions, government buildings) via google map or gather information's from reliable source such as the district council. Besides that, SIA/ EIA also will be used as some of the reliable sources for stakeholder's identifications. The appropriate radius will be determined by how close the settlements and other sensitive receptors are to the project activity instance's boundary.

Step 2: List all individuals or groups who may have an impact on or be affected by the project. This also includes identification of any relevant government department/agencies that is related to the nature of the project activity.

Step 3: Information about the project activity and details of stakeholder consultations will be published and invitation letters shall be distributed to the identified stakeholders. Individuals who believe they are project stakeholders are encouraged to contact the project proponent directly to participate in the consultation process

The process to identify stakeholders for PAI1 is by extending a radius of 10 km from the project site (Tian Siang Oil Mill (Triang) Sdn Bhd.), assuming that this is the area likely to be impacted by the project. The area within 10km boundary of the project site was found to include local residents, schools, government agencies, housing estates and neighboring plantation estates,

The list of stakeholders identified to be included in the invitation list are:

- Department Of Environment (DOE) Pahang
- Labour Department (Kuantan),
- Department of Occupational Safety and Health (DOSH)

	• Pejabat Daerah Dan Tanah Bera • Communities residing within MRSM Bera • Communities residing within SK LKTP Triang II • Communities residing within SJK (C) Mengkuang • Nearby local residents – Felda Triang II • Felda Triang 01 and Felda Mengkuang millworkers and community residing within the mills • Tian Siang Oil Mill (Triang) Sdn Bhd workers and community residing within the mill • Bera town and housing estates residents The stakeholders identified above are considered adequate to fulfil the criteria to represent communities affected by the project activity within 10km radius of the project site.									
Legal or customary tenure/access rights	The land for the project activity is owned by Tian Siang Oil Mill (Triang) Sdn Bhd. Surrounding area also has plantation area owned by Felda plantation estate and its settlement. There is a small town, Bera located at the north-eastern part of the 10-km perimeter border of the area. Stakeholders within the area are comprised of different ethnicities having diverse cultural diversity, synonym with Malaysian mix of different racial groups. Though having different diversity of culture with different economic and social groups who make up the population, there is no known conflict amongst the community and therefore interactions between the stakeholder groups are harmonious. The nearest indigenous people settlements are located near Tasek Bera which is 40km away from the site⁸. Thus, there is no legal or customary tenure/access rights involved or expected.									
Stakeholder diversity and changes over time	Triang is under Bera District Council. The population estimation according to ethnic group in Bera District Council for 2020⁹ are as below: <table><tr><th>Ethnic Group</th><th>Total</th><th>Percentage (%)</th></tr><tr><td>Bumiputera</td><td>63,400</td><td>63</td></tr><tr><td>Chinese</td><td>25,400</td><td>25</td></tr></table>	Ethnic Group	Total	Percentage (%)	Bumiputera	63,400	63	Chinese	25,400	25
Ethnic Group	Total	Percentage (%)								
Bumiputera	63,400	63								
Chinese	25,400	25								

⁸ <https://www.mdbera.gov.my/en/mdbera/profile/background>
⁹ <https://statsgeo.mycensus.gov.my/geostats/mapv2.php#>

Indian	3,900	4
Others	500	1
Foreigner	6900	7
Total	100,100	100

The population estimation according to ethnic groups in 2005¹⁰ are as below:

Ethnic Group	Total	Percentage (%)
Bumiputera	48,184	54.2
Chinese	24,981	28.1
Indian	4978	5.6
Others	10,757	12.1
Total	88,900	100

There was an increase of 13% of the population from 2005 to 2020. Based on the Population Census 2020, the population in Mukim Triang consist of 5,322 people with the distribution number of male and female, 2,748 and 2,574, respectively.

Within the period of 15 years between the two census periods, there seem to be an increase of the ethic group 'others'. The ethnicity 'others' make up for people who are not categorized in the major three ethnicities in Malaysia, i.e. Malay, Chinese or Indian. These could be people from other races, i.e. foreigners who have been brought into Malaysia for employment as laborers and plantation workers. This is due to increase in numbers of workers to meet the demand of plantation jobs¹¹ and is a trend within Malaysia.

Expected changes in well-being

No major changes to the well-being and stakeholder characteristics for Bera district since the district where the project is located, is an agricultural area located within the central of Peninsular Malaysia. Due to the location of the town being in the rural part of Peninsular Malaysia, it is not expected to show significant development

¹⁰ <https://www.mdbera.gov.my/en/mdbera/profile/background>

¹¹ <https://documents1.worldbank.org/curated/en/892721588859396364/pdf/Who-is-Keeping-Score-Estimating-the-Number-of-Foreign-Workers-in-Malaysia.pdf>

	compared to other more centrally located cities within Peninsular Malaysia in the next decade. Over time, though there may be minimal development, the area where the project is located will still remain to be less-developed compared to cities. Therefore, the project will provide a positive impact to the stakeholders by bringing in investment and more work opportunity to the local community.
Location of stakeholders	The majority of the stakeholders consist of local communities and institutions who inhabit or are located nearby the project activity site. The project activity is surrounded by palm oil plantation, and the nearest large communities identified are located approximately at the edge of the 10km perimeter from the project site.
Location of resources	The area identified within the 10km radius of the proposed project are located in the district of Bera. The area is largely surrounded by palm oil plantation except the north-eastern part where the town and housing estates of residents are located. Felda palm oil plantation are owned by a public listed company. The housing estates are privately owned properties. Therefore, there is no customary access land within the area.

2.1.2 Stakeholder Consultation and Ongoing Communication

Date of stakeholder consultation	31-Oct-2023
Stakeholder engagement process	The stakeholder meeting announcement was opened to the public and widely promoted through postings on Green Lagoon company website (https://glt.my/updates-media/) on 23rd October 2023, and also social medias of GLTSB (Green Lagoon Technology Sdn Bhd) in Facebook and LinkedIn on 21st October 2023. In addition to public announcement, invitation by letter to a number of identified organizations and individuals relevant to the project was also issued. The invitations were either delivered by hand or e-mailed. Examples of the invitation means are available as evidence.
Consultation outcome	<u>Duration of the meeting</u> The stakeholder consultation meeting started at 10.00am and adjourned at 1130am. <u>Location of the meeting</u>

The stakeholder consultation was held in Pahang Rest House @ Bera.

Meeting participants

The meeting was attended by local residents and representatives from the following stakeholder categories:

- i. Local residents,
- ii. Government representatives,
- iii. Schools representative,
- iv. Felda plantation staff, and
- v. Tian Siang POM employees.

A total of 23 participants attended the meetings.

Language

Documentation and meeting were held in both English and local Malay language.

Meeting procedure

The agenda of the meeting include the followings:

- I. Welcoming remarks
- II. Overview of biogas plant (project design, implementation, regulations and benefits etc.)
- III. Introduction to climate change and GHGs
- IV. Introduction to VCS project activities, validation & verification process
- V. Questions and Answers session

Meeting documents and protocols

All public comments received during the meetings were compiled and documented.

The overall response to the project from all invited stakeholders, was encouraging and positive to local communities, seen as improving social-economic conditions. By recovering the wastewater in a more efficient way to eliminate the odour of emissions from the lagoons, the project will also create job opportunities for the locals.

Overall, there were no adverse reactions, comments or clarifications received during the Stakeholder consultation process. There were also no comments or issues raised on the needs to make any changes to the project design.

As a current practice, the project proponent encourages the locals/ participants to actively provide feedback or complaints regarding operation of the project activity.

Ongoing communication	The ongoing communication for the PAI1 were via continuous grievance mechanism. The details for this mechanism were shared with the attendees during the stakeholder consultation. The project proponent shall make sure any current feedback is provided by the locals in fair and transparent manner. Any grievances/ communication between project proponents and all stakeholders can be reachable via the company website: www.glt.my. or via phone call at +60327332960.								
Stakeholder input	Most of the questions were related to the environmental effect of the project activity and benefits from project to the locals. The comment received from the stakeholders consultation is tabulated below: <table border="1"> <thead> <tr> <th>Comments received</th><th>Action taken</th></tr> </thead> <tbody> <tr> <td>Will biogas generated be connected to Tenaga National Berhad (TNB) grid or micro-grid?</td><td>It will be connected to the grid after the voltage goes through step-down. This is a normal process to reduce voltage before connecting to a grid.</td></tr> <tr> <td>What is the type of gas released during flaring?</td><td>Burning of methane by flaring in high temperature >500 degree centigrade will release carbon dioxide</td></tr> <tr> <td>Where is the location of the project and when will it be expected to start operation? Will the project be open for public/school children to visit?</td><td> The biogas plant is located in Triang and is expected to start operation in April 2024. As for the possibility of visiting the plant, they (the stakeholders or any interested parties) are more than welcome to visit. However, prior approval is required and therefore they need to contact the plant manager. There is no safety issue that the public need to worry because methane capture projects is common these days, and this is not the first project in Malaysia. No untoward incidence has been recorded so far in Malaysia </td></tr> </tbody> </table>	Comments received	Action taken	Will biogas generated be connected to Tenaga National Berhad (TNB) grid or micro-grid?	It will be connected to the grid after the voltage goes through step-down. This is a normal process to reduce voltage before connecting to a grid.	What is the type of gas released during flaring?	Burning of methane by flaring in high temperature >500 degree centigrade will release carbon dioxide	Where is the location of the project and when will it be expected to start operation? Will the project be open for public/school children to visit?	The biogas plant is located in Triang and is expected to start operation in April 2024. As for the possibility of visiting the plant, they (the stakeholders or any interested parties) are more than welcome to visit. However, prior approval is required and therefore they need to contact the plant manager. There is no safety issue that the public need to worry because methane capture projects is common these days, and this is not the first project in Malaysia. No untoward incidence has been recorded so far in Malaysia
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	Is the project status under Verra Registry?	Not yet. The PD is still currently under development, pending the information related to stakeholder consultation before being uploaded to VERRA website
There has been no request to project design update brought up during the meeting. All issues raised are summarised above, and no further actions on the need to update project design are required		

2.1.3 Free Prior and Informed Consent

Obtaining consent	The project activity does not affect any IPs, LCs, and customary rights holders. Therefore, there is no ongoing or unresolved conflicts in relation to the project activity The nearest settlements of IPs, LCs are located near Tasek Bera, which is 40km from the project site .
Outcome of FPIC	No consent is needed from the IPs, LCs and customary right holders as the project activity does not involve any encroachment of Indigenous Customary land rights. The project activity was constructed within the existing Tian Siang POM existing boundary.

2.1.4 Grievance Redress Procedure

Development process	The grievance redress procedure applies to comment/feedback/complaints received from stakeholders and publics at grouped project level and respective PAI level, which will be channeled directly to GLT for further action. Any comments/complaints can be made verbally/written to any of the GLT personnel and the complaints will be channeled to the Management Representatives (MR) for further and necessary action.					
Grievance redress procedure	<table><tr><th>Flow/ Process</th><th>Description</th></tr><tr><td>1. Receives and Receipt of comment/ feedback</td><td> The comment/feedback received from stakeholders and publics shall be acknowledged with the respondent, prior to channeling to the Management Representatives (MR). </td></tr></table>	Flow/ Process	Description	1. Receives and Receipt of comment/ feedback	The comment/feedback received from stakeholders and publics shall be acknowledged with the respondent, prior to channeling to the Management Representatives (MR).	
Flow/ Process	Description					
1. Receives and Receipt of comment/ feedback	The comment/feedback received from stakeholders and publics shall be acknowledged with the respondent, prior to channeling to the Management Representatives (MR).					

	2. Record Complaints and Communicate	- Complaints/ comments/ feedback received verbally or in writing shall be kept in a register namely as ‘Complaint Record’. - Complaint Log shall be updated to capture the total number of complaints to date.
	3. Review Feedback and Request Information	- Investigating parties shall review the complaint and request from complainant if the information is insufficient to proceed with the investigation. - All preliminary investigation results shall be recorded in the “Complaint Record” - If the complaint is justifiable for any corrective actions, discussions with relevant parties will be initiated to obtain different views on the subject matter. - If complaint is non-justifiable, inform and explain to the complaining party and that will end the process.
	4. Initiative Corrective	“Corrective/Preventive Action Report” to be initiated if the complaint is valid and justifiable and to be submitted to MR for subsequent actions. - Copy of the ‘Corrective/Preventive Action Report’ shall be extended to the MR for subsequent follow up.
	5. Follow- up & Verify actions & effectiveness	- Follow-up with the implementation of the actions proposed and the effectiveness. - Verification results shall be recorded in: - Corrective/Preventive Action Report - Complaint Record

		Complaint Summary
	6. Update QMS & Risks/ Opportunities	Updating relevant documentation, including the risk(s) and/or opportunities for the respective System Process Procedure (only if deemed necessary)
	7. Feedback to respondent	GLT shall revert to the respondent via post/fax/email, the completed "Corrective/Preventive Action Report" and/or the "Complaint Record", unless the respondent had specifically required GLT to reply via pre-designed forms supplied by the latter.
	8. Compile Data	- Based on the information recorded in "Complaint Summary" - Summarize the data using suitable statistical techniques such as histograms and charts, to provide information of overall status of complaints - Compute the achievement of the objective(s)
	9. Annual Review & Suggests Improvements	- During scheduled Management Review Meeting(s), the trend charts to be presented to the management to determine the current status has any impact on the integrity of the QMS. - Any improvement needed during the Management Review Meeting(s).

2.1.5 Public Comments

One (1) public comment was received on 15-February-2024 during the comment period. The description of the comment and actions taken is tabulated below:..

Comments received	Actions taken
Public comment requested more details about the baseline scenario description	The details description for the baseline scenario is included under section 1.1 of the VCS PD

2.2 Risks to Stakeholders and the Environment

2.2.1 Management Experience

Green Lagoon Technology Sdn Bhd is a Malaysian business entity which was established in May 2010. The Company's goal is to play a role as an environmental solutions provider, contributing to the environmental movement. The Company initially participated in Clean Development Mechanism ("CDM") projects in the palm oil mill industry by providing design, construction, operations, and management services. This laid the foundation for the company to expand into new green products and business models. By converting biogas from palm oil milling into various uses, such as domestic consumption and electricity sales through Malaysia's Feed-in-Tariff (FiT) program, the Company can monetize its slogan, "Powering Waste into Profit." With involvement in more than 60 projects, including investment and ownership in these power assets, the company has become one of Malaysia's top biogas developers.

2.2.2 Risk Assessment

	Risks identified	Mitigation or preventative measure(s) taken
Natural and human-induced risks to stakeholders' wellbeing	Low risk identified	Natural and man-made hazards include droughts, desertification, floods, landslides, wildfires. Earthquakes and tsunamis. The project activity is not expected to cause any natural and human-induced risks to the stakeholder's wellbeing. GLT has formulated Environmental Policy to minimize the impact of the operation activity.
Risks to stakeholder participation	No risk identified	Conduct briefing on the project activity to the involved stakeholder to inform on the project. All stakeholders have been informed during the stakeholder consultation meeting on the safety concern of the project activity
Working conditions	Low risk identified	Policy related to safety of working conditions was established. Emergency Plan and First Aid Management is available in the biogas plant
Safety of women and girls	No risk identified	GLT abide to the Anti-Sexual Harassment Act 2022. General safety (eg, posters) on sexual harassment can be found on-site.
Safety of minority and marginalized groups, including children	No risk identified	The project requires skilled employees to operate, maintain, and manage the biogas plant. Therefore, the project does not employ and is not complicit in any form of child labor. Based on the GLT list of employees, the project does not employ any child

Risks identified		Mitigation or preventative measure(s) taken
		labor. Safety procedures for visitors including children to the biogas plant are also available and therefore safety issues are not a concern for visitors.
Pollutants (air, noise, discharges to water, generation of waste, and release of hazardous materials and chemical pesticides and fertilizers)	Low risk identified	It is expected that the by-product of the project activity shall not release any harmful pollutants that might degrade the environment. During construction and operation phase, the disposal of scheduled waste shall comply with the Environmental Quality (Scheduled Wastes) Regulations 2005. Meanwhile emission of noise shall comply with the Environmental Quality Act 1974.

2.3 Respect for Human Rights and Equity

2.3.1 Labor and Work

Risks identified ¹²		Mitigation or preventative measure(s) taken
Discrimination	Low risk identified	There are no discrimination or sexual harassment occurred or expected to occur in the project site. In the GLT Employee Policy, it has been outlined that there will be no discrimination in employment opportunities.
Sexual harassment	Low risk identified	There are no discrimination or sexual harassment occurred or expected to occur in the project activity. As to date, there is no woman working in the biogas plant. GLT has strictly abide to the Anti-Sexual Harassment Act 2022. General safety reminder (e.g. posters) on sexual harassment can be found on-site.
Equal pay for equal work	No risk identified	In the GLT Employee Policy, has stated that equal pay shall be given to each of the employees based on merit, qualifications and abilities. The payment

¹² The identified risks and commensurate mitigation or preventative measure(s) for forced labor, child labor, and human trafficking, must be inclusive of staff and contracted workers employed by third parties.

Risks identified ¹²		Mitigation or preventative measure(s) taken
		of salary and wages of GLT employees are in accordance with the Employment Act 1955.
Gender equity in labor and work	No risk identified	There is no gender discrimination in terms of labor in this project activity. Equal job opportunities were provided to both genders and demonstrated equal pay to both. In GLT Headquarter, the staff ratio of 60% to 40% for male to female. Overall, at the Group level, a gender ratio of 95% to 5% for male to female, with males being dominant in biogas plants operation. Men work in upstream and technical activities, such as putting feedstock into the digester, cleaning clogged feedstock channels and biogas pipe distribution. Besides, most of the operational plants are located in rural areas where safety is one of the utmost concerns. That is why all the trained personnel that work at operational plants are men.
Forced labor	No risk identified	GLT comply with the Employment Act 1955 to protect the legitimate rights and interests of workers and prohibit forced labor
Child labor	No risk identified	Employment regulations are described in the Malaysia Employment Act 1955. The Project requires skilled employees to operate, maintain and manage the biogas plant. Therefore, the project does not employ and is not complicit in any form of child labor. Based on GLT's list of employees, it can be confirmed that the project does not employ any child labor.
Human trafficking	No risk identified	The project activity will not employ victims of human trafficking in their operation. GLT abide with the Employment Act 1955.

2.3.2 Human Rights

Risks identified	Mitigation or preventative measure(s) taken
Low risk identified	This project does not discriminate with regards to participation and inclusion. The project abides by the rules of equality accordingly and does not involve and is not complicit in any form of discrimination. GLT Employee Policy has an outline on equality and discrimination.

2.3.3 Indigenous Peoples and Cultural Heritage

Risks identified	Mitigation(s) or preventative measure taken
No risk identified	There is no risk identified as the project activity does not involve any indigenous peoples and also cultural heritage in or within the project area. PAI1 is sited within the Tian Siang Palm Oil Mill boundary.

2.3.4 Property Rights

Risks identified	Mitigation or preventative measure(s) taken
No risk identified	The PAI1 does not involve any legal or customary tenure/access rights to territories, property, and resources, including collective and/or conflicting rights, held by stakeholders. The land for the PAI1 is owned by the mill.

2.3.5 Benefit Sharing

Process used to design the benefit sharing plan	No risk identified for PAI1 as the project does not impact any property rights
Summary of the benefit sharing plan	No risk identified for PAI1 as the project does not impact any property rights
Approval and dissemination of benefit sharing plan	No risk identified for PAI1 as the project does not impact any property rights

2.4 Ecosystem Health

	Risks identified	Mitigation or preventative measure(s) taken
Impacts on biodiversity and ecosystems	Low risk identified	There will be low risk on the impacts on biodiversity and ecosystems since the sitting of the PA1 is located on the existing mill, i.e. has been developed many years ago. Furthermore, there were no new areas explored for PAI1. Based on the on-site observation and interviews done with the mill workers prior to the site clearing of the PAI1, there were no appearances of endangered species/ big mammals, only snakes are commonly seen. A regular monitoring is advisable to be conducted to observe any appearance of endangered species within the PAI1 boundary.
Soil degradation and soil erosion	Low risk identified	There will be low risk on soil degradation and soil erosion since the sitting of the PA1 is located on the existing mill, i.e. has been developed many years ago. Furthermore, there were no new areas explored for PAI1. A potential risk which is low is hydroseeding/ turfing which will be implemented at any erosion-prone area within the PAI1 boundary. GLT has included the hydroseeding layout in their engineering drawings.
Water consumption and stress	Low risk identified	Low impact on water stress and consumption as the water is only used for daily usage in the office and not for operation wise.

2.4.1 Rare, Threatened, and Endangered Species

Is the project located in or adjacent to habitats for rare, threatened, or endangered species?

☐ Yes

☒ No

Based on interviews done with the mill worker, there was no presence of rare, threatened, and endangered species observed within the project boundary

2.4.2 Introduction of Species

Not applicable as the Grouped Project and PAI1 is not introducing any species.

2.4.3 Ecosystem Conversion

Not applicable as the Grouped Project and PAI1 does not involve ARR, ALM, WRC or ACoGS projects.

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

Grouped Project

Type (methodology, tool or module).	Reference ID, if applicable	Title	Version
Methodology	AMS-III.H	AMS-III.H.: Methane recovery in wastewater treatment	19.0
Tool	TOOL03	Tool to calculate project or leakage CO ₂ emissions from fossil fuel combustion	03.0
Tool	TOOL04	Emissions from solid waste disposal sites	08.1
Tool	TOOL05	Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation	03.0
Tool	TOOL06	Project emissions from flaring	04.0
Tool	TOOL32	Positive lists of technologies	04.0
Methodology	AMS-I.D	AMS-I.D: Grid connected renewable electricity generation	18.0
Tool	TOOL03	Tool to calculate project or leakage CO ₂ emissions from fossil fuel combustion	03.0
Tool	TOOL07	Tool to calculate the emission factor for an electricity system	07.0
Tool	TOOL10	Tool to determine the remaining lifetime of an equipment	01
Tool	TOOL16	Project and leakage emissions from biomass	05.0
Tool	TOOL11	Assessment of the validity of the original/current baseline and update of the baseline at the renewal of the crediting period	03.0.1

Note that PAIs within the Grouped Project may not require the use of all the above approved methodologies and methodological tools. Relevance and/or applicability should be assessed on a case by-case basis.

PAI1

Type (methodology, tool or module).	Reference ID, if applicable	Title	Version
Methodology	AMS-III.H	AMS-III.H.: Methane recovery in wastewater treatment	19.0
Tool	TOOL05	Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation	03.0
Tool	TOOL06	Project emissions from flaring	04.0
Tool	TOOL32	Positive lists of technologies	04.0
Methodology	AMS-I.D	AMS-I.D: Grid connected renewable electricity generation	18.0
Tool	TOOL07	Tool to calculate the emission factor for an electricity system	07.0
Tool	TOOL10	Tool to determine the remaining lifetime of an equipment	01

3.2 Applicability of Methodology

The project includes small-scale activities for which methodologies of type AMS-III.H, and AMS-I.D are applicable as well as the methodological tool referred in the methodology. The possible combination of the methodologies for new PAIs must demonstrate the respective eligibility criteria tabulated below

Grouped Project

Methodology ID	Applicability condition	Justification of compliance
AMS III H	As per paragraph 2 of AMS-III.H. The methodology comprises measures that recover biogas from biogenic organic matter in	<u>Potentially Applicable</u> If applicable to a PAI within the Grouped Project, PAIs to be implemented under this Grouped Project must recover biogas from

Methodology ID	Applicability condition	Justification of compliance
	wastewater by means of one, or a combination, of the following options: (a) Substitution of aerobic wastewater or sludge treatment systems with anaerobic systems with biogas recovery and combustion; (b) Introduction of anaerobic sludge treatment system with biogas recovery and combustion to a wastewater treatment plant without sludge treatment; (c) Introduction of biogas recovery and combustion to a sludge treatment system; (d) Introduction of biogas recovery and combustion to an anaerobic wastewater treatment system such as anaerobic reactor, lagoon, septic tank or an on-site industrial plant; (e) Introduction of anaerobic wastewater treatment with biogas recovery and combustion, with or without anaerobic sludge treatment to an untreated wastewater stream; (f) Introduction of a sequential stage of wastewater treatment with biogas recovery and combustion, with or without sludge treatment, to an anaerobic wastewater treatment system without biogas recovery (e.g. introduction of treatment in an anaerobic reactor with biogas recovery as a sequential treatment step for the wastewater that is presently being treated in an anaerobic lagoon without methane recovery).	wastewater treatment system which would have otherwise been vented to the atmosphere.
AMS III H	As per paragraph 3 of AMS-III.H. In cases where baseline system is anaerobic lagoon, the methodology is applicable if: (a) The lagoons are ponds with a depth greater than two meters,	<u>Potentially Applicable</u> If applicable to a PAI within the Grouped Project, the baseline system for the PAIs to be implemented under this Grouped Project is anaerobic

Methodology ID	Applicability condition	Justification of compliance
	without aeration. The value for depth is obtained from engineering design documents, or through direct measurement, or by dividing the surface area by the total volume. If the lagoon filling level varies seasonally, the average of the highest and lowest levels may be taken; (b) Ambient temperature above 15°C, at least during part of the year on a monthly average basis; (c) The minimum interval between two consecutive sludge removal events shall be 30 days.	lagoon of which must satisfy the following: (a) The lagoons are deeper than two meters and without aeration; (b) Ambient temperature above 15°C, at least during part of the year on a monthly average basis, for Malaysia as having tropical climate¹³; (c) The minimum interval between two consecutive sludge removal events is more than 30 days¹⁴.
AMS III H	As per paragraph 4 of AMS-III.H. The recovered biogas from the above measures may also be utilised for the following applications instead of combustion/flaring: (a) Thermal or mechanical, electrical energy generation directly; (b) Thermal or mechanical, electrical energy generation after bottling of upgraded biogas, in this case, additional guidance provided in Annex 1 shall be followed; or (c) Thermal or mechanical, electrical energy generation after upgrading and distribution, in this case, additional guidance provided in Annex 1 shall be followed: (i) Upgrading and injection of biogas into a natural gas distribution grid with no significant transmission constraints; (ii) Upgrading and transportation of biogas via a dedicated piped network to a group of end users; or	<u>Potentially Applicable</u> If applicable to a PAI within the Grouped Project, PAIs to be implemented under this Grouped Project must involve facilities to utilize biogas generated for renewable energy and that excess gas is burned via flaring.

¹³ Malaysia climate: <https://climateknowledgeportal.worldbank.org/country/malaysia/climate-data-historical#:~:text=Malaysia%20has%20a%20tropical%20climate,25.9%C2%B0C%20in%20May>.

¹⁴ Refer to the desludging record from respective PAI's POM.

Methodology ID	Applicability condition	Justification of compliance
	(iii) Upgrading and transportation of biogas (e.g. by trucks) to distribution points for end users. (d) Hydrogen production. (e) Use as fuel in transportation applications after upgrading.	
AMS III H	As per paragraph 5 of AMS-III.H. If the recovered biogas is used for project activities covered under paragraph 4 (a), that component of the project activity can use a corresponding methodology under Type I.	<u>Potentially Applicable</u> If applicable to a PAI within the Grouped Project, PAIs to be implemented under this Grouped Project must have methane captured utilized for renewable electricity generation of which the approved methodologies AMS-I.D is used.
AMS III H	As per paragraph 6 of AMS-III.H. For project activities covered under paragraph 4 (b), if bottles with upgraded biogas are sold outside the project boundary, the end use of the biogas shall be ensured via a contract between the bottled biogas vendor and the end-user. No emission reductions may be claimed from the displacement of fuels from the end use of bottled biogas in such situations. If, however, the end use of the bottled biogas is included in the project boundary and is monitored during the crediting period, CO₂ emissions avoided by the displacement of fossil fuel can be claimed under the corresponding Type I methodology, e.g. AMS-I.C Thermal energy production with or without electricity.	<u>Potentially Applicable</u> If applicable to a PAI within the Grouped Project, bottled upgraded biogas is sold outside of the project boundary, end-use of the biogas will be ensured via a contract between bottled biogas vendor and end user. End users will not claim emission reduction for fuel displacement. For end-use within project boundary, emission from displacement of fuel can be claimed.
AMS III H	As per paragraph 7 of AMS-III.H.	<u>Potentially Applicable</u>

Methodology ID	Applicability condition	Justification of compliance
	For project activities covered under paragraph 4 (c) (i), emission reductions from the displacement of the use of natural gas are eligible under this methodology, provided the geographical extent of the natural gas distribution grid is within the host country boundaries.	If applicable to a PAI within the Grouped Project, emission reduction displacement of use of natural gas are eligible as it is located within the host country boundary distribution grid.
AMS III H	As per paragraph 8 of AMS-III.H. For project activities covered under paragraph 4 (c) (ii), emission reductions for the displacement of the use of fuels can be claimed following the provision in the corresponding Type I methodology, e.g. AMS-I.C.	<u>Not Applicable</u> For PAIs to be implemented under this Grouped Project, there will be no activities of transportation of natural gas distribution via a dedicated piped network to a group of end user(s), for the PAIs. Thus is not claimed as in AMS-I.C.
AMS III H	As per paragraph 9 of AMS-III.H. In particular, for the case of 4 (b) and (c) (iii), the physical leakage during storage and transportation of upgraded biogas as well as the emissions from fossil fuel consumed by vehicles for transporting biogas shall be considered. Relevant procedures in paragraph 18 of the appendix of AMS-III.H Methane recovery in wastewater treatment shall be followed in this regard.	<u>Potentially Applicable</u> If applicable to a PAI within the Grouped Project, for bottling, upgrading and transportation to distribution points, physical leakage will be considered in accordance with paragraph 18 or AMS-III.H.
AMS III H	As per paragraph 10 of AMS-III.H. For project activities covered under paragraph 4 (b) and (c), this methodology is applicable if the upgraded methane content of the biogas is in accordance with relevant national regulations (where these	<u>Potentially Applicable</u> If applicable to a PAI within the Grouped Project, for PAIs to be implemented under this Grouped Project, as relevant national regulations do not exist, the minimum

Methodology ID	Applicability condition	Justification of compliance
	exist) or in the absence of national regulations, a minimum of 96% (by volume).	requirement of 96% (by volume) applies.
AMS III H	As per paragraph 11 of AMS-III.H. If the recovered biogas is utilized for the production of hydrogen (project activities covered under paragraph 4 (d)), that component of the project activity shall use the corresponding methodology AMS-III.O Hydrogen production using methane extracted from biogas.	<u>Potentially Applicable</u> If applicable to a PAI within the Grouped Project, the recovered biogas is utilized for the production of hydrogen that component of the project activity shall use the corresponding methodology AMS-III.O Hydrogen production using methane extracted from biogas
AMS III H	As per paragraph 12 of AMS-III.H. If the recovered biogas is used for project activities covered under paragraph 4 (e), that component of the project activity shall use the corresponding methodology AMS-III.AQ Introduction of Bio-CNG in road transportation.	<u>Not Applicable</u> For PAIs to be implemented under this Grouped Project, there will be no activities that use the upgraded biogas as fuel in transportation applications for the PAIs.
AMS III H	As per paragraph 13 of AMS-III.H. New facilities (Greenfield projects) and project activities involving a change of equipment resulting in a capacity addition of the wastewater or sludge treatment system compared to the designed capacity of the baseline treatment system are only eligible to apply this methodology if they comply with the relevant requirements in the General guidelines to SSC CDM methodologies. In addition, the requirements for demonstrating the remaining lifetime of the equipment	<u>Not Applicable</u> PAIs to be implemented under this Grouped Project is not a Greenfield project and the project activities do not result in capacity addition of the treatment system.

Methodology ID	Applicability condition	Justification of compliance
	replaced as described in the general guidelines shall be followed.	
	As per paragraph 2.2.14 of AMS-III.H. The location of the wastewater treatment plant as well as the source generating the wastewater shall be uniquely defined and described in the Project Design Document (PDD).	<u>Potentially Applicable</u> If applicable to a PAI within the Grouped Project, the location of each PAIs will be identified by specific reference code, address, map and GPS coordinates. The unique location of the wastewater generating source will be defined and described in Section 1.12. of Project Description (PD), and also shown in the baseline and project activity diagram
AMS III H	As per paragraph 2.2.15 of AMS-III.H Measures are limited to those that result in aggregate emission reductions of less than or equal to 60 ktCO₂ equivalent annually from all Type III components of the project activity.	<u>Potentially Applicable</u> The project activity instance under grouped project activity will result in less than or equal to 60ktCO₂ equivalent annually from all Type III, as demonstrated in section 1.10 of PD.
AMS I D	As per paragraph 2 of AMS-I.D This methodology comprises renewable energy generation units such as photovoltaic, hydro, tidal/wave, wind, geothermal and renewable biomass: (a) Supplying electricity to a national or a regional grid; or (b) Supplying electricity to an identified consumer facility via national/regional grid through a contractual arrangement such as wheeling.	<u>Potentially Applicable</u> If applicable to a PAI within the Grouped Project, the PAIs comprises renewable energy generation units such as photovoltaic, hydro, tidal/wave, wind, geothermal and renewable biomass: (a) Supplying electricity to a national or a regional grid; or (b) Supplying electricity to an identified consumer facility via national/regional grid through a contractual arrangement such as wheeling.
AMS I D	As per paragraph 4 of AMS-I.D This methodology is applicable to project activities that:	<u>Potentially Applicable</u> If applicable to a PAI within the Grouped Project, the PAIs will satisfy:

Methodology ID	Applicability condition	Justification of compliance
	(a) Install a Greenfield plant; (b) Involve a capacity addition in (an) existing plant(s); (c) Involve a retrofit of (an) existing plant(s); (d) Involve a rehabilitation of (an) existing plant(s)/ unit(s); or (e) Involve a replacement of (an) existing plant(s).	(a) Install a Greenfield plant; (b) Involve a capacity addition in (an) existing plant(s); (c) Involve a retrofit of (an) existing plant(s); (d) Involve a rehabilitation of (an) existing plant(s)/ unit(s); or (e) Involve a replacement of (an) existing plant(s).
AMS I D	As per paragraph 5 of AMS-I.D Hydro power plants with reservoirs that satisfy at least one of the following conditions are eligible to apply this methodology: (a) The project activity is implemented in an existing reservoir with no change in the volume of reservoir; (b) The project activity is implemented in an existing reservoir where the volume of reservoir is increased and the power density of the project activity, as per definitions given in the project emissions section, is greater than 4W/m ² ; or (c) The project activity results in new reservoirs and the power density of the power plant, as per definitions given in the project emissions section, is greater than 4W/m ² .	<u>Not Applicable</u> Project activity instances under grouped project activity are not hydro power plant projects.
AMS I D	As per paragraph 6 of AMS-I.D If the new unit has both renewable and non-renewable components (e.g. a wind/diesel unit), the eligibility limit of 15MW for a small-scale CDM project activity applies only to the	<u>Potentially Applicable</u> If applicable to a PAI within the Grouped Project, the PAIs will have both renewable and non-renewable components (e.g. a wind/diesel unit), the eligibility limit of 15MW for a

Methodology ID	Applicability condition	Justification of compliance
	renewable component. If the new unit co-fires fossil fuel, the capacity of the entire unit shall not exceed the limit of 15MW.	small-scale CDM project activity applies only to the renewable component. If the new unit co-fires fossil fuel, the capacity of the entire unit shall not exceed the limit of 15MW.
AMS I D	As per paragraph 7 of AMS-I.D Combined heat and power (co-generation) systems are not eligible under this category.	<u>Not Applicable</u> PAI under the grouped project activity will not involve combined heat and power (co-generation) systems.
AMS I D	As per paragraph 8 of AMS-I.D In the case of project activities that involve the capacity addition of renewable energy generation units at an existing renewable power generation facility, the added capacity of the units added by the project should be lower than 15MW and should be physically distinct from the existing units.	<u>Not Applicable</u> PAI under the grouped project activity will not involve the capacity addition of renewable power generation facility.
AMS I D	As per paragraph 9 of AMS-I.D In the case of retrofit, rehabilitation or replacement, to qualify as a small-scale project, the total output of the retrofitted, rehabilitated or replacement power plant/ unit shall not exceed the limit of 15MW.	<u>Not Applicable</u> PAI under the grouped project activity will not retrofit, rehabilitation or replacement facility.
AMS I D	As per paragraph 10 of AMS-I.D In the case of landfill gas, waste gas, wastewater treatment and agro-industries projects, recovered methane emissions are eligible under a relevant Type III category. If the recovered methane is used for electricity generation for supply to the grid then the baseline for the	<u>Potentially Applicable</u> If applicable to a PAI within the Grouped Project, Project activity instances under grouped project activity are wastewater treatment projects, and eligible under Type III category and the component is in accordance with type III methodology.

Methodology ID	Applicability condition	Justification of compliance
	electricity component shall be in accordance with procedure prescribed under this methodology. If the recovered methane is used for heat generation or cogeneration other applicable Type-I methodologies such as 'AMS-I.C: Thermal energy production with or without electricity' shall be explored.	
AMS I D	As per paragraph 11 of AMS-I.D In case biomass is sourced from dedicated plantations, the applicability criteria in the tool 'Project emissions from cultivation of biomass' shall apply.	<u>Not Applicable</u> PAI under the grouped project activity will not using biomass sourced from dedicated plantations (i.e. grown specifically for the grouped project utilization).

Tool ID	Applicability condition	Justification of compliance
TOOL03	This tool provides procedures to calculate project and/or leakage CO ₂ emissions from the combustion of fossil fuels. It can be used in cases where CO ₂ emissions from fossil fuel combustion are calculated based on the quantity of fuel combusted and its properties. Methodologies using this tool should specify to which combustion process j this tool is being applied.	<u>Potentially Applicable</u> If applicable to a PAI within the Grouped Project, PAIs to be implemented will account for CO ₂ emissions from fossil fuel combustion based on the quantity of fuel combusted and its properties.
TOOL04	The tool can be used to determine emissions for the following types of applications: (a) Application A: The CDM project activity mitigates methane emissions from a specific existing SWDS. Methane emissions are mitigated by	<u>Potentially Applicable</u> If applicable to a PAI within the Grouped Project, PAIs to be implemented will account for the methane emissions from anaerobic decay of sludge produced.

Tool ID	Applicability condition	Justification of compliance
	capturing and flaring or combusting the methane (e.g. “ACM0001: Flaring or use of landfill gas”). The methane is generated from waste disposed in the past, including prior to the start of the CDM project activity. In these cases, the tool is only applied for an ex ante estimation of emissions in the project design document (CDM-PDD). The emissions will then be monitored during the crediting period using the applicable approaches in the relevant methodologies (e.g. measuring the amount of methane captured from the SWDS); (b) Application B: The CDM project activity avoids or involves the disposal of waste at a SWDS. An example of this application of the tool is ACM0022, in which municipal solid waste (MSW) is treated with an alternative option, such as composting or anaerobic digestion, and is then prevented from being disposed of in a SWDS. The methane is generated from waste disposed or avoided from disposal during the crediting period. In these cases, the tool can be applied for both ex ante and ex post estimation of emissions. These project activities may apply the simplified approach detailed in 0 when calculating baseline emissions.	

Tool ID	Applicability condition	Justification of compliance
	In the case that: (a) different types of residual waste are disposed or prevented from disposal; or that (b) both MSW and residual waste(s) are prevented from disposal, then the tool should be applied separately to each residual waste and to the MSW.	<u>Not Applicable</u> PAIs to be implemented under this Grouped Project will not prevent or involve disposal of any type of residual waste or MSW at a SWDS.
TOOL05	If emissions are calculated for electricity consumption, the tool is only applicable if one out of the following three scenarios applies to the sources of electricity consumption: (a) Scenario A: Electricity consumption from the grid. The electricity is purchased from the grid only, and either no captive power plant(s) is/are installed at the site of electricity consumption or, if any captive power plant exists on site, it is either not operating or it is not physically able to provide electricity to the electricity consumer; (b) Scenario B: Electricity consumption from (an) off-grid fossil fuel fired captive power plant(s). One or more fossil fuel fired captive power plants are installed at the site of the electricity consumer and supply the consumer with electricity. The captive power plant(s) is/are not connected to the electricity grid; or (c) Scenario C: Electricity consumption from the grid and (a) fossil fuel fired captive power	<u>Potentially Applicable</u> If applicable to a PAI within the Grouped Project, PAIs to be implemented will account for its electricity consumption emission, based on one out of the following three scenarios applies to its sources of electricity consumption.

Tool ID	Applicability condition	Justification of compliance
	plant(s). One or more fossil fuel fired captive power plants operate at the site of the electricity consumer. The captive power plant(s) can provide electricity to the electricity consumer. The captive power plant(s) is/are also connected to the electricity grid. Hence, the electricity consumer can be provided with electricity from the captive power plant(s) and the grid.	
	This tool can be referred to in methodologies to provide procedures to monitor amount of electricity generated in the project scenario, only if one out of the following three project scenarios applies to the recipient of the electricity generated: (a) Scenario I: Electricity is supplied to the grid; (b) Scenario II: Electricity is supplied to consumers/electricity consuming facilities; or (c) Scenario III: Electricity is supplied to the grid and consumers/electricity consuming facilities.	<u>Potentially Applicable</u> If applicable to a PAI within the Grouped Project, PAIs to be implemented will monitor amount of electricity generated, based on one out of the following three project scenarios applies to its recipient of the electricity generated.
	This tool is not applicable in cases where captive renewable power generation technologies are installed to provide electricity in the project activity, in the baseline scenario or to sources of leakage. The tool only accounts for CO₂ emissions.	<u>Potentially Applicable</u> If applicable to a PAI within the Grouped Project, PAIs to be implemented will install captive renewable power generation technologies to provide electricity in the project activity, in the baseline

Tool ID	Applicability condition	Justification of compliance
		scenario or to sources of leakage. The tool only accounts for CO ₂ emissions.
TOOL06	This tool provides procedures to calculate project emissions from flaring of a residual gas. The tool is applicable to enclosed or open flares and project participants should document in the CDM-PDD the type of flare used in the project activity.	<u>Potentially Applicable</u> If applicable to a PAI within the Grouped Project, PAIs to be implemented will account for its project emissions from flaring of a residual gas.
	This tool is applicable to the flaring of flammable greenhouse gases where: (a) Methane is the component with the highest concentration in the flammable residual gas; and (b) The source of the residual gas is coal mine methane or a gas from a biogenic source (e.g. biogas, landfill gas or wastewater treatment gas).	<u>Potentially Applicable</u> If applicable to a PAI within the Grouped Project, PAIs to be implemented will involve the flaring of flammable greenhouse gases where: (a) Methane is the component with the highest concentration in the flammable residual gas; and (b) The source of the residual gas is coal mine methane or a gas from a biogenic source (e.g. biogas, landfill gas or wastewater treatment gas).
	The tool is not applicable to the use of auxiliary fuels and therefore the residual gas must have sufficient flammable gas present to sustain combustion. In the case of an enclosed flare, there shall be operating specifications provided by the manufacturer of the flare and these shall be followed by the project participant.	<u>Potentially Applicable</u> If applicable to a PAI within the Grouped Project, PAIs to be implemented will involve the use of auxiliary fuels and therefore the residual gas must have sufficient flammable gas present to sustain combustion. In the case of an enclosed flare, there shall be operating specifications provided by the manufacturer of the flare and these shall be followed by the project participant.

Tool ID	Applicability condition	Justification of compliance
TOOL07	This tool may be applied to estimate the OM, BM and/or CM when calculating baseline emissions for a project activity that substitutes grid electricity that is where a project activity supplies electricity to a grid or a project activity that results in savings of electricity that would have been provided by the grid (e.g. demand-side energy efficiency projects).	<u>Potentially Applicable</u> If applicable to a PAI within the Grouped Project, PAIs to be implemented will estimate the OM, BM and/or CM when calculating baseline emissions for a project activity that substitutes grid electricity that is where a project activity supplies electricity to a grid or a project activity that results in savings of electricity that would have been provided by the grid (e.g. demand-side energy efficiency projects).
	Under this tool, the emission factor for the project electricity system can be calculated either for grid power plants only or, as an option, can include off-grid power plants. In the latter case, two sub-options under the step 2 of the tool are available to the project participants, i.e. option IIa and option IIb. If option IIa is chosen, the conditions specified in “Appendix 1: Procedures related to off-grid power generation” should be met. Namely, the total capacity of off-grid power plants (in MW) should be at least 10 per cent of the total capacity of grid power plants in the electricity system; or the total electricity generation by off-grid power plants (in MWh) should be at least 10 per cent of the total electricity generation by grid power plants in the electricity system; and that factors which negatively affect the reliability and stability of the grid are primarily due	<u>Potentially Applicable</u> If applicable to a PAI within the Grouped Project, PAIs to be implemented will calculate either for grid power plants only or, as an option, can include off-grid power plants.

Tool ID	Applicability condition	Justification of compliance
	to constraints in generation and not to other aspects such as transmission capacity.	
	In case of CDM projects the tool is not applicable if the project electricity system is located partially or totally in an Annex I country.	<u>Not Applicable</u> PAIs to be implemented under this Grouped Project will not locate partially or totally in an Annex I country.
TOOL10	The tool provides guidance to determine the remaining lifetime of baseline or project equipment. The tool may, for example, be used for project activities which involve the replacement of existing equipment with new equipment or which retrofit existing equipment as part of energy efficiency improvement activities. Methodologies referring to this tool should clearly specify for which equipment the remaining lifetime should be determined. The remaining lifetime of relevant equipment shall be determined prior to the implementation of the project activity. Project participants using this tool shall document transparently in the CDM-PDD how the remaining lifetime of applicable equipment has been determined, including (references to) all documentation used. Under this tool, impacts on the lifetime of the equipment due to policies and regulations (e.g. environmental regulations) or changes in the services needed (e.g. increased energy demand) are not	<u>Not Applicable</u> PAIs to be implemented under this Grouped Project will be a Greenfield power generation, therefore it is not applicable to determine the remaining lifetime of baseline or project equipment

Tool ID	Applicability condition	Justification of compliance
	considered. Methodologies referring to this tool shall, where applicable, provide specific guidance on how regulations that warrant the replacement of the equipment before it has reached the end of its technical lifetime should be addressed.	
TOOL11	This tool provides a stepwise procedure to assess the continued validity of the baseline and to update the baseline at the renewal of a crediting period, as required by paragraph 49 (a) of the modalities and procedures of the clean development mechanism. The tool consists of two steps. The first step provides an approach to evaluate whether the current baseline is still valid for the next crediting period. The second step provides an approach to update the baseline in case that the current baseline is not valid anymore for the next crediting period.	<u>Potentially Applicable</u> If applicable to a PAI within the Grouped Project, PAIs to be implemented will assess the continued validity of the baseline and to update the baseline at the renewal of a crediting period
TOOL16	This tool shall be applied in conjunction with the methodology that refers to this tool. For project activities which include biomass cultivation: (a) The land in which biomass is cultivated: (i) Does not contain wetlands; (ii) Does not contain organic soils as defined in (iii) Is not subjected to flood irrigation.	<u>Potentially Applicable</u> If applicable to a PAI within the Grouped Project, PAIs to be implemented will determine its project and leakage emissions from biomass. PAIs to be implemented that include biomass cultivation will follow the conditions as follow: (a) The land in which biomass is cultivated: (i) Does not contain wetlands;

Tool ID	Applicability condition	Justification of compliance
	(b) The land in which biomass is cultivated: (i) Does not contain forest nor contained forest since 31 December 1989; or (ii) Contains a forest plantation that before the start of the project will be harvested and the land would be neither reforested nor will regenerate on its own into a forest in the absence of the project activity. (c) Desalination is not a substantial source of water in the host country.	(ii) Does not contain organic soils as defined in (iii) Is not subjected to flood irrigation. (b) The land in which biomass is cultivated: (i) Does not contain forest nor contained forest since 31 December 1989; or (ii) Contains a forest plantation that before the start of the project will be harvested and the land would be neither reforested nor will regenerate on its own into a forest in the absence of the project activity. (c) Desalination is not a substantial source of water in the host country.
	In case the land contains a forest plantation, the project participants shall demonstrate that before the start of the project activity the plantation will be finally harvested and regeneration to forestland (according to the respective national definition) will not take place. In doing so, the project proponent shall: (a) Identify realistic and credible alternatives with regard to the possible land use scenarios that would occur in the absence of the project activity, including but not limited to: (i) The forest plantation continues under the	<u>Potentially Applicable</u> If applicable to a PAI within the Grouped Project, PAIs to be implemented will involve a land containing forest plantation and follow the conditions that the project participants shall demonstrate that before the start of the project activity the plantation will be finally harvested and regeneration to forestland (according to the respective national definition) will not take place. In doing so, the project proponent shall: (a) Identify realistic and credible alternatives with regard to the possible land use scenarios that would occur in

Tool ID	Applicability condition	Justification of compliance
	current management practice; (ii) The forest plantation is harvested and the land is replanted; (iii) The forest plantation is harvested and the land is abandoned; (b) Assess the economic attractiveness of the existing forest plantation by applying Step 2 of the “TOOL01: Tool for the demonstration and assessment of additionality”; (c) Confirm, based on the plantation management practices in the region for the considered species, that the situation referred to in paragraph 7(b)(ii) is the common practice; and (d) Use relevant credible evidence, including but not limited to official land use maps, satellite images/aerial photographs, cadastral information, official land use records.	the absence of the project activity, including but not limited to: (i) The forest plantation continues under the current management practice; (ii) The forest plantation is harvested and the land is replanted; (iii) The forest plantation is harvested and the land is abandoned; (b) Assess the economic attractiveness of the existing forest plantation by applying Step 2 of the “TOOL01: Tool for the demonstration and assessment of additionality”; (c) Confirm, based on the plantation management practices in the region for the considered species, that the situation referred to in paragraph 7(b)(ii) is the common practice; and (d) Use relevant credible evidence, including but not limited to official land use maps, satellite images/aerial photographs, cadastral information, official land use records.
	The tool is also applicable if biomass residues are consumed in a CDM project activity, and the biomass residues can be utilized after processing or without processing. These could be:	<u>Potentially Applicable</u> If applicable to a PAI within the Grouped Project, PAIs to be implemented will consume biomass residues and the biomass residues

Tool ID	Applicability condition	Justification of compliance
	(a) Procured by the project proponents; or (b) The result of an agro-industrial process under the control of the project proponents.	can be utilized after processing or without processing. These could be: (a) Procured by the project proponents; or (b) The result of an agro-industrial process under the control of the project proponents.
TOOL32	The use of this methodological tool is not mandatory for the project participants of a CDM project activity or CDM PoA for demonstrating their additionality. This methodological tool shall be applied in conjunction with a small-scale or large-scale methodology which refers to this tool.	<u>Potentially Applicable</u> If applicable to a PAI within the Grouped Project, PAIs to be implemented will use the tool to demonstrate their additionality in conjunction with a small-scale methodology which refers to this tool.
	The positive lists as contained in section 5 of this tool are valid up to 10 March 2025. Notwithstanding the provisions on the validity of new, revised and previous versions of methodologies and methodological tools in the “Procedure: Development, revision and clarification of baseline and monitoring methodologies and methodological tools”, there will be no grace period for the application of this tool and the validity of the positive list after this date, including in cases where further technologies are added to the positive list through revisions of this tool before this date.	<u>Potentially Applicable</u> If applicable to a PAI within the Grouped Project, PAIs to be implemented will use the tool to demonstrate their additionality as the positive lists are valid up to 10 March 2025. The grouped project has completed validation and submitted a request for registration on 03 March 2025. Thus, this applicability condition is met. As per the Clarification to CDM TOOL32 Positive Lists of Technologies published by Verra on 18 March 2025, this tool remains active for use in the VCS Program until 30 September 2025.

PAI1

Methodology ID	Applicability condition	Justification of compliance
AMS III H	As per paragraph 2 of AMS-III.H. The methodology comprises measures that recover biogas from biogenic organic matter in wastewater by means of one, or a combination, of the following options: (a) Substitution of aerobic wastewater or sludge treatment systems with anaerobic systems with biogas recovery and combustion; (b) Introduction of anaerobic sludge treatment system with biogas recovery and combustion to a wastewater treatment plant without sludge treatment; (c) Introduction of biogas recovery and combustion to a sludge treatment system; (d) Introduction of biogas recovery and combustion to an anaerobic wastewater treatment system such as anaerobic reactor, lagoon, septic tank or an on-site industrial plant; (e) Introduction of anaerobic wastewater treatment with biogas recovery and combustion, with or without anaerobic sludge treatment to an untreated wastewater stream; (f) Introduction of a sequential stage of wastewater treatment with biogas recovery and combustion, with or without sludge treatment, to an anaerobic wastewater treatment system without biogas recovery (e.g. introduction of treatment in an anaerobic reactor with biogas recovery as a sequential treatment step for the wastewater that is presently being treated in an anaerobic lagoon without methane recovery).	<u>Applicable</u> Option (f) as in the methodology is applicable to PAI1.

Methodology ID	Applicability condition	Justification of compliance
AMS III H	As per paragraph 3 of AMS-III.H. In cases where baseline system is anaerobic lagoon, the methodology is applicable if: (d) The lagoons are ponds with a depth greater than two meters, without aeration. The value for depth is obtained from engineering design documents, or through direct measurement, or by dividing the surface area by the total volume. If the lagoon filling level varies seasonally, the average of the highest and lowest levels may be taken; (e) Ambient temperature above 15°C, at least during part of the year on a monthly average basis; (f) The minimum interval between two consecutive sludge removal events shall be 30 days.	<u>Applicable</u> The baseline system of PAI1 is an anaerobic lagoon of which: (a) The lagoons are deeper than 2 meters and without aeration¹⁵. The value is obtained from an engineering design document; (b) The average ambient temperature in Pahang, Malaysia, is reported to be 30°C¹⁶ which is way above 15°C; (c) The sludge removal interval is more than 30 days¹⁷.
AMS III H	As per paragraph 4 of AMS-III.H. The recovered biogas from the above measures may also be utilised for the following applications instead of combustion/flaring: (a) Thermal or mechanical, electrical energy generation directly; (b) Thermal or mechanical, electrical energy generation after bottling of upgraded biogas, in this case, additional guidance provided in Annex 1 shall be followed; or (c) Thermal or mechanical, electrical energy generation after upgrading and distribution, in this case, additional guidance provided in Annex 1 shall be followed:	<u>Applicable</u> PAI1 uses the captured biogas for direct generation of electrical energy.

¹⁵ The values are obtained from an engineering design document provided by PAI1.

¹⁶ <https://www.accuweather.com/en/my/temerloh/229429/june-weather/229429>

¹⁷ Sludge removal record received from PAI1

Methodology ID	Applicability condition	Justification of compliance
	(i) Upgrading and injection of biogas into a natural gas distribution grid with no significant transmission constraints; (ii) Upgrading and transportation of biogas via a dedicated piped network to a group of end users; or (iii) Upgrading and transportation of biogas (e.g. by trucks) to distribution points for end users. (d) Hydrogen production. (e) Use as fuel in transportation applications after upgrading.	
AMS III H	As per paragraph 5 of AMS-III.H. If the recovered biogas is used for project activities covered under paragraph 4 (a), that component of the project activity can use a corresponding methodology under Type I.	<u>Applicable</u> PAI1 uses the recovered biogas for direct generation of electrical energy, AMS-I.D is used in the calculation.
AMS III H	As per paragraph 6 of AMS-III.H. For project activities covered under paragraph 4 (b), if bottles with upgraded biogas are sold outside the project boundary, the end use of the biogas shall be ensured via a contract between the bottled biogas vendor and the end-user. No emission reductions may be claimed from the displacement of fuels from the end use of bottled biogas in such situations. If, however, the end use of the bottled biogas is included in the project boundary and is monitored during the crediting period, CO₂ emissions avoided by the displacement of fossil fuel can be	<u>Not applicable</u> PAI1 does not involve bottling of upgraded biogas

Methodology ID	Applicability condition	Justification of compliance
	claimed under the corresponding Type I methodology, e.g. AMS-I.C Thermal energy production with or without electricity.	
AMS III H	As per paragraph 7 of AMS-III.H. For project activities covered under paragraph 4 (c) (i), emission reductions from the displacement of the use of natural gas are eligible under this methodology, provided the geographical extent of the natural gas distribution grid is within the host country boundaries.	<u>Not applicable</u> PAI1 does not involve upgrading and injection of biogas into a natural gas distribution grid.
AMS III H	As per paragraph 8 of AMS-III.H. For project activities covered under paragraph 4 (c) (ii), emission reductions for the displacement of the use of fuels can be claimed following the provision in the corresponding Type I methodology, e.g. AMS-I.C.	<u>Not applicable</u> PAI1 does not involve the transportation of natural gas distribution via a dedicated piped network to a group of end-user (s).
AMS III H	As per paragraph 9 of AMS-III.H. In particular, for the case of 4 (b) and (c) (iii), the physical leakage during storage and transportation of upgraded biogas as well as the emissions from fossil fuel consumed by vehicles for transporting biogas shall be considered. Relevant procedures in paragraph 18 of the appendix of AMS-III.H Methane recovery in wastewater treatment shall be followed in this regard.	<u>Not Applicable</u> PAI1 does not involve thermal or mechanical electrical energy generation after bottling of upgraded biogas or upgrading and transportation of biogas (e.g. by trucks) to distribution points for end users
AMS III H	As per paragraph 10 of AMS-III.H.	<u>Not Applicable</u>

Methodology ID	Applicability condition	Justification of compliance
	For project activities covered under paragraph 4 (b) and (c), this methodology is applicable if the upgraded methane content of the biogas is in accordance with relevant national regulations (where these exist) or in the absence of national regulations, a minimum of 96% (by volume).	PAI1 does not involve bottling, upgrading or transportation of biogas.
AMS III H	As per paragraph 11 of AMS-III.H. If the recovered biogas is utilized for the production of hydrogen (project activities covered under paragraph 4 (d)), that component of the project activity shall use the corresponding methodology AMS-III.O Hydrogen production using methane extracted from biogas.	<u>Not applicable</u> PAI1 does not involve hydrogen production
AMS III H	As per paragraph 12 of AMS-III.H. If the recovered biogas is used for project activities covered under paragraph 4 (e), that component of the project activity shall use the corresponding methodology AMS-III.AQ Introduction of Bio-CNG in road transportation.	<u>Not applicable</u> PAI1 does not involve use of recovered biogas as fuel in transportation applications after upgrading.
AMS III H	As per paragraph 13 of AMS-III.H. New facilities (Greenfield projects) and project activities involving a change of equipment resulting in a capacity addition of the wastewater or sludge treatment system compared to the designed capacity of the baseline treatment system are only eligible to apply this	<u>Not Applicable</u> PAI1 is not a Greenfield project. The project activity is implemented in an existing wastewater treatment plant. There is also no capacity addition of wastewater treatment system compared to the designed capacity of the baseline treatment system.

Methodology ID	Applicability condition	Justification of compliance
	methodology if they comply with the relevant requirements in the General guidelines to SSC CDM methodologies. In addition, the requirements for demonstrating the remaining lifetime of the equipment replaced as described in the general guidelines shall be followed.	
	As per paragraph 2.2.14 of AMS-III.H. The location of the wastewater treatment plant as well as the source generating the wastewater shall be uniquely defined and described in the Project Design Document (PDD).	<u>Applicable</u> The location of the wastewater treatment plant as well as the source generating the wastewater is uniquely defined and described in the PD
AMS III H	As per paragraph 2.2.15 of AMS-III.H Measures are limited to those that result in aggregate emission reductions of less than or equal to 60 ktCO ₂ equivalent annually from all Type III components of the project activity.	<u>Applicable</u> The estimated average emission reductions from Type III component (methane avoidance) over the crediting period is 35,217 tCO ₂ -equivalent/year. This is lower than the 60 ktCO ₂ e annual threshold limit for Type III component (methane avoidance).
TOOL03	This tool provides procedures to calculate project and/or leakage CO ₂ emissions from the combustion of fossil fuels. It can be used in cases where CO ₂ emissions from fossil fuel combustion are calculated based on the quantity of fuel combusted and its properties. Methodologies using this tool should specify to which combustion process j this tool is being applied.	<u>Not Applicable</u> PAI 1 does not involve any combustion of fossil fuels.
TOOL04	The tool can be used to determine emissions for the following types of applications:	<u>Not Applicable</u> PAI 1 does not involve any sludge treatment activity.

Methodology ID	Applicability condition	Justification of compliance
	(a) Application A: The CDM project activity mitigates methane emissions from a specific existing SWDS. Methane emissions are mitigated by capturing and flaring or combusting the methane (e.g. “ACM0001: Flaring or use of landfill gas”). The methane is generated from waste disposed in the past, including prior to the start of the CDM project activity. In these cases, the tool is only applied for an ex ante estimation of emissions in the project design document (CDM-PDD). The emissions will then be monitored during the crediting period using the applicable approaches in the relevant methodologies (e.g. measuring the amount of methane captured from the SWDS); (b) Application B: The CDM project activity avoids or involves the disposal of waste at a SWDS. An example of this application of the tool is ACM0022, in which municipal solid waste (MSW) is treated with an alternative option, such as composting or anaerobic digestion, and is then prevented from being disposed of in a SWDS. The methane is generated from waste disposed or avoided from disposal during the crediting period. In these cases, the tool can be	

Methodology ID	Applicability condition	Justification of compliance
	applied for both ex ante and ex post estimation of emissions. These project activities may apply the simplified approach detailed in 0 when calculating baseline emissions.	
	In the case that: (a) different types of residual waste are disposed or prevented from disposal; or that (b) both MSW and residual waste(s) are prevented from disposal, then the tool should be applied separately to each residual waste and to the MSW.	<u>Not Applicable</u> PAIs to be implemented under this Grouped Project will not prevent or involve disposal of any type of residual waste or MSW at a SWDS.
TOOL05	If emissions are calculated for electricity consumption, the tool is only applicable if one out of the following three scenarios applies to the sources of electricity consumption: (a) Scenario A: Electricity consumption from the grid. The electricity is purchased from the grid only, and either no captive power plant(s) is/are installed at the site of electricity consumption or, if any captive power plant exists on site, it is either not operating or it is not physically able to provide electricity to the electricity consumer; (b) Scenario B: Electricity consumption from (an) off-grid fossil fuel fired captive power plant(s). One or more fossil fuel fired captive power plants are installed at the site of the electricity consumer and supply the consumer with electricity. The captive power plant(s) is/are not connected to the electricity grid; or (c) Scenario C: Electricity consumption from the grid and (a) fossil fuel fired captive power	<u>Not Applicable</u> PAI 1 does not involve any electricity consumption during the ex ante estimation, however, the power consumption from the grid will be calculated during the ex post if applicable.

Methodology ID	Applicability condition	Justification of compliance
	plant(s). One or more fossil fuel fired captive power plants operate at the site of the electricity consumer. The captive power plant(s) can provide electricity to the electricity consumer. The captive power plant(s) is/are also connected to the electricity grid. Hence, the electricity consumer can be provided with electricity from the captive power plant(s) and the grid.	
	This tool can be referred to in methodologies to provide procedures to monitor amount of electricity generated in the project scenario, only if one out of the following three project scenarios applies to the recipient of the electricity generated: (a) Scenario I: Electricity is supplied to the grid; (b) Scenario II: Electricity is supplied to consumers/electricity consuming facilities; or (c) Scenario III: Electricity is supplied to the grid and consumers/electricity consuming facilities.	<u>Not Applicable</u> PAI 1 does not apply this tool as not refer in methodology to monitor the amount of electricity generation.
	This tool is not applicable in cases where captive renewable power generation technologies are installed to provide electricity in the project activity, in the baseline scenario or to sources of leakage. The tool only accounts for CO ₂ emissions.	<u>Not Applicable</u> PAI 1 does not involve any installation of captive renewable power generation technologies as the power generation is supplied to grid.
TOOL06	This tool provides procedures to calculate project emissions from flaring of a residual gas. The tool is applicable to enclosed or open flares and project participants should document in the CDM-PDD the type of flare used in the project activity.	<u>Applicable</u> PAI1 accounts for its project emissions from flaring of a residual gas as it involves open flaring.
	This tool is applicable to the flaring of flammable greenhouse gases where:	<u>Applicable</u>

Methodology ID	Applicability condition	Justification of compliance
	(a) Methane is the component with the highest concentration in the flammable residual gas; and (b) The source of the residual gas is coal mine methane or a gas from a biogenic source (e.g. biogas, landfill gas or wastewater treatment gas).	PAI1 involves the open flaring of flammable greenhouse gases where: (a) Methane is the component with the highest concentration in the flammable residual gas; and (b) The source of the residual gas is a gas from a biogenic source that is biogas from wastewater treatment.
	The tool is not applicable to the use of auxiliary fuels and therefore the residual gas must have sufficient flammable gas present to sustain combustion. In the case of an enclosed flare, there shall be operating specifications provided by the manufacturer of the flare and these shall be followed by the project participant.	<u>Not Applicable</u> PAI1 does not use auxiliary fuels and its residual gas has sufficient flammable gas present to sustain combustion. PA1 involves open flaring instead of enclosed flare.
TOOL32	The use of this methodological tool is not mandatory for the project participants of a CDM project activity or CDM PoA for demonstrating their additionality. This methodological tool shall be applied in conjunction with a small-scale or large-scale methodology which refers to this tool.	<u>Applicable</u> PAI 1 uses the tool to demonstrate their additionality in conjunction with small scale methodology AMS-III.H. Version 19.0 which refers to this tool.
	The positive lists as contained in section 5 of this tool are valid up to 10 March 2025. Notwithstanding the provisions on the validity of new, revised and previous versions of methodologies and methodological tools in the “Procedure: Development, revision and clarification of baseline and monitoring methodologies and methodological tools”, there will be no grace period for the application of this tool and the validity of the positive list after this date, including	<u>Applicable</u> PAI 1 will use the tool to demonstrate their additionality as the positive lists are valid up to 10 March 2025. The grouped project has completed validation and submitted a request for registration on 03 March 2025. Thus, this applicability condition is met. As per the Clarification to CDM TOOL32 Positive Lists of Technologies published by Verra on 18 March 2025, this tool remains active for use

Methodology ID	Applicability condition	Justification of compliance
	in cases where further technologies are added to the positive list through revisions of this tool before this date.	in the VCS Program until 30 September 2025.
AMS I D	As per paragraph 2 of AMS-I.D This methodology comprises renewable energy generation units such as photovoltaic, hydro, tidal/wave, wind, geothermal and renewable biomass: (a) Supplying electricity to a national or a regional grid; or (b) Supplying electricity to an identified consumer facility via national/regional grid through a contractual arrangement such as wheeling.	<u>Applicable</u> PAI1 utilizes the captured biogas from Palm Oil Mill Effluent for electricity generation for own use and upload to grid.
AMS I D	As per paragraph 4 of AMS-I.D This methodology is applicable to project activities that: (a) Install a Greenfield plant; (b) Involve a capacity addition in (an) existing plant(s); (c) Involve a retrofit of (an) existing plant(s); (d) Involve a rehabilitation of (an) existing plant(s)/ unit(s); or (e) Involve a replacement of (an) existing plant(s).	<u>Applicable</u> (a) PAI1 is a new power generation plant that qualifies as greenfield plant.
AMS I D	As per paragraph 5 of AMS-I.D Hydro power plants with reservoirs that satisfy at least one of the following conditions are eligible to apply this methodology: (a) The project activity is implemented in an existing reservoir with no change in the volume of reservoir;	<u>Not Applicable</u> PAI1 is not hydro power plant project.

Methodology ID	Applicability condition	Justification of compliance
	(b) The project activity is implemented in an existing reservoir where the volume of reservoir is increased and the power density of the project activity, as per definitions given in the project emissions section, is greater than 4W/m²; or (c) The project activity results in new reservoirs and the power density of the power plant, as per definitions given in the project emissions section, is greater than 4W/m².	
AMS I D	As per paragraph 6 of AMS-I.D If the new unit has both renewable and non-renewable components (e.g. a wind/diesel unit), the eligibility limit of 15MW for a small-scale CDM project activity applies only to the renewable component. If the new unit co-fires fossil fuel, the capacity of the entire unit shall not exceed the limit of 15MW.	<u>Applicable</u> The capacity of the renewable component for PAI1 is 1.2MW which is less than 15MW
AMS I D	As per paragraph 7 of AMS-I.D Combined heat and power (co-generation) systems are not eligible under this category.	<u>Not Applicable</u> PAI1 is not co-generation project
AMS I D	As per paragraph 8 of AMS-I.D In the case of project activities that involve the capacity addition of renewable energy generation units at an existing renewable power generation facility, the added capacity of the units added by the project should be lower than 15MW	<u>Not Applicable</u> PAI1 is not capacity addition of renewable energy generation units.

Methodology ID	Applicability condition	Justification of compliance
	and should be physically distinct from the existing units.	
AMS I D	As per paragraph 9 of AMS-I.D In the case of retrofit, rehabilitation or replacement, to qualify as a small-scale project, the total output of the retrofitted, rehabilitated or replacement power plant/ unit shall not exceed the limit of 15MW.	<u>Not Applicable</u> PAI1 does not involve retrofit, rehabilitation or replacement of existing facilities.
AMS I D	As per paragraph 10 of AMS-I.D In the case of landfill gas, waste gas, wastewater treatment and agro-industries projects, recovered methane emissions are eligible under a relevant Type III category. If the recovered methane is used for electricity generation for supply to the grid then the baseline for the electricity component shall be in accordance with procedure prescribed under this methodology. If the recovered methane is used for heat generation or cogeneration other applicable Type-I methodologies such as 'AMS-I.C: Thermal energy production with or without electricity' shall be explored.	<u>Applicable</u> PAI1 is wastewater treatment methane recovery project, and eligible under Type III category.
AMS I D	As per paragraph 11 of AMS-I.D In case biomass is sourced from dedicated plantations, the applicability criteria in the tool 'Project emissions from cultivation of biomass' shall apply.	<u>Not Applicable</u> PAI1 does not use biomass sourced from dedicated plantations.
TOOL03	This tool provides procedures to calculate project and/or leakage CO₂ emissions from the combustion of	<u>Not Applicable</u> PAI 1 does not involve any combustion of fossil fuels.

Methodology ID	Applicability condition	Justification of compliance
	fossil fuels. It can be used in cases where CO ₂ emissions from fossil fuel combustion are calculated based on the quantity of fuel combusted and its properties. Methodologies using this tool should specify to which combustion process j this tool is being applied.	
TOOL07	This tool may be applied to estimate the OM, BM and/or CM when calculating baseline emissions for a project activity that substitutes grid electricity that is where a project activity supplies electricity to a grid or a project activity that results in savings of electricity that would have been provided by the grid (e.g. demand-side energy efficiency projects).	<u>Applicable</u> PAI1 applies reference that estimates the OM, BM and CM when calculating baseline emissions for a project activity that substitutes grid electricity that is where a project activity supplies electricity to a grid.
	Under this tool, the emission factor for the project electricity system can be calculated either for grid power plants only or, as an option, can include off-grid power plants. In the latter case, two sub-options under the step 2 of the tool are available to the project participants, i.e. option IIa and option IIb. If option IIa is chosen, the conditions specified in “Appendix 1: Procedures related to off-grid power generation” should be met. Namely, the total capacity of off-grid power plants (in MW) should be at least 10 per cent of the total capacity of grid power plants in the electricity system; or the total electricity generation by off-grid power plants (in MWh) should be at least 10 per	<u>Applicable</u> PAI1 applies reference that calculates the emission factor for the project electricity system for grid power plants only.

Methodology ID	Applicability condition	Justification of compliance
	cent of the total electricity generation by grid power plants in the electricity system; and that factors which negatively affect the reliability and stability of the grid are primarily due to constraints in generation and not to other aspects such as transmission capacity.	
	In case of CDM projects the tool is not applicable if the project electricity system is located partially or totally in an Annex I country.	<u>Not Applicable</u> PAI1 will not locate partially or totally in an Annex I country.
TOOL10	The tool provides guidance to determine the remaining lifetime of baseline or project equipment. The tool may, for example, be used for project activities which involve the replacement of existing equipment with new equipment or which retrofit existing equipment as part of energy efficiency improvement activities. Methodologies referring to this tool should clearly specify for which equipment the remaining lifetime should be determined. The remaining lifetime of relevant equipment shall be determined prior to the implementation of the project activity. Project participants using this tool shall document transparently in the CDM-PDD how the remaining lifetime of applicable equipment has been determined, including (references to) all documentation used. Under this tool, impacts on the lifetime of the equipment due to	<u>Not Applicable</u> PAI1 is a Greenfield power generation, therefore it is not applicable to determine the remaining lifetime of baseline or project equipment.

Methodology ID	Applicability condition	Justification of compliance
	policies and regulations (e.g. environmental regulations) or changes in the services needed (e.g. increased energy demand) are not considered. Methodologies referring to this tool shall, where applicable, provide specific guidance on how regulations that warrant the replacement of the equipment before it has reached the end of its technical lifetime should be addressed.	
TOOL16	This tool shall be applied in conjunction with the methodology that refers to this tool. For project activities which include biomass cultivation: (a) The land in which biomass is cultivated: (i) Does not contain wetlands; (ii) Does not contain organic soils as defined in (iii) Is not subjected to flood irrigation. (b) The land in which biomass is cultivated: (i) Does not contain forest nor contained forest since 31 December 1989; or (ii) Contains a forest plantation that before the start of the project will be harvested and the land would be neither reforested nor will regenerate on its own	<u>Not Applicable</u> PAI 1 does not involves biomass cultivation.

Methodology ID	Applicability condition	Justification of compliance
	into a forest in the absence of the project activity. (c) Desalination is not a substantial source of water in the host country.	
	In case the land contains a forest plantation, the project participants shall demonstrate that before the start of the project activity the plantation will be finally harvested and regeneration to forestland (according to the respective national definition) will not take place. In doing so, the project proponent shall: (a) Identify realistic and credible alternatives with regard to the possible land use scenarios that would occur in the absence of the project activity, including but not limited to: (i) The forest plantation continues under the current management practice; (ii) The forest plantation is harvested and the land is replanted; (iii) The forest plantation is harvested and the land is abandoned; (b) Assess the economic attractiveness of the existing forest plantation by applying Step 2 of the “TOOL01: Tool for the demonstration and assessment of additionality”;	<u>Not Applicable</u> PAI 1 does not involves forest land.

Methodology ID	Applicability condition	Justification of compliance
	(c) Confirm, based on the plantation management practices in the region for the considered species, that the situation referred to in paragraph 7(b)(ii) is the common practice; and (d) Use relevant credible evidence, including but not limited to official land use maps, satellite images/aerial photographs, cadastral information, official land use records.	
	The tool is also applicable if biomass residues are consumed in a CDM project activity, and the biomass residues can be utilized after processing or without processing. These could be: (a) Procured by the project proponents; or (b) The result of an agro-industrial process under the control of the project proponents.	<u>Not Applicable</u> PAI1 does not consume biomass residues.
TOOL11	This tool provides a stepwise procedure to assess the continued validity of the baseline and to update the baseline at the renewal of a crediting period, as required by paragraph 49 (a) of the modalities and procedures of the clean development mechanism. The tool consists of two steps. The first step provides an approach to evaluate whether the current	<u>Applicable</u> PAI1 will assess the continued validity of the baseline and to update the baseline at the renewal of a crediting period.

Methodology ID	Applicability condition	Justification of compliance
	baseline is still valid for the next crediting period. The second step provides an approach to update the baseline in case that the current baseline is not valid anymore for the next crediting period.	

3.3 Project Boundary

Grouped Project

The following GHG sources for the baseline and project scenarios should be assessed on relevance for additional project activity instances added to this Grouped Project

Table 2: Emission sources included or excluded for AMS-III.H in the project boundary

	Source	Gas	Included?	Justification/Explanation
Baseline	Emissions on account of electricity or fossil fuel used	CO ₂	Yes	Main emission source
		CH ₄	No	Excluded for simplification
		N ₂ O	No	Excluded for simplification
		Other	No	Excluded for simplification
	Methane emissions from baseline wastewater treatment systems	CO ₂	No	Excluded for simplification
		CH ₄	Yes	Main emission source
		N ₂ O	No	Excluded for simplification
		Other	No	Excluded for simplification
	Methane emissions from baseline sludge treatment systems	CO ₂	No	Excluded for simplification
		CH ₄	Yes	Main emission source
		N ₂ O	No	Excluded for simplification
		Other	No	Excluded for simplification
	Methane emissions on account of inefficiencies in the baseline wastewater treatment systems	CO ₂	No	Excluded for simplification
		CH ₄	Yes	Main emission source
		N ₂ O	No	Excluded for simplification

	Source	Gas	Included?	Justification/Explanation
	and presence of degradable organic carbon in the treated wastewater discharged into river/lake/sea	Other	No	Excluded for simplification
	Methane emissions from the decay of the final sludge generated by the baseline treatment systems	CO ₂	No	Excluded for simplification
		CH ₄	Yes	Main emission source
		N ₂ O	No	Excluded for simplification
		Other	No	Excluded for simplification
	Emissions from injection of upgraded biogas into a natural gas distribution grid	CO ₂	No	Excluded for simplification
		CH ₄	Yes	Main emission source
		N ₂ O	No	Excluded for simplification
		Other	No	Excluded for simplification
Project	CO ₂ emissions from electricity and fuel used by the project facilities	CO ₂	Yes	Main emission source
		CH ₄	No	Excluded for simplification
		N ₂ O	No	Excluded for simplification
		Other	No	Excluded for simplification
	Methane emissions from wastewater treatment systems affected by the project activity, and not equipped with biogas recovery in the project scenario	CO ₂	No	Excluded for simplification
		CH ₄	Yes	Main emission source
		N ₂ O	No	Excluded for simplification
		Other	No	Excluded for simplification
	Methane emissions from sludge treatment systems affected by the project activity, and not equipped with biogas recovery in the project situation	CO ₂	No	Excluded for simplification
		CH ₄	Yes	Main emission source
		N ₂ O	No	Excluded for simplification
		Other	No	Excluded for simplification
	Methane emissions on account of inefficiency of the project activity wastewater treatment systems and presence	CO ₂	No	Excluded for simplification
		CH ₄	Yes	Main emission source
		N ₂ O	No	Excluded for simplification

Source	Gas	Included?	Justification/Explanation
of degradable organic carbon in treated wastewater	Other	No	Excluded for simplification
Methane emissions from the decay of the final sludge generated by the project activity treatment systems	CO ₂	No	Excluded for simplification
	CH ₄	Yes	Main emission source
	N ₂ O	No	Excluded for simplification
	Other	No	Excluded for simplification
Methane fugitive emissions due to inefficiencies in capture systems	CO ₂	No	Excluded for simplification
	CH ₄	Yes	Main emission source
	N ₂ O	No	Excluded for simplification
	Other	No	Excluded for simplification
Methane emissions due to incomplete flaring	CO ₂	No	Excluded for simplification
	CH ₄	Yes	Main emission source
	N ₂ O	No	Excluded for simplification
	Other	No	Excluded for simplification
Methane emissions from biomass stored under anaerobic conditions which would not have occurred in the baseline situation	CO ₂	No	Excluded for simplification
	CH ₄	Yes	Main emission source
	N ₂ O	No	Excluded for simplification
	Other	No	Excluded for simplification
Emissions from electricity and fuel used by upgrading facilities	CO ₂	No	Excluded for simplification
	CH ₄	Yes	Main emission source
	N ₂ O	No	Excluded for simplification
	Other	No	Excluded for simplification
Methane emissions from the discharge of the upgrading equipment	CO ₂	No	Excluded for simplification
	CH ₄	Yes	Main emission source
	N ₂ O	No	Excluded for simplification
	Other	No	Excluded for simplification
	CO ₂	No	Excluded for simplification

Source		Gas	Included?	Justification/Explanation
	Fugitive methane emissions from leaks in compression equipment	CH ₄	Yes	Main emission source
		N ₂ O	No	Excluded for simplification
		Other	No	Excluded for simplification
	Emissions on account of vent gases from upgrading equipment	CO ₂	No	Excluded for simplification
		CH ₄	Yes	Main emission source
		N ₂ O	No	Excluded for simplification
		Other	No	Excluded for simplification

Table 3: Emission sources included or excluded for AMS-I.D. in the project boundary

Source		Gas	Included?	Justification/Explanation
Baseline	CO ₂ emissions from electricity generation in power plants that are displaced due to the project activity	CO ₂	Yes	Main emission source
		CH ₄	No	Excluded for simplification
		N ₂ O	No	Excluded for simplification
		Other	No	Excluded for simplification
Project	CO ₂ emissions from on-site consumption of fossil fuel due to project activity	CO ₂	Yes/No	Main emission source
		CH ₄	No	Excluded for simplification
		N ₂ O	No	Excluded for simplification
		Other	No	Excluded for simplification

PAI1

The following boundaries apply to PAI1, as per AMS-III.H. and AMS I.D.

Table 4: Emission sources included for AMS-III.H. for PAI1

Source		Gas	Included?	Justification/Explanation
Baseline	Methane emissions from baseline wastewater treatment systems	CO ₂	No	Excluded for simplification
		CH ₄	Yes	Main emission source
		N ₂ O	No	Excluded for simplification
		Other	No	Excluded for simplification
		CO ₂	No	Excluded for simplification

Source		Gas	Included?	Justification/Explanation
	Methane emissions on account of inefficiencies in the baseline wastewater treatment systems and presence of degradable organic carbon in the treated wastewater discharged into river/lake/sea	CH ₄	Yes	Main emission source
		N ₂ O	No	Excluded for simplification
		Other	No	Excluded for simplification
Project	CO ₂ emissions from electricity and fuel used by the project facilities	CO ₂	Yes	Main emission source
		CH ₄	No	Excluded for simplification
		N ₂ O	No	Excluded for simplification
		Other	No	Excluded for simplification
	Methane emissions from wastewater treatment systems affected by the project activity, and not equipped with biogas recovery in the project scenario	CO ₂	No	Excluded for simplification
		CH ₄	Yes	Main emission source
		N ₂ O	No	Excluded for simplification
		Other	No	Excluded for simplification
	Methane emissions on account of inefficiency of the project activity wastewater treatment systems and presence of degradable organic carbon in treated wastewater	CO ₂	No	Excluded for simplification
		CH ₄	Yes	Main emission source
		N ₂ O	No	Excluded for simplification
		Other	No	Excluded for simplification
	Methane fugitive emissions due to inefficiencies in capture systems	CO ₂	No	Excluded for simplification
		CH ₄	Yes	Main emission source
		N ₂ O	No	Excluded for simplification
		Other	No	Excluded for simplification
	Methane emissions due to incomplete flaring	CO ₂	No	Excluded for simplification
		CH ₄	Yes	Main emission source
		N ₂ O	No	Excluded for simplification
		Other	No	Excluded for simplification

Table 5: Emission sources included for AMS- I.D. for PAI1

Source		Gas	Included?	Justification/Explanation
Baseline	CO ₂ emissions from electricity generation in power plants that are displaced due to the project activity	CO ₂	Yes	Main emission source
		CH ₄	No	Excluded for simplification
		N ₂ O	No	Excluded for simplification
		Other	No	Excluded for simplification
Project	CO ₂ emissions from on-site consumption of fossil fuel due to project activity	CO ₂	Yes	Main emission source
		CH ₄	No	Excluded for simplification
		N ₂ O	No	Excluded for simplification
		Other	No	Excluded for simplification

As per methodology AMS-III.H. version 19.0, paragraph 22 states that Implementation of the project activity at a wastewater and/or sludge treatment system will affect certain sections of the treatment systems while others may remain unaffected. The treatment systems not affected by the project activity, i.e. sections operating in the project scenario under the same operational conditions as in the baseline scenario (e.g. wastewater inflow and COD content, temperature, retention time, etc.), shall be described in the PDD, but emissions from those sections do not have to be accounted for in the baseline and project emission calculations (since the same emissions would occur in both baseline and project scenarios). The assessment and identification of the systems affected by the project activity will be undertaken ex ante, and the PDD shall justify the exclusion of sections or components of the system. The treatment systems (lagoons, reactors, digesters, etc.) that will be covered and/or equipped with biogas recovery by the project activity, but continue to operate with the same quality of feed inflow, volume (retention time), and temperature (heating) as in the baseline scenario, may be considered as not affected i.e. the methane generation potential remains unaltered.

Therefore, a clear delineation of treatment systems which are present in the baseline and the ones that will be added as part of the project as well as project boundary is illustrated in figure below:

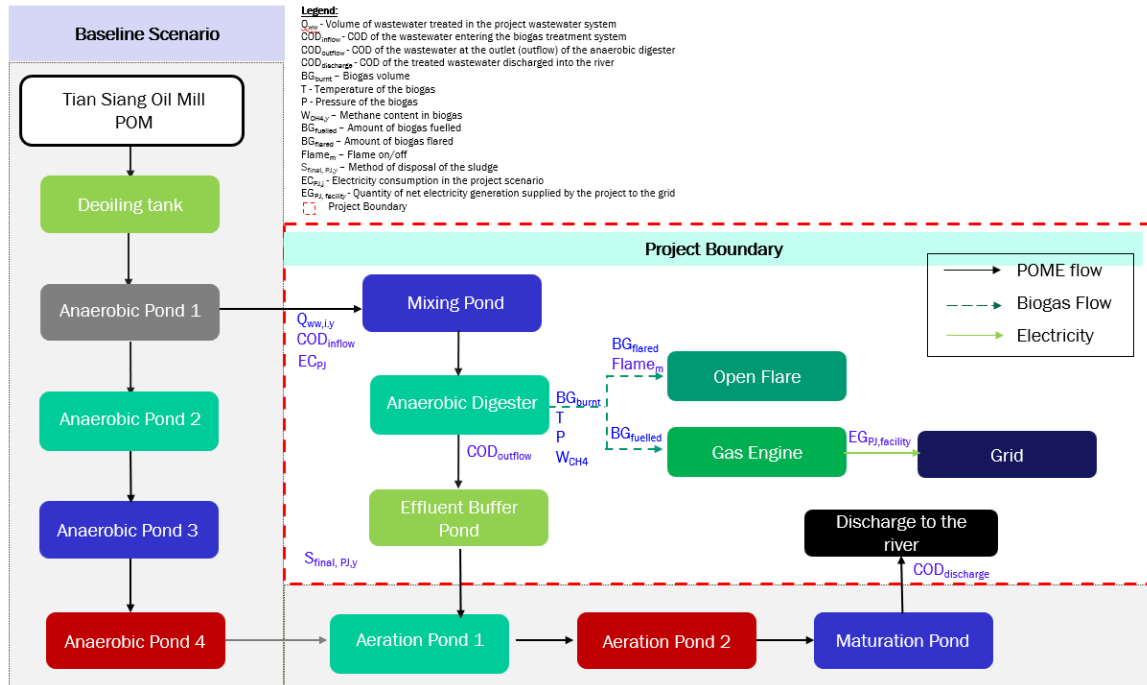


Figure 5: PAI1 Baseline Scenario and Project Boundary

3.4 Baseline Scenario

Grouped Project

According to AMS-III.H version 19.0, the baseline scenario involves a baseline system with anaerobic wastewater treatment system without biogas recovery, which is the case the baseline scenario is determined to be the continuation of the existing open lagoon-based treatment of wastewater without methane recovery as commonly practiced because it only requires small investment and maintenance cost where the treatment system only consume less energy for pumping and mixing.

According to AMS-I.D version 18.0, the baseline scenario for greenfield power plant, electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources into the grid.

Therefore, as a result of the combination of baseline analysis for all the above methodologies, it is concluded that the baseline scenario of the proposed Grouped Project is, in the absence of the project activity:

- For wastewater related to palm oil mill would have been treated in anaerobic open lagoons; and
- For electricity generation, it would have been generated by operation of grid connected power plants and by addition of new generation sources into the grid

PAI1

In the Baseline Scenario, this wastewater is treated in a series of lagoons. Lagoons 1-4 of this system are anaerobic lagoons, with each of the depth deeper than two (2) meters and the average atmospheric temperature in the project location is 31¹⁸ degree Celsius. These conditions create an anaerobic environment in lagoons 1-4, resulting in methane generation. Once the wastewater has completed processing through the baseline treatment system, this wastewater is within the acceptable legal limits applicable to Malaysia and is then discharged into a river.

The sludge removal from lagoons was conducted on an “as-needed” basis where the gap between the two (2) consecutive sludge removals is more than 30 days. The sludge removed is discarded on the nearby land of the palm oil mill and, as a result, does not produce any emissions of greenhouse gases.

In the absence of PAI1, the baseline scenario will be the continuation of the present wastewater treatment, including ongoing methane emissions to the atmosphere from the anaerobic lagoons.

The baseline scenarios for PAI1 are presented below:

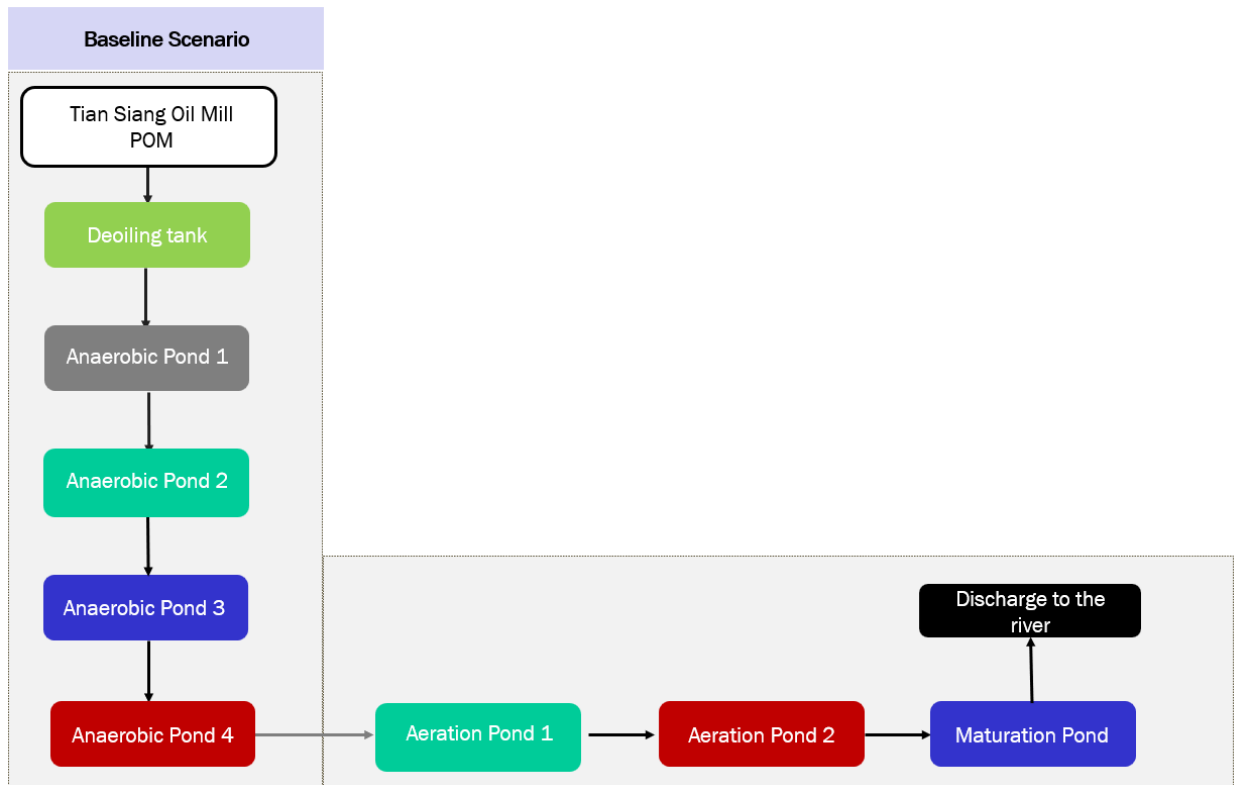


Figure 6: PAI1 Baseline Scenario

¹⁸ <https://www.weatheravenue.com/en/asia/my/pahang/triang-weather.html>

3.5 Additionality

As per tools Positive lists of technologies - Version 04.0, A positive list of technology and project activity types defined as automatically additional if the PAIs apply such technologies and meet specified conditions.

The positive list of technology requirements for ‘waste handling and disposal’ is tabulated below:

Table 6: Additionality for PAIs

Requirements for Additionality	PAI Scenario
As per paragraph 12(a): The existing treatment system is an anaerobic lagoon and the wastewater discharged meets the host country regulation	The existing treatment system is an anaerobic lagoon and the wastewater discharged meets the regulations of Malaysia ¹⁹ .
As per paragraph 12(b): No regulation in the host country requires the management of biogas from domestic, industrial and agricultural sites	Yes There is no regulation in Malaysia that requires the management of biogas from a site such as the project site ²⁰ . Condition in the criteria and guidelines for an application for licensing by the Malaysian Palm Oil Board (MPOB) require new palm-oil mills constructed after 01-Jan-2014 to install wastewater treatment systems with methane capture. The requirement was also placed on existing mills who applied for capacity extensions after 01-Jan-2014, although an exemption effective 22-May-2021 was given to those existing mills who applied for capacity extensions below 270,000 MT FFB per year, noting this threshold has been revised several times. Tian Siang POM was established in 2002 and was not a mill constructed after 01-Jan-2014 and has not undergone a capacity expansion since 01-Jan-2014. Therefore, the new criteria do not apply and there is no obligation to recover the methane and/or use the biogas
As per paragraph 12(c):	Yes

¹⁹ Engineering diagrams for the anaerobic lagoon and documentation relevant to wastewater discharge

²⁰ MPOB criteria for new licenses and specifics that relate PAI1 (eg. license for mill).

Requirements for Additionality	PAI Scenario
There is no capacity increase in the wastewater treatment system	There has been no capacity increase in the wastewater treatment system ²¹
As per paragraph 12(d): No other alternative economic activity is expected to be undertaken on the land of the existing lagoon	Yes No other alternative economic activity is expected to be undertaken on the land of the existing lagoon.
As per paragraph 12(e): The biogas is used to generate electricity in one or more power plants, and the total nameplate	Yes The biogas is used to generate electricity in a power plant and the total nameplate capacity is below 5MW ²² .

As PAI1 meets all five (5) conditions, PAI1 is automatically additional.

For PAIs that do not apply the technologies and meet specified conditions as outlined in the tool, the PAIs are required to use the tool “Demonstration of additionality of small-scale project activities”, Version 13.1 for the demonstration of additionality.

As per the abovementioned tool, project participants shall provide an explanation to show that the project activity would not have occurred anyway due to at least one (1) of the following barriers²³:

- (a) Investment barrier: a financially more viable alternative to the project activity would have led to higher emissions
- (b) Technological barrier: a less technologically advanced alternative to the project activity involves lower risks due to the performance uncertainty or low market share of the new technology adopted for the project activity and so would have led to higher emissions
- (c) Barrier due to prevailing practice: prevailing practice or existing regulatory or policy requirements would have led to implementation of a technology with higher emissions
- (d) Other barriers: without the project activity, for another specific reason identified by the project participant, such as institutional barriers or limited information, managerial resources, organizational capacity, financial resources, or capacity to absorb new technologies, emissions would have been higher.

Documentation of barriers, as per the paragraph above, is not required for the positive list of technologies and project activity types that are defined as automatically additional²⁴ for project sizes up to and including the small-scale CDM thresholds (e.g. installed capacity up to 15 MW).

²¹ License for mill and supporting documentation.

²² Technical specifications for the biogas engine

²³ Project participant may like to refer to “Non-binding best practice examples to demonstrate additionality for SSC project activities” (EB 35 Annex 34)

²⁴ The appendix provides a flow chart to guide the users to help navigate provisions for automatic additionality across “Tool for Demonstration of additionality of small-scale project activities” and “Tool for Demonstration of additionality of microscale project”

3.5.1 Regulatory Surplus

Is the project located in an UNFCCC Annex 1 or Non-Annex 1 country?

- ☐ Annex 1 country ☒ Non-Annex 1 country

Are the project activities mandated by any law, statute, or other regulatory framework?

- ☐ Yes ☒ No

If the project is located inside a Non-Annex 1 country and the project activities are mandated by a law, statute, or other regulatory framework, are such laws, statutes, or regulatory frameworks systematically enforced?

- ☐ Yes ☐ No

Not applicable. The project is located inside a Non-Annex 1 country, but the project activities are not mandated by any law, statute, or other regulatory framework.

3.5.2 Additionality Methods

Grouped Project

Future PAIs may be able to consider the Positive Lists of Technologies or any other suitable demonstration of additionality. Suitable evidence to demonstrate additionality is required for each PAI. If the PAIs in the Grouped Project need to apply to demonstrate additionality and the procedure in the applied methodology or tool involves several steps. The PAIs will describe how each step is applied and document the outcome of each step.

PAI1

Not applicable to PAI1. As per tools Positive lists of technologies - Version 04.0, A positive list of technology and project activity types is defined as automatically additional if the PAIs apply such technologies and meet specified conditions. PAI1 meets all five (5) conditions. As a result, PAI1 is automatically additional.

3.6 Methodology Deviations

Grouped Project

If a project activity instance, added to this Grouped Project, deviates from the methodology an explanation shall be provided in this section.

PAI1

Not applicable as no deviations from the methodology were made.

4 QUANTIFICATION OF ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

Grouped Project

AMS III H, Version 19.0

According to the approved methodology AMS III H, Version 19.0, the baseline emissions for instances in the Grouped Project are determined as follows:

Baseline emissions for the systems affected by the project activity may consist of:

- Emissions on account of electricity or fossil fuel used ($BE_{power,y}$);
- Methane emissions from baseline wastewater treatment systems ($BE_{ww,treatment,y}$);
- Methane emissions from baseline sludge treatment systems ($BE_{s,treatment,y}$);
- Methane emissions on account of inefficiencies in the baseline wastewater treatment systems and presence of degradable organic carbon in the treated wastewater discharged into river/lake/sea ($BE_{ww,discharge,y}$);
- Methane emissions from the decay of the final sludge generated by the baseline treatment systems ($BE_{s,final,y}$).

$$BE_y = \{BE_{power,y} + BE_{ww,treatment,y} + BE_{s,treatment,y} + BE_{ww,discharge,y} + BE_{s,final,y}\}$$

Where:

Parameter	Description	Unit
BE_y	Baseline emissions in year y	tCO ₂ e
$BE_{power,y}$	Emissions on account of electricity or fossil fuel used	tCO ₂ e
$BE_{ww,treatment,y}$	Methane emissions from baseline wastewater treatment systems	tCO ₂ e
$BE_{s,treatment,y}$	Methane emissions from baseline sludge treatment systems	tCO ₂ e
$BE_{ww,discharge,y}$	Methane emissions on account of inefficiencies in the baseline wastewater treatment systems and presence of	tCO ₂ e

Parameter	Description	Unit
	degradable organic carbon in the treated wastewater discharged into river/lake/sea	
$BE_{s,final,y}$	Methane emissions from the decay of the final sludge generated by the baseline treatment systems	tCO _{2e}

Baseline emissions from electricity and fossil fuel consumption ($BE_{power,y}$) are determined as per the procedures described in the “Tool to calculate baseline, project and/or leakage emissions from electricity consumption” and “Tool to calculate project or leakage CO₂ emissions from fossil fuel combustion”, respectively. The energy consumption shall include all equipment/devices in the baseline wastewater and sludge treatment facility. If recovered biogas in the baseline is used to power auxiliary equipment it should be taken into account, accordingly, using zero as its emission factor

Methane emissions from the baseline wastewater treatment systems affected by the project ($BE_{ww,treatment,y}$) are determined using the COD removal efficiency of the baseline plant:

$$BE_{ww,treatment,y} = \sum_i (Q_{ww,i,y} \times COD_{inflow,i,y} \times \eta_{COD,BL,i} \times MCF_{ww,treatment,BL,i}) \times B_{o,ww} \times UF_{BL} \times GWP_{CH_4}$$

Where:

Parameter	Description	Unit
$Q_{ww,i,y}$	Volume of wastewater treated in baseline wastewater treatment system <i>i</i> in year <i>y</i> (m ³). For ex ante estimation, forecasted wastewater generation volume or the designed capacity of the wastewater treatment facility can be used. However, the ex post emissions reduction calculation shall be based on the actual monitored volume of treated wastewater	m ³
$COD_{inflow,i,y}$	Chemical oxygen demand of the wastewater inflow to the baseline treatment system <i>i</i> in year <i>y</i> (t/m ³). Average value may be used through sampling with the confidence/precision level 90/10	t/m ³
$\eta_{COD,BL,i}$	COD removal efficiency of the baseline treatment system <i>i</i> , determined as per the paragraphs 35, 36 or 37 of AMS-III.H.	%

Parameter	Description	Unit
$MCF_{ww,treatment,BL,i}$	Methane correction factor for baseline wastewater treatment systems i (MCF values as per Table 2 of AMS-III.H.)	-
i	Index for baseline wastewater treatment system	-
$B_{o,ww}$	Methane producing capacity of the wastewater (IPCC value of 0.25 kg CH ₄ /kg COD) ²⁵	kg CH ₄ /kg COD
UF_{BL}	Model correction factor to account for model uncertainties (0.89) ²⁶	-
GWP_{CH_4}	Global Warming Potential for methane	tCO _{2e} /tCH ₄

Methane emissions from the baseline sludge treatment systems affected by the project activity are determined using the methane generation potential of the sludge treatment systems:

$$BE_{treatment,s,y} = \sum_j S_{j,BL,y} \times MCF_{s,treatment,BL,j} \times DOC_s \times UF_{BL} \times DOC_F \times F \times 16/12 \times GWP_{CH_4}$$

Where:

Parameter	Description	Unit
$S_{j,BL,y}$	Amount of dry matter in the sludge that would have been treated by the sludge treatment system j in the baseline scenario (t). For ex ante estimation, forecasted sludge generation volume or the designed capacity of the sludge treatment facility can be used. However, the ex post emissions reduction calculation shall be based on the actual monitored volume of treated sludge	t
j	Index for baseline sludge treatment system	-

²⁵ Project activities may use the default value of 0.6 kg CH₄/kg BOD, if the parameter BOD_{5,20} is used to determine the organic content of the wastewater. In this case, baseline and project emissions calculations shall use BOD instead of COD in the equations, and the monitoring of the project activity shall be based in direct measurements of BOD_{5,20}, i.e. the estimation of BOD values based on COD measurements is not allowed.

²⁶ Reference: FCCC/SBSTA/2003/10/Add.2, page 25.

Parameter	Description	Unit
DOC_s	Degradable organic content of the untreated sludge generated in the year y (fraction, dry basis). Default values of 0.5 for domestic sludge and 0.257 for industrial sludge ²⁷ shall be used	-
$MCF_{s,treatment,BL,j}$	Methane correction factor for the baseline sludge treatment system j (MCF values as per Table 2 of AMS-III.H.)	-
UF_{BL}	Model correction factor to account for model uncertainties (0.89)	-
DOC_F	Fraction of DOC dissimilated to biogas (IPCC default value of 0.5)	-
F	Fraction of CH ₄ in biogas (IPCC default of 0.5)	-

If the sludge is composted, the following equation shall be applied:

$$BE_{s,treatment,y} = \sum_j S_{j,BL,y} \times EF_{composting} \times GWP_{CH4}$$

Where:

Parameter	Description	Unit
$EF_{composting}$	Emission factor for composting organic waste (tCH ₄ /t waste treated). Emission factors can be based on facility/site-specific measurements, country specific values or IPCC default values (Table 4.1, chapter 4, Volume 5, 2006 IPCC Guidelines for National Greenhouse Gas Inventories). IPCC default value is 0.01 t CH ₄ / t sludge treated on a dry weight basis	tCH ₄ /t waste treated

If the baseline wastewater treatment system is different from the treatment system in the project scenario, the sludge generation rate (amount of sludge generated per unit of COD removed) in the baseline may differ significantly from that of the project scenario. For example, it is known that the amount of sludge generated in aerobic wastewater systems is larger than in anaerobic systems, for the same COD removal efficiency. Therefore, for these cases, the monitored values of the

²⁷ The IPCC default values of 0.05 for domestic sludge (wet basis, considering a default dry matter content of 10 per cent) or 0.09 for industrial sludge (wet basis, assuming dry matter content of 35 per cent), were corrected for dry basis.

amount of sludge generated during the crediting period will be used to estimate the amount of sludge generated in the baseline, as follows:

$$S_{j,BL,y} = S_{l,PJ,y} \times \frac{SGR_{BL}}{SGR_{PJ}}$$

Where:

Parameter	Description	Unit
$S_{l,PJ,y}$	Amount of dry matter in the sludge treated by the sludge treatment system i in year y in the project scenario (t)	t
SGR_{BL}	Sludge generation ratio of the wastewater treatment plant in the baseline scenario (tonne of dry matter in sludge/t COD removed). This ratio will be determined as per paragraphs 35, 36 or 37 of AMS-III.H.	tonne of dry matter in sludge/t COD removed
SGR_{PJ}	Sludge generation ratio of the wastewater treatment plant in the project scenario (tonne of dry matter in sludge/t COD removed). Calculated using the monitored values of COD removal (i.e. $COD_{inflow,i}$ minus $COD_{outflow,i}$) and sludge generation in the project scenario	tonne of dry matter in sludge/t COD removed

Methane emissions from degradable organic carbon in treated wastewater discharged in e.g. a river, sea or lake in the baseline situation are determined as follows:

$$BE_{ww,discharge,y} = Q_{ww,y} \times GWP_{CH_4} \times B_{o,ww} \times UF_{BL} \times COD_{ww,discharge,BL,y} \times MCF_{ww,BL,discharge}$$

Where:

Parameter	Description	Unit
$Q_{ww,y}$	Volume of treated wastewater discharged in year y (m ³)	m ³
UF_{BL}	Model correction factor to account for model uncertainties (0.89)	-

Parameter	Description	Unit
$COD_{ww,discharge,BL,y}$	Chemical oxygen demand of the treated wastewater discharged into sea, river or lake in the baseline situation in the year y (t/m ³). If the baseline scenario is the discharge of untreated wastewater, the COD of untreated wastewater shall be used	t/m ³
$MCF_{ww,BL,discharge}$	Methane correction factor based on discharge pathway in the baseline situation (e.g. into sea, river or lake) of the wastewater (fraction) (MCF values as per Table 2 of AMS-III.H.)	Equation (1)-

To determine $COD_{ww,discharge,BL,y}$: if the baseline treatment system(s) is different from the treatment system(s) in the project scenario, the monitored values of the COD inflow during crediting period will be used to calculate the baseline emissions ex post. The outflow COD of the baseline systems will be estimated using the removal efficiency of the baseline treatment systems, estimated as per paragraphs 35, 36 or 37 of AMS-III.H.

Methane emissions from anaerobic decay of the final sludge produced are determined as follows:

$$BE_{s,final,y} = S_{final,BL,y} \times DOC_s \times UF_{BL} \times MCF_{s,BL,final} \times DOC_F \times F \times 16/12 \times GWP_{CH_4}$$

Where:

Parameter	Description	Unit
$S_{final,BL,y}$	Amount of dry matter in the final sludge generated by the baseline wastewater treatment systems in the year y (t). If the baseline wastewater treatment system is different from the project system, it will be estimated using the monitored amount of dry matter in the final sludge generated by the project activity ($S_{final,PJ,y}$) corrected for the sludge generation ratios of the project and baseline systems as per equation (5) above	t
$MCF_{s,BL,final}$	Methane correction factor of the disposal site that receives the final sludge in the baseline situation, estimated as per the procedures described in the methodological tool "Emissions from solid waste disposal sites"	-
UF_{BL}	Model correction factor to account for model uncertainties (0.89)	-

For each PAI in the Grouped Project, the baseline emissions will be identified according to the procedures set out in *AMS-III.H.: Methane recovery in wastewater treatment --- Version 19.0*. The table below summarises the baseline emissions that are considered for each PAI.

Source	Included/Excluded	Justification
Baseline emissions from electricity or fuel consumption in year y ($BE_{power,y}$)	Included/Excluded	Electricity or fuel may or may not be used in the baseline scenario
Baseline emissions of the wastewater treatment systems affected by the project activity in year y ($BE_{ww,treatment,y}$)	Included	Main emissions source in the baseline from open lagoons
Baseline emissions of the sludge treatment systems affected by the project activity in year y ($BE_{s,treatment,y}$)	Excluded	No sludge treatment activity involved in each PAI
Baseline methane emissions from degradable organic carbon in treated wastewater discharged into sea/river/lake in year y ($BE_{ww,discharge,y}$)	Included/Excluded	The treated effluent from open anaerobic lagoons may or may not be discharged into sea/river/lake
Baseline methane emissions from anaerobic decay of the final sludge produced in year y ($BE_{s,final,y}$)	Excluded	No sludge treatment activity involved in each PAI

AMS I D, Version 18.0

According to the approved methodology AMS I D, Version 18.0, the baseline emissions include only CO₂ emissions from electricity generation in power plants that are displaced due to the project activity:

$$BE_y = EG_{PJ,y} \times EF_{grid,y}$$

Where:

Parameter	Description	Unit
$EG_{PJ,y}$	Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y	MWh
$EF_{grid,y}$	Combined margin CO ₂ emission factor for grid connected power generation in year y calculated using the latest version of the "Tool to calculate the emission factor for an electricity system"	tCO ₂ /MWh

The table below summarises the baseline emissions that are considered for each PAI.

Source	Included/ Excluded	Justification
Baseline emissions from electricity generation in power plants that are displaced due to the project activity (BE_y)	Included/ Excluded	Electricity generation in power plants that are displaced due to the project activity may or may not be applied in the baseline scenario

Baseline Emissions for the PAI1

AMS III H, Version 19.0

The table below summarises the baseline emissions that are considered for the PAIs:

Source	Included/Excluded	Justification
$BE_{power,y}$	Excluded	No electricity or fossil fuel used
$BE_{ww,treatment,y}$	Included	Main emissions source from open lagoons
$BE_{s,treatment,y}$	Excluded	No sludge treatment activity involved
$BE_{ww,discharge,y}$	Included	The effluent is discharged into river
$BE_{s,final,y}$	Excluded	No sludge treatment activity involved

AMS I D, Version 18.0

The table below summarises the baseline emissions that are considered for the PAIs:

Source	Included/Excluded	Justification
BE_y	Included	Power generation from the PAI1 will be uploaded to the grid

The summary of baseline emission for the PAI1 is tabulated below:

Year	Total BE_y	AMS IIIH		AMS ID
		$BE_{ww,treatment,y}$	$BE_{ww,discharge,y}$	BE_y
24-Apr-2024 - 31-Dec-2024 (252 days)	27,762	25,594	70	2,098
01-Jan-2025 - 31-Dec-2025	41,784	37,071	101	4,612

Year	Total BE _y	AMS IIIH		AMS ID
		BE _{ww,treatment,y}	BE _{ww,discharge,y}	BE _y
01-Jan-2026 – 31-Dec-2026	41,784	37,071	101	4,612
01-Jan-2027 – 31-Dec-2027	41,784	37,071	101	4,612
01-Jan-2028 – 31-Dec-2028	41,784	37,071	101	4,612
01-Jan-2029 – 31-Dec-2029	41,784	37,071	101	4,612
01-Jan-2030 – 31-Dec-2030	41,784	37,071	101	4,612
01-Jan-2031 – 23-Apr-2031 (113 days)	12,936	11,477	31	1,428
Total	291,404	259,498	708	31,198

The detailed calculation for baseline emission is tabulated below:

AMS III H

BE_{ww,treatment,y} - Baseline emission of the wastewater treatment systems affected by the project activity

Parameter	24-Apr-2024 – 31-Dec-2024	01-Jan-2025 – 31-Dec-2025	01-Jan-2026 – 31-Dec-2026	01-Jan-2027 – 31-Dec-2027	01-Jan-2028 – 31-Dec-2028	01-Jan-2029 – 31-Dec-2029	01-Jan-2030 – 31-Dec-2030	01-Jan-2031 – 23-Apr-2031
$\Sigma Q_{ww,i,y}$	92,002	133,257	133,257	133,257	133,257	133,257	133,257	41,255
COD _{inflow,i,y}	0.0657	0.0657	0.0657	0.0657	0.0657	0.0657	0.0657	0.0657
$h_{COD, BL, i}$	85%	85%	85%	85%	85%	85%	85%	85%
$MCF_{ww,treatment,BL,i}$	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.8
$B_{o,ww}$	0.25	0.25	0.25	0.25	0.25	0.25	0.25	0.25
UF _{BL}	0.89	0.89	0.89	0.89	0.89	0.89	0.89	0.89
GWP _{CH4}	28	28	28	28	28	28	28	28
BE_{ww,treatment,y}	25,594	37,071	37,071	37,071	37,071	37,071	37,071	11,477

BE_{ww,discharge,y} - Baseline methane emissions from degradable organic carbon in treated wastewater discharged into river

Parameter	24-Apr-2024 – 31-Dec-2024	01-Jan-2025 – 31-Dec-2025	01-Jan-2026 – 31-Dec-2026	01-Jan-2027 – 31-Dec-2027	01-Jan-2028 – 31-Dec-2028	01-Jan-2029 – 31-Dec-2029	01-Jan-2030 – 31-Dec-2030	01-Jan-2031 – 23-Apr-2031
Q _{ww,y}	92,002	133,257	133,257	133,257	133,257	133,257	133,257	41,255
GWP _{CH4}	28	28	28	28	28	28	28	28
B _{o,ww}	0.25	0.25	0.25	0.25	0.25	0.25	0.25	0.25
UF _{BL}	0.89	0.89	0.89	0.89	0.89	0.89	0.89	0.89
COD _{ww,discharge,BL,y}	0.00122	0.00122	0.00122	0.00122	0.00122	0.00122	0.00122	0.00122
MCF _{ww,BL,discharge}	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1
BE_{ww,discharge,y}	70	101	101	101	101	101	101	31

AMS ID

BE_y - Baseline emissions from electricity generation in power plants that are displaced due to the project activity

Parameter	19-July-2024 – 31-Dec-2024	01-Jan-2025 – 31-Dec-2025	01-Jan-2026 – 31-Dec-2026	01-Jan-2027 – 31-Dec-2027	01-Jan-2028 – 31-Dec-2028	01-Jan-2029 – 31-Dec-2029	01-Jan-2030 – 31-Dec-2030	01-Jan-2031 – 23-Apr-2031
EG _{BLy}	3,586	7884	7884	7884	7884	7884	7884	2,441
EF _{CO2}	0.585	0.585	0.585	0.585	0.585	0.585	0.585	0.585
BE_y	2,098	4,612	4,612	4,612	4,612	4,612	4,612	1,428

4.2 Project Emissions

Grouped Project

AMS III H, Version 19.0

According to the approved methodology AMS III H, Version 19.0, the project emissions for instances in the Grouped Project are determined as follows:

Project emissions consist of:

- CO₂ emissions from electricity and fuel used by the project facilities ($PE_{power,y}$);
- Methane emissions from wastewater treatment systems affected by the project activity, and not equipped with biogas recovery in the project scenario ($PE_{ww,treatment,y}$);
- Methane emissions from sludge treatment systems affected by the project activity, and not equipped with biogas recovery in the project situation ($PE_{s,treatment,y}$);
- Methane emissions on account of inefficiency of the project activity wastewater treatment systems and presence of degradable organic carbon in treated wastewater ($PE_{ww,discharge,y}$);
- Methane emissions from the decay of the final sludge generated by the project activity treatment systems ($PE_{s,final,y}$);
- Methane fugitive emissions due to inefficiencies in capture systems ($PE_{fugitive,y}$);
- Methane emissions due to incomplete flaring ($PE_{flaring,y}$);
- Methane emissions from biomass stored under anaerobic conditions which would not have occurred in the baseline situation ($PE_{biomass,y}$);

$$PE_y = \left\{ \begin{array}{l} PE_{power,y} + PE_{ww,treatment,y} + PE_{s,treatment,y} + PE_{ww,discharge,y} + PE_{s,final,y} + \\ PE_{fugitive,y} + PE_{biomass,y} + PE_{flaring,y} \end{array} \right\}$$

Where:

Parameter	Description	Unit
PE_y	Project emissions in year y	tCO ₂ e
$PE_{power,y}$	Emissions from electricity or fuel consumption in the year y (t CO ₂ e). These emissions shall be calculated as per paragraph 26 of AMS III H, for the situation of the project scenario, using energy consumption data of all equipment/devices used in the project activity wastewater and sludge treatment systems and systems for biogas recovery and flaring/gainful use	tCO ₂ e
$PE_{ww,treatment,y}$	Methane emissions from wastewater treatment systems affected by the project activity, and not equipped with biogas recovery, in year y (t CO ₂ e). These emissions shall	tCO ₂ e

Parameter	Description	Unit
	be calculated as per equation (2) in paragraph 27 of AMS-III.H, using an uncertainty factor of 1.12 and data applicable to the project situation ($MCF_{ww,treatment,PJ,k}$ and $\eta_{PJ,k,y}$) and with the following changed definition of parameters: $MCF_{ww,treatment,PJ,k}$ Methane correction factor for project wastewater treatment system k (MCF values as per Table 2) $\eta_{PJ,k,y}$ Chemical oxygen demand removal efficiency of the project wastewater treatment system k in year y (t/m³), measured based on inflow COD and outflow COD in system k	
$PE_{s,treatment,y}$	Methane emissions from sludge treatment systems affected by the project activity, and not equipped with biogas recovery, in year y (t CO₂e). These emissions shall be calculated as per equations (3) and (4) in paragraphs 30 and 31 of AMS-III.H, using an uncertainty factor of 1.12 and data applicable to the project situation ($S_{i,PJ,y}$, $MCF_{s,treatment,l}$) and with the following changed definition of parameters: $S_{i,PJ,y}$ Amount of dry matter in the sludge treated by the sludge treatment system l in the project scenario in year y (t) $MCF_{s,treatment,l}$ Methane correction factor for the project sludge treatment system l (MCF values as per Table 2 of AMS-III.H)	tCO ₂ e
$PE_{ww,discharge,y}$	Methane emissions from degradable organic carbon in treated wastewater in year y (tCO₂e). These emissions shall be calculated as per equation (6) in paragraph 33 of AMS-III.H, using an uncertainty factor of 1.12 and data applicable to the project conditions ($COD_{ww,discharge,PJ,y}$, $MCF_{ww,PJ,discharge}$) and with the following changed definition of parameters: $COD_{ww,discharge,PJ,y}$ Chemical oxygen demand of the treated wastewater discharged into the sea, river or lake in the project scenario in year y (t/m³)	tCO ₂ e

Parameter	Description	Unit
	$MCF_{ww,PJ,discharge}$ Methane correction factor based on the discharge pathway of the wastewater in the project scenario (e.g. into sea, river or lake) (MCF values as per Table 2)	
$PE_{s,final,y}$	Methane emissions from anaerobic decay of the final sludge produced in year y (t CO₂e). These emissions shall be calculated as per equation (7) in paragraph 35 of AMS-III.H, using an uncertainty factor of 1.12 and data applicable to the project conditions ($MCF_{s,PJ,final}$, $S_{final,PJ,y}$). If the sludge is controlled combusted, disposed in a landfill with biogas recovery, or used for soil application in aerobic conditions in the project activity, this term shall be neglected, and the sludge treatment and/or use and/or final disposal shall be monitored during the crediting period with the following revised definition of the parameters: $MCF_{s,PJ,final}$ Methane correction factor of the disposal site that receives the final sludge in the project situation, estimated as per the procedures described in the methodological tool “Emissions from solid waste disposal sites” $S_{final,PJ,y}$ Amount of dry matter in final sludge generated by the project wastewater treatment systems in the year y (t)	tCO ₂ e
$PE_{fugitive,y}$	Methane emissions from biogas release in capture systems in year y, calculated as per paragraph 4 (tCO ₂ e)	tCO ₂ e
$PE_{biomass,y}$	Methane emissions due to incomplete flaring in year y (t CO ₂ e). For ex ante estimation, baseline emission calculation for wastewater and/or sludge treatment (i.e. equation (2) and/or equation (3)) can be used but without the consideration of GWP for CH ₄ . However, the ex post emission reduction shall be calculated as per methodological tool “Project emissions from flaring”	tCO ₂ e
$PE_{flaring,y}$	Methane emissions from biomass stored under anaerobic conditions. If storage of biomass under anaerobic conditions takes place in the project and does not occur in the baseline, methane emissions due to anaerobic decay of this biomass shall be considered and be determined as	tCO ₂ e

Parameter	Description	Unit
	per the procedure in the methodological tool “Emissions from solid waste disposal sites” (t CO ₂ e)	

Project emissions from methane release in capture systems are determined as follows:

Based on the methane emission potential of wastewater and/or sludge:

$$PE_{fugitive,y} = PE_{fugitive,ww,y} + PE_{fugitive,s,y}$$

Where:

Parameter	Description	Unit
$PE_{fugitive,ww,y}$	Fugitive emissions through capture inefficiencies in the anaerobic wastewater treatment systems in the year y	tCO ₂ e
$PE_{fugitive,s,y}$	Fugitive emissions through capture inefficiencies in the anaerobic sludge treatment systems in the year y	tCO ₂ e

$$PE_{fugitive,ww,y} = (1 - CFE_{ww}) \times MEP_{ww,treatment,y} \times GWP_{CH_4}$$

Where:

Parameter	Description	Unit
CFE_{ww}	Capture efficiency of the biogas recovery equipment in the wastewater treatment systems (a default value of 0.9 shall be used)	-
$MEP_{ww,treatment,y}$	Methane emission potential of wastewater treatment systems equipped with biogas recovery system in year y (t)	-

$$MEP_{ww,treatment,y} = Q_{ww,y} \times B_{o,ww} \times UF_{PJ} \times \sum_k COD_{removed,PJ,k,y} \times MCF_{ww,treatment,PJ,k}$$

Where:

Parameter	Description	Unit
$COD_{removed,PJ,k,y}$	The chemical oxygen demand removed ²⁸ by the treatment system k of the project activity equipped with biogas recovery in the year y	t/m ³

²⁸ Difference between the inflow COD and the outflow COD.

Parameter	Description	Unit
$MCF_{ww,treatment,PJ,k}$	Methane correction factor for the project wastewater treatment system k equipped with biogas recovery equipment (MCF values as per Table 2 above)	-
UF_{PJ}	Model correction factor to account for model uncertainties (1.12)	-

$$PE_{fugitive,s,y} = (1 - CFE_s) \times MEP_{s,treatment,y} \times GWP_{CH4}$$

Where:

Parameter	Description	Unit
CFE_s	Capture efficiency of the biogas recovery equipment in the sludge treatment systems (a default value of 0.9 shall be used)	-
$MEP_{s,treatment,y}$	Methane emission potential of the sludge treatment systems equipped with a biogas recovery system in year y	t

$$MEP_{s,treatment,y} = \sum_l (S_{l,PJ,y} \times MCF_{s,treatment,PJ,l}) \times DOC_s \times UF_{PJ} \times DOC_F \times F \times 16/12$$

Where:

Parameter	Description	Unit
$S_{l,PJ,y}$	Amount of sludge treated in the project sludge treatment system l equipped with a biogas recovery system (on a dry basis) in year y	t
$MCF_{s,treatment,PJ,l}$	Methane correction factor for the sludge treatment system equipped with biogas recovery equipment (MCF values as per Table 2 above)	-
UF_{PJ}	Model correction factor to account for model uncertainties (1.12)	-

For each project activity instance in the Grouped Project, the project emissions will be identified according to the procedures set out in *AMS-III.H.: Methane recovery in wastewater treatment --- Version 19.0*. The table below summarises the project emissions that will be considered for each project activity instance in the Grouped Project.

Source	Included/ Excluded	Justification
CO ₂ emissions from electricity and fuel used by the project facilities ($PE_{power,y}$)	Included/ Excluded	Electricity or fuel may or may not be used in the project scenario
Methane emissions from wastewater treatment systems affected by the project activity, and not equipped with biogas recovery in the project scenario ($PE_{ww,treatment,y}$)	Included/ Excluded	Each project activity instance to determine based on the project design
Methane emissions from sludge treatment systems affected by the project activity, and not equipped with biogas recovery in the project situation ($PE_{s,treatment,y}$)	Excluded	No sludge treatment activity involved in each PAI
Methane emissions on account of inefficiency of the project activity wastewater treatment systems and presence of degradable organic carbon in treated wastewater ($PE_{ww,discharge,y}$)	Included/ Excluded	The treatment effluent from open anaerobic lagoons may or may not be discharged into sea/river/lake
Methane emissions from the decay of the final sludge generated by the project activity treatment systems ($PE_{s,final,y}$)	Excluded	No sludge treatment activity involved in each PAI
Methane fugitive emissions due to inefficiencies in capture systems ($PE_{fugitive,y}$)	Included	Project emissions due to potential inefficiencies of biogas capture system.
Methane emissions due to incomplete flaring ($PE_{flaring,y}$)	Included/ Excluded	The project may or may not emit GHG from incomplete flaring.
Methane emissions from biomass stored under anaerobic conditions which would not have occurred in the baseline situation ($PE_{biomass,y}$)	Excluded	There is no storage of additional biomass under anaerobic conditions in project activity instances.

Project Emissions for the Project Activity Instances 1

AMS III H, Version 19.0

The table below summarises the project emissions that are considered for the PAI1

Source	Included/Excluded	Justification
$PE_{power,y}$	Excluded	No electricity or fossil fuel used in the project scenario
$PE_{ww,treatment,y}$	Excluded	The wastewater from the anaerobic digester is discharged to aerobic pond
$PE_{s,treatment,y}$	Excluded	No sludge treatment activity involved in the project activity
$PE_{ww,discharge,y}$	Included	The effluent is discharged into river
$PE_{s,final,y}$	Excluded	No sludge treatment activity involved in the project activity
$PE_{fugitive,y}$	Included	Project emissions due to potential inefficiencies of biogas capture system
$PE_{flaring,y}$	Included	Project emission due to incomplete flaring
$PE_{biomass,y}$	Excluded	No storage of additional biomass under anaerobic conditions in project activity instances

The project emissions calculated for the PAI1 is calculated below:

Year	PE _y	PE _{ww,discharge,y}	PE _{fugitive,y}	PE _{flaring,y}
24-Apr-2024 – 31-Dec-2024 (252 days)	4,427	291	3,222	914
01-Jan-2025 – 31-Dec-2025	6,412	422	4,667	1,324
01-Jan-2026 – 31-Dec-2026	6,412	422	4,667	1,324
01-Jan-2027 – 31-Dec-2027	6,412	422	4,667	1,324
01-Jan-2028 – 31-Dec-2028	6,412	422	4,667	1,324
01-Jan-2029 – 31-Dec-2029	6,412	422	4,667	1,324
01-Jan-2030 – 31-Dec-2030	6,412	422	4,667	1,324
01-Jan-2031 – 23-Apr-2031 (113 days)	1,985	131	1,445	410
Total	44,886	2,951	32,667	9,268

The detailed calculation for project emission is tabulated below:

AMS III H

PE_{ww,discharge,y} - Methane emissions from degradable organic carbon in treated wastewater

Parameter	24-Apr-2024 – 31-Dec-2024	01-Jan-2025 – 31-Dec-2025	01-Jan-2026 – 31-Dec-2026	01-Jan-2027 – 31-Dec-2027	01-Jan-2028 – 31-Dec-2028	01-Jan-2029 – 31-Dec-2029	01-Jan-2030 – 31-Dec-2030	01-Jan-2031 – 23-Apr-2031
$\Sigma Q_{ww,i,y}$	92,002	133,257	133,257	133,257	133,257	133,257	133,257	41,255
COD _{ww,discharge,PJ,y}	0.004035	0.004035	0.004035	0.004035	0.004035	0.004035	0.004035	0.004035
MCF _{ww,discharge,PJ,k}	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1
B _{o,ww}	0.25	0.25	0.25	0.25	0.25	0.25	0.25	0.25
UF _{PJ}	1.12	1.12	1.12	1.12	1.12	1.12	1.12	1.12
GWP _{CH4}	28	28	28	28	28	28	28	28
PE_{ww,discharge,y}	291	422	422	422	422	422	422	131

PE_{fugitive,y} - Emissions on account of inefficiencies in capture systems

Parameter	24-Apr-2024 – 31-Dec-2024	01-Jan-2025 – 31- Dec-2025	01-Jan-2026 – 31- Dec-2026	01-Jan-2027 – 31- Dec-2027	01-Jan-2028 – 31- Dec-2028	01-Jan-2029 – 31- Dec-2029	01-Jan-2030 – 31- Dec-2030	01-Jan-2031 – 23-Apr-2031
CFE _{ww}	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9
GWP _{CH4}	28	28	28	28	28	28	28	28
Q _{ww,y}	92,002	133,257	133,257	133,257	133,257	133,257	133,257	41,255
B _{o,ww}	0.25	0.25	0.25	0.25	0.25	0.25	0.25	0.25
UF _{PJ}	1.12	1.12	1.12	1.12	1.12	1.12	1.12	1.12
COD _{removed,PJ,k,y}	0.0558	0.0558	0.0558	0.0558	0.0558	0.0558	0.0558	0.0558
MCF _{ww,treatment,PJ,k}	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.8
MEP _{ww,treatment,y}	1,256	1,667	1,667	1,667	1,667	1,667	1,667	411
PE_{fugitive,y}	3,222	4,667	4,667	4,667	4,667	4,667	4,667	1,445

PE_{flaring,y} - Methane emissions due to incomplete flaring

Parameter	241-Apr-2024 – 31-Dec-2024	01-Jan-2025 – 31-Dec-2025	01-Jan-2026 – 31-Dec-2026	01-Jan-2027 – 31-Dec-2027	01-Jan-2028 – 31-Dec-2028	01-Jan-2029 – 31-Dec-2029	01-Jan-2030 – 31-Dec-2030	01-Jan-2031 – 23-Apr-2031
$\Sigma Q_{ww,i,y}$	92,002	133,257	133,257	133,257	133,257	133,257	133,257	41,255
COD _{inflow,i,y}	0.0657	0.0657	0.0657	0.0657	0.0657	0.0657	0.0657	0.0657
h _{COD,BL,i}	85.0%	85.0%	85.0%	85.0%	85.0%	85.0%	85.0%	85.0%
MCF _{ww,treatment,BL,i}	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.8
B _{o,ww}	0.25	0.25	0.25	0.25	0.25	0.25	0.25	0.25
UF _{BL}	0.89	0.89	0.89	0.89	0.89	0.89	0.89	0.89
GWP _{CH4}	28	28	28	28	28	28	28	28
PE _{flaring,y}	914	1,324	1,324	1,324	1,324	1,324	1,324	410

According to AMS III H, version 19, PE_{flaring}, for ex ante estimation, baseline emission calculation for wastewater and/or sludge treatment (i.e. equation (2) and/or equation (3)) can be used but without the consideration of GWP for CH₄.

AMS I.D, Version 18.0

According to AMS-I.D paragraph 39, the project emission is 0, and CO₂ emissions from on-site consumption of fossil fuel due to project activity shall be calculated using the latest version of “Tool to calculate project or leakage CO₂ emissions from fossil fuel combustion

4.3 Leakage Emissions

Grouped Project

AMS-III.H Version 19.0

According to AMS III-H Version 19.0, paragraph 41, if the technology is using equipment transferred from another activity, leakage effects at the site of the other activity are to be considered and estimated (LE_y). There are no relevant leakage effects to be considered for all the PAIs.

PAI1

No leakage emission for PAI1

4.4 Estimated GHG Emission Reductions and Carbon Dioxide Removals

Grouped Project

As per the AMS-III.H. methodology, Version 19.0, emission reductions for the Grouped Project will be calculated as follows.

For all options in paragraph 2 of AMS-III.H, emission reductions shall be estimated ex ante in the PD using the equations provided in the baseline, project and leakage emissions sections above. Emission reductions shall be estimated ex ante as follows:

$$ER_{y,ex\ ante} = BE_{y,ex\ ante} - (PE_{y,ex\ ante} + LE_{y,ex\ ante})$$

Where:

Parameter	Description	Unit
$ER_{y,ex\ ante}$	Ex-ante emission reduction in year y	tCO ₂ e
$LE_{y,ex\ ante}$	Ex-ante leakage emissions in year y	tCO ₂ e
$PE_{y,ex\ ante}$	Ex-ante project emissions in year y calculated as paragraph 39 of AMS-III.H	tCO ₂ e
$BE_{y,ex\ ante}$	Ex-ante baseline emissions in year y calculated as per paragraph 25 of AMS-III.H	tCO ₂ e

For cases 2(b), 2(c), 2(d) and 2(f), ex post emission reductions shall be based on the lowest value of the following, as per paragraph 44:

- The amount of biogas recovered and fuelled or flared (MD_y) during the crediting period, that is monitored ex post;

- (b) Ex post calculated baseline, project and leakage emissions based on actual monitored data for the project activity.

For cases 2(b), 2(c), 2(d) and 2(f): it is possible that the project activity involves wastewater and sludge treatment systems with higher methane conversion factors (*MCF*) or with higher efficiency than the treatment systems used in the baseline situation. Therefore, the emission reductions achieved by the project activity is limited to the ex post calculated baseline emissions minus project emissions using the actual monitored data for the project activity. The emission reductions achieved in any year are the lowest value of the following:

$$ER_{y,ex\ post} = \min \left((BE_{y,ex\ post} - PE_{y,ex\ post} - LE_{y,ex\ post}), (MD_y - PE_{power,y} - PE_{biomass,y} - LE_{y,ex\ post}) \right)$$

Where:

Parameter	Description	Unit
$ER_{y,ex\ post}$	Emission reductions achieved by the project activity based on monitored values for year y	tCO ₂ e
$BE_{y,ex\ post}$	Baseline emissions calculated as per paragraph 25 of AMS-III.H using ex post monitored values	tCO ₂ e
$PE_{y,ex\ post}$	Project emissions calculated as per paragraph 39 using ex post monitored values	tCO ₂ e
MD_y	Methane captured and destroyed/gainfully used by the project activity in the year y	tCO ₂ e

In the case of flaring/combustion MD_y will be measured using the conditions of the flaring process:

$$MD_y = BG_{burnt,y} \times w_{CH_4,y} \times D_{CH_4} \times FE \times GWP_{CH_4}$$

Where:

Parameter	Description	Unit
$BG_{burnt,y}$	Biogas flared/combusted in year y	m ³
$w_{CH_4,y}$	Methane content of the biogas in the year y	Volume fraction
D_{CH_4}	Density of methane at the temperature and pressure of the biogas in the year y	t/m ³
FE	Flare efficiency in year y (fraction). If the biogas is combusted for gainful purposes, e.g. fed to an engine, an efficiency of 100 per cent may be applied	-

For the cases 2 (a) and (e) the emission reduction achieved by the project activity (ex post) will be the difference between the baseline emissions and the sum of the project emissions and leakage.

$$ER_y = BE_{y,ex\ post} - (PE_{y,ex\ post} + LE_{y,ex\ post})$$

The historical records of electricity and fuel consumption, the COD content of untreated and treated wastewater, and the quantity of sludge produced by the replaced units will be used for the baseline calculation.

In case (a), if the volumetric flow and the characteristic properties (e.g. COD) of the inflow and outflow of the wastewater are equivalent in the project and the baseline scenarios (i.e. the project and baseline systems have the same efficiency for COD removal for wastewater treatment), then the higher energy consumption and sludge generation in the baseline scenario are the only significant differences contributing to emissions reductions in the project case. In this case, the emission reductions can be calculated as the difference between the historical energy consumption of the replaced unit and the recorded energy consumption of the new system, plus the difference in emissions from sludge treatment and/or disposal. Project emissions from fugitive emissions and incomplete flaring ($PE_{fugitive,y}$, $PE_{flaring,y}$) shall also be considered in the calculation of the emission reductions, however the emissions from the wastewater outflow and sludge ($PE_{ww,discharge,y}$, $PE_{s,final,y}$) may be disregarded, if they are equivalent in the baseline and project scenarios.

Emissions Reduction for the PAI1

The estimated emissions reduction for all the PAIs are calculated as per AMS-III.H, Version 19.0, which is tabulated as below:

Vintage period	Estimated baseline emissions (tCO ₂ e)	Estimated project emissions (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated removal VCU (tCO ₂ e)	Estimated total VCUs (tCO ₂ e)
24-Apr-2024 – 31-Dec-2024	27,762	4,427	0	0	23,335
01-Jan-2025 – 31-Dec-2025	41,784	6,412	0	0	35,372
01-Jan-2026 – 31-Dec-2026	41,784	6,412	0	0	35,372
01-Jan-2027 – 31-Dec-2027	41,784	6,412	0	0	35,372

Vintage period	Estimated baseline emissions (tCO ₂ e)	Estimated project emissions (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated removal VCU (tCO ₂ e)	Estimated total VCUs (tCO ₂ e)
01-Jan-2028 – 31-Dec-2028	41,784	6,412	0	0	35,372
01-Jan-2029 – 31-Dec-2029	41,784	6,412	0	0	35,372
01-Jan-2030 – 31-Dec-2030	41,784	6,412	0	0	35,372
01-Jan-2031 – 23-Apr-2031	12,936	1,985	0	0	10,951
Total	291,404	44,886	0	0	246,518

5 MONITORING

5.1 Data and Parameters Available at Validation

Grouped Project

AMS-III.H, version 19.0

Data / Parameter	GWP _{CH₄}
Data unit	tCO ₂ /tCH ₄
Description	Global warming potential for methane
Source of data	IPCC
Value applied	28
Justification of choice of data or description of measurement methods and procedures applied	Default value, latest as per IPCC
Purpose of data	Calculation of baseline and project emissions
Comments	-

Data / Parameter	B _{0,ww}
Data unit	kg CH ₄ /kg COD
Description	Methane producing capacity of the wastewater
Source of data	IPCC default value as per methodology AMS III-H version 19
Value applied	0.25
Justification of choice of data or description of measurement methods and procedures applied	Default value as per methodology AMS III-H version 19
Purpose of data	Calculation of baseline and project emissions
Comments	-

Data / Parameter	UF _{BL}
Data unit	-
Description	Model correction factor to account for model uncertainties
Source of data	AMS-III.H version 19
Value applied	0.89
Justification of choice of data or description of measurement methods and procedures applied	Default value as per methodology AMS III-H version 19
Purpose of data	Calculation of baseline emissions
Comments	-

Data / Parameter	UF _{PJ}
Data unit	-
Description	Model correction factor to account for model uncertainties
Source of data	AMS-III.H version 19
Value applied	1.12
Justification of choice of data or description of measurement methods and procedures applied	Default value as per methodology AMS III-H version 19
Purpose of data	Calculation of project emissions
Comments	-

Data / Parameter	MCF _{ww, treatment, BL, i}
Data unit	-
Description	Methane correction factor for the baseline anaerobic wastewater treatment system j
Source of data	Table 2 from AMS-III.H, Version 19, IPCC default values for Methane Correction Factor (MCF)

Value applied		
	Type of wastewater treatment and discharge pathway or system	MCF value
	Discharge of wastewater to sea, river or lake	0.1
	Land application	0.1
	Aerobic treatment, well managed	0.0
	Aerobic treatment, poorly managed or overloaded	0.3
	Anaerobic digester for sludge without methane recovery	0.8
	Anaerobic reactor without methane recovery	0.8
	Anaerobic shallow lagoon (depth less than 2 metres)	0.2
	Anaerobic deep lagoon (depth more than 2 metres)	0.8
	Septic system	0.5
	Land application	0.1
Justification of choice of data or description of measurement methods and procedures applied	Default value according to table 2 from AMS-III.H, Version 19, IPCC default values for Methane Correction Factor (MCF)	
Purpose of data	Calculation of baseline emissions	
Comments	-	

Data / Parameter	$MCF_{ww, treatment, PJ, k}$	
Data unit	-	
Description	Methane correction factor for the project anaerobic wastewater treatment system k	
Source of data	Table 2 from AMS-III.H, Version 19, IPCC default values for Methane Correction Factor (MCF)	
Value applied		
	Type of wastewater treatment and discharge pathway or system	MCF value

	Discharge of wastewater to sea, river or lake	0.1
	Land application	0.1
	Aerobic treatment, well managed	0.0
	Aerobic treatment, poorly managed or overloaded	0.3
	Anaerobic digester for sludge without methane recovery	0.8
	Anaerobic reactor without methane recovery	0.8
	Anaerobic shallow lagoon (depth less than 2 metres)	0.2
	Anaerobic deep lagoon (depth more than 2 metres)	0.8
	Septic system	0.5
	Land application	0.1
Justification of choice of data or description of measurement methods and procedures applied	Default value according to table 2 from AMS-III.H, Version 19, IPCC default values for Methane Correction Factor (MCF)	
Purpose of data	Calculation of project emissions	
Comments	-	

Data / Parameter	CFE _{ww}
Data unit	-
Description	Capture efficiency of the biogas recovery equipment in the wastewater treatment systems
Source of data	AMS-III.H, Version 19
Value applied	0.9
Justification of choice of data or description of measurement methods and procedures applied	Default value as per methodology AMS III-H version 19
Purpose of data	Calculation of project emissions
Comments	-

Data / Parameter	$\eta_{\text{COD,BL},i}$
Data unit	%
Description	COD removal efficiency of the baseline system
Source of data	Historical records of at least one (1) year prior to the project implementation or 10-day baseline measurement campaign for the existing wastewater system
Value applied	-
Justification of choice of data or description of measurement methods and procedures applied	COD removal efficiency is based on either historical records of at least one (1) year prior to the project implementation or 10-day baseline measurement campaign for the existing wastewater system.
Purpose of data	Calculation of baseline emissions
Comments	-

Data / Parameter	$\eta_{\text{COD,PJ},i}$
Data unit	%
Description	COD removal efficiency of the project system
Source of data	Technology provider
Value applied	-
Justification of choice of data or description of measurement methods and procedures applied	COD removal efficiency is based on the expertise of the technology provider.
Purpose of data	Ex ante estimation of project emissions.
Comments	Note that ex-post calculation of project estimations is based on the monitored data for actual COD reductions, as relevant to the PAI.

Data / Parameter	EF _{equipment}	
Data unit	Kg/hour/source	
Description	Methane emission factors for equipment ²⁹	
Source of data	AMS-III.H Appendix para 12	
Value applied	Equipment type	Emission factor for methane
	Valves	4.5E-0.3
	Pump seals	2.4E-0.3
	Others ³⁰	8.8E-0.3
	Connectors	2.0E-0.4
	Flanges	3.9E-0.4
	Open ended lines	2.0E-0.3
Justification of choice of data or description of measurement methods and procedures applied	Taken from Protocol from Equipment Leak estimates, published by EPA (United States Environmental Protection Agency)	
Purpose of data	For the purpose of estimation of methane emissions in Project emissions	
Comments	For PAIs that apply AMS-III.H (para 4(b) and 4(c))	

Data / Parameter	LR _{pipeline}	
Data unit	Gg/10 ⁶ m ³	
Description	Physical leakage rate from the dedicated piped network	
Source of data	AMS-III.H Appendix para 17	
Value applied	0.0125	

²⁹ Please refer to the document US EPA-453/R-95-017 Table 2.4, page 2-15, accessed on 23/10/2007

³⁰ The emission factor for “other” equipment type was derived from compressors, diaphragms, drains, dump arms, hatches, instruments, meters, pressure relief valves, polished rods, relief valves and vents. This “other” equipment type should be applied for any equipment type other than connectors, flanges, open-ended lines, pumps or valves.

Justification of choice of data or description of measurement methods and procedures applied	2006 IPCC Guidelines for National Greenhouse Gas Inventories, Volume 2, chapter 4, Table 4.2.5 provides default values for fugitive emissions from gas operations in developing countries. The default values provided for fugitive emissions for the distribution of natural gas to end users range from 1.1 E-3 to 2.5 E-3 Gg per 106 m ³ of utility sales. The uncertainty in this value is -20 per cent to 500 per cent. A conservative value of 2.5 E-3 * 500% = 0.0125 Gg per 106 m ³ of utility sales shall be taken.
Purpose of data	To calculate physical leakage of upgraded biogas from dedicated piped network
Comments	For PAIs that apply AMS-III.H (para 4(b) and 4(c))

Data / Parameter	SPEC _{flare}
Data unit	Temperature - °C Flow rate or heat flux - kg/h or m ³ /h Maintenance schedule - number of days
Description	Manufacturer's flare operating specifications for temperature, flow rate and maintenance schedule
Source of data	Flare manufacturer
Value applied	-
Justification of choice of data or description of measurement methods and procedures applied	-
Purpose of data	Calculation of project emissions
Comments	Only applicable in case of enclosed flares. The maintenance schedule is not required if Option A is selected to determine flare efficiency of an enclosed flare.

Data / Parameter	Q _{ww,i,y}
Data unit	m ³ /month
Description	Volume of wastewater treated in the project wastewater system in year y
Source of data	Historical data from the mill

Value applied	To be provided by PAIs
Justification of choice of data or description of measurement methods and procedures applied	Measured continuously using flow meter
Purpose of data	Calculation of baseline and project emissions
Comments	-

Data / Parameter	COD _{ww,untreated,y} or COD _{inflow,y}
Data unit	tCOD/m ³
Description	Chemical oxygen demand of the wastewater inflow to the baseline treatment system i in year y
Source of data	10 day COD campaign
Value applied	To be provided by PAIs
Justification of choice of data or description of measurement methods and procedures applied	Laboratory analysis
Purpose of data	Calculation of baseline and project emissions
Comments	-

Data / Parameter	COD _{ww,treated,y} or COD _{outflow,y}
Data unit	tCOD/m ³
Description	Chemical oxygen demand of the wastewater outflow from the baseline treatment system i in year y
Source of data	10 day COD campaign
Value applied	To be provided by PAIs
Justification of choice of data or description of measurement methods and procedures applied	Laboratory analysis

Purpose of data	Calculation of baseline and project emissions
Comments	-

Data / Parameter	COD _{ww,discharged,y}
Data unit	tCOD/m ³
Description	Chemical oxygen demand of the treated wastewater discharged into the sea, river or lake in the project scenario in year y
Source of data	10 day COD campaign
Value applied	To be provided by PAIs
Justification of choice of data or description of measurement methods and procedures applied	Laboratory analysis
Purpose of data	Calculation of baseline and project emissions
Comments	-

Data / Parameter	COD _{removed,PJ,k,y}
Data unit	tCOD/m ³
Description	Chemical oxygen demand removed by the treatment system k of the project activity equipped with biogas recovery in the year y
Source of data	10 day COD campaign
Value applied	To be provided by PAIs
Justification of choice of data or description of measurement methods and procedures applied	Laboratory analysis
Purpose of data	Calculation of baseline and project emissions
Comments	Difference between the inflow COD and the outflow COD

AMS-I.D, version 18.0

Data / Parameter	EF _{grid,CM,y}								
Data unit	tCO ₂ /MWh								
Description	Grid Emission Factor for Peninsula Malaysia, Sabah, and Sarawak								
Source of data	Ministry of Energy, Science, Technology, Environment and Climate Change								
Value applied	<table border="1"> <thead> <tr> <th>Region</th><th>Combined Margin (CM) (tCO₂/MWh)</th></tr> </thead> <tbody> <tr> <td>Peninsula Malaysia</td><td>0.585</td></tr> <tr> <td>Sabah</td><td>0.525</td></tr> <tr> <td>Sarawak</td><td>0.330</td></tr> </tbody> </table> During actual monitoring, any updated GEFs officially published by the Government will be assessed. For each monitoring period, a direct comparison will be made between the latest available official GEF and the 2017 GEF. The lower of the two values will be applied in the ex-post calculation of emission reductions,	Region	Combined Margin (CM) (tCO ₂ /MWh)	Peninsula Malaysia	0.585	Sabah	0.525	Sarawak	0.330
Region	Combined Margin (CM) (tCO ₂ /MWh)								
Peninsula Malaysia	0.585								
Sabah	0.525								
Sarawak	0.330								
Justification of choice of data or description of measurement methods and procedures applied	Calculated by official sources according to the “Tool to calculate the emission factor for an electricity system” ³¹								
Purpose of data	Calculation of baseline emissions								
Comments	Year 2017								

Data / Parameter	EG _{PJ,facility,y}
Data unit	MWh/year
Description	Quantity of net electricity generation supplied by the project to the grid in year y
Source of data	Measurement
Value applied	To be provided by PAIs

³¹ <https://www.mgta.gov.my/wp-content/uploads/2019/12/2017-CDM-Electricity-Baseline-Final-Report-Publication-Version.pdf>

Justification of choice of data or description of measurement methods and procedures applied	Measured continuously by electricity meter
Purpose of data	Calculation of baseline emissions
Comments	-

Tool 05: Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation, version 3.0

Data / Parameter	$EC_{PJ,j,y}$
Data unit	MWh/yr
Description	Quantity of electricity consumed by the project electricity consumption source j in year y
Source of data	Direct measurement or calculated based on measurements from more than one electricity meters
Description of measurement methods and procedures to be applied	Use electricity meters installed at the electricity consumption sources
Frequency of monitoring/recording	Continuous measurement and at least monthly recording
Value applied	Value to be monitored ex-post
Monitoring equipment	Transformer power meter
QA/QC procedures to be applied	In cases where electricity meters are regulated (e.g. the electricity is supplied by the electric grid), the electricity meter will be subject to regular maintenance and testing in accordance with the stipulation of the meter supplier and/or as per the requirements set by the grid operators or national requirements. The calibration of meters, including the frequency of calibration, should be done in accordance with national standards or requirements set by the meter supplier or requirements set by the grid operators. The accuracy class of the meters should be in accordance with the stipulation of the meter supplier and/or as per the requirements set by the grid operators or national requirements.

	In cases where electricity meters are not regulated (e.g. the electricity is supplied by captive power plants), the electricity meter will be subject to regular maintenance and testing in accordance with the stipulation of the meter supplier or national requirements. The calibration of meters, including the frequency of calibration, should be done in accordance with national standards or requirements set by the meter supplier. The accuracy class of the meters should be in accordance with the stipulation of the meter supplier or national requirements. If these standards are not available, and meter supplier does not specify, calibrate the meters every 3 years and use the meters with at least 0.5 accuracy class (e.g. a meter with 0.2 accuracy class is more accurate and thus it is accepted).
Purpose of data	Calculation of project emissions.
Calculation method	-
Comments	-

PAI1

AMS-III.H, version 19.0

Data / Parameter	GWP _{CH4}
Data unit	tCO ₂ /tCH ₄
Description	Global warming potential for methane
Source of data	IPCC
Value applied	28
Justification of choice of data or description of measurement methods and procedures applied	Default value, latest as per IPCC
Purpose of data	Calculation of baseline and project emissions
Comments	-

Data / Parameter	Bo,ww
Data unit	kg CH ₄ /kg COD
Description	Methane producing capacity of the wastewater

Source of data	IPCC default value as per methodology AMS III-H version 19
Value applied	0.25
Justification of choice of data or description of measurement methods and procedures applied	Default value as per methodology AMS III-H version 19
Purpose of data	Calculation of baseline and project emissions
Comments	-

Data / Parameter	UF _{BL}
Data unit	-
Description	Model correction factor to account for model uncertainties
Source of data	AMS-III.H version 19
Value applied	0.89
Justification of choice of data or description of measurement methods and procedures applied	Default value as per methodology AMS III-H version 19
Purpose of data	Calculation of baseline emissions
Comments	-

Data / Parameter	UF _{PJ}
Data unit	-
Description	Model correction factor to account for model uncertainties
Source of data	AMS-III.H version 19
Value applied	1.12
Justification of choice of data or description of measurement methods and procedures applied	Default value as per methodology AMS III-H version 19

Purpose of data	Calculation of project emissions
Comments	-

Data / Parameter	$MCF_{ww, treatment, BL, i}$
Data unit	-
Description	Methane correction factor for the baseline anaerobic wastewater treatment system j
Source of data	Table 2 from AMS-III.H, Version 19, IPCC default values for Methane Correction Factor (MCF)
Value applied	0.8
Justification of choice of data or description of measurement methods and procedures applied	The baseline wastewater treatment system consists of a succession of anaerobic deep lagoons (depth more than 2 metres) therefore the MCF value is chosen as 0.8
Purpose of data	Calculation of baseline emissions
Comments	-

Data / Parameter	$MCF_{ww, treatment, PJ, k}$
Data unit	-
Description	Methane correction factor for the project anaerobic wastewater treatment system k
Source of data	Table 2 from AMS-III.H, Version 19, IPCC default values for Methane Correction Factor (MCF)
Value applied	0.8
Justification of choice of data or description of measurement methods and procedures applied	The baseline wastewater treatment system consists of a succession of anaerobic deep lagoons (depth more than 2 metres) therefore the MCF value is chosen as 0.8
Purpose of data	Calculation of project emissions
Comments	-

Data / Parameter	CFE _{ww}
Data unit	-
Description	Capture efficiency of the biogas recovery equipment in the wastewater treatment systems
Source of data	AMS-III.H, Version 19
Value applied	0.9
Justification of choice of data or description of measurement methods and procedures applied	Default value as per methodology AMS III-H version 19
Purpose of data	Calculation of project emissions
Comments	-

Data / Parameter	$\eta_{\text{COD,BL},i}$																																																
Data unit	%																																																
Description	COD removal efficiency of the baseline system																																																
Source of data	10-day baseline measurement campaign for the existing wastewater system																																																
Value applied	95.5%																																																
Justification of choice of data or description of measurement methods and procedures applied	COD removal efficiency is based on COD values for the baseline system before implementation of project activity. <table><tr><th>DATE</th><th>Point A - Raw POME</th><th>Point C - Treated POME</th><th>Point D - Final Discharge</th></tr><tr><td>5/7/2023</td><td>57,400</td><td>4,550</td><td>1,100</td></tr><tr><td>6/7/2023</td><td>86,700</td><td>5,300</td><td>1,070</td></tr><tr><td>7/7/2023</td><td>60,700</td><td>3,670</td><td>1,090</td></tr><tr><td>11/7/2023</td><td>59,800</td><td>2,150</td><td>1,400</td></tr><tr><td>12/7/2023</td><td>57,400</td><td>1,920</td><td>1,100</td></tr><tr><td>15/7/2023</td><td>65,000</td><td>2,200</td><td>1,180</td></tr><tr><td>18/7/2023</td><td>32,500</td><td>2,520</td><td>1,160</td></tr><tr><td>21/7/2023</td><td>93,700</td><td>2,210</td><td>1,200</td></tr><tr><td>22/7/2023</td><td>71,600</td><td>2,180</td><td>1,420</td></tr><tr><td>25/7/2023</td><td>72,100</td><td>3,040</td><td>1,460</td></tr><tr><td>Average</td><td>65,690</td><td>2,974</td><td>1,218</td></tr></table>	DATE	Point A - Raw POME	Point C - Treated POME	Point D - Final Discharge	5/7/2023	57,400	4,550	1,100	6/7/2023	86,700	5,300	1,070	7/7/2023	60,700	3,670	1,090	11/7/2023	59,800	2,150	1,400	12/7/2023	57,400	1,920	1,100	15/7/2023	65,000	2,200	1,180	18/7/2023	32,500	2,520	1,160	21/7/2023	93,700	2,210	1,200	22/7/2023	71,600	2,180	1,420	25/7/2023	72,100	3,040	1,460	Average	65,690	2,974	1,218
DATE	Point A - Raw POME	Point C - Treated POME	Point D - Final Discharge																																														
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Average	65,690	2,974	1,218																																														

	tCOD/m³	0.0657	0.0030	0.0012
	COD Removal (tCOD/m³)	0.0627	1,756	64,472
	Removal Rate	95.5%	59.0%	98.1%
Purpose of data	Calculation of baseline emissions			
Comments	-			

Data / Parameter	$\eta_{\text{COD,PJ,i}}$
Data unit	%
Description	COD removal efficiency of the project system
Source of data	Technology provider
Value applied	90%
Justification of choice of data or description of measurement methods and procedures applied	COD removal efficiency is based on the expertise of the technology provider.
Purpose of data	Ex ante estimation of project emissions.
Comments	Note that ex-post calculation of project estimations is based on the monitored data for actual COD reductions, as relevant to the PAI.

Data / Parameter	$Q_{\text{ww,i,y}}$
Data unit	m ³ /month
Description	Volume of wastewater treated in the project wastewater system in year y
Source of data	Historical data from the mill
Value applied	133,257 m ³ /year
Justification of choice of data or description of measurement methods and procedures applied	Measured continuously using flow meter
Purpose of data	Calculation of baseline and project emissions

Comments	-																										
Data / Parameter	COD _{ww,untreated,y} or COD _{inflow,y}																										
Data unit	tCOD/m ³																										
Description	Chemical oxygen demand of the wastewater inflow to the baseline treatment system i in year y																										
Source of data	10 day COD campaign																										
Value applied	COD values obtained from the 10 day COD campaign <table border="1"> <thead> <tr> <th>DATE</th><th>Point A - Raw POME</th></tr> </thead> <tbody> <tr><td>5/7/2023</td><td>57,400</td></tr> <tr><td>6/7/2023</td><td>86,700</td></tr> <tr><td>7/7/2023</td><td>60,700</td></tr> <tr><td>11/7/2023</td><td>59,800</td></tr> <tr><td>12/7/2023</td><td>57,400</td></tr> <tr><td>15/7/2023</td><td>65,000</td></tr> <tr><td>18/7/2023</td><td>32,500</td></tr> <tr><td>21/7/2023</td><td>93,700</td></tr> <tr><td>22/7/2023</td><td>71,600</td></tr> <tr><td>25/7/2023</td><td>72,100</td></tr> <tr><td>Average</td><td>65,690</td></tr> <tr><td>tCOD/m³</td><td>0.0657</td></tr> </tbody> </table>	DATE	Point A - Raw POME	5/7/2023	57,400	6/7/2023	86,700	7/7/2023	60,700	11/7/2023	59,800	12/7/2023	57,400	15/7/2023	65,000	18/7/2023	32,500	21/7/2023	93,700	22/7/2023	71,600	25/7/2023	72,100	Average	65,690	tCOD/m ³	0.0657
DATE	Point A - Raw POME																										
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Average	65,690																										
tCOD/m ³	0.0657																										
Justification of choice of data or description of measurement methods and procedures applied	Laboratory analysis																										
Purpose of data	Calculation of baseline and project emissions																										
Comments	-																										

Data / Parameter	COD _{ww,treated,y} or COD _{outflow,y}
Data unit	tCOD/m ³
Description	Chemical oxygen demand of the wastewater outflow from the baseline treatment system i in year y
Source of data	10 day COD campaign
Value applied	COD values obtained from the 10 day COD campaign.

	DATE	Point C - Treated POME
	5/7/2023	4,550
	6/7/2023	5,300
	7/7/2023	3,670
	11/7/2023	2,150
	12/7/2023	1,920
	15/7/2023	2,200
	18/7/2023	2,520
	21/7/2023	2,210
	22/7/2023	2,180
	25/7/2023	3,040
	Average	2,974
	tCOD/m ³	0.0030
Justification of choice of data or description of measurement methods and procedures applied	Laboratory analysis	
Purpose of data	Calculation of baseline and project emissions	
Comments	-	

Data / Parameter	COD _{ww,discharged,y}
Data unit	tCOD/m ³
Description	Chemical oxygen demand of the treated wastewater discharged into the sea, river or lake in the project scenario in year y
Source of data	10 day COD campaign
Value applied	COD values obtained from the 10 day COD campaign.

DATE	Point D - Final Discharge
5/7/2023	1,100
6/7/2023	1,070
7/7/2023	1,090
11/7/2023	1,400
12/7/2023	1,100
15/7/2023	1,180
18/7/2023	1,160
21/7/2023	1,200
22/7/2023	1,420
25/7/2023	1,460
Average	1,218

	tCOD/m ³	0.0012
Justification of choice of data or description of measurement methods and procedures applied	Laboratory analysis	
Purpose of data	Calculation of baseline and project emissions	
Comments	-	

Data / Parameter	COD _{removed,PJ,k,y}																											
Data unit	tCOD/m ³																											
Description	Chemical oxygen demand removed by the treatment system k of the project activity equipped with biogas recovery in the year y																											
Source of data	10 day COD campaign																											
Value applied	COD values obtained from the 10 day COD campaign. <table><tr><th>DATE</th><th>COD Removed</th></tr><tr><td>5/7/2023</td><td>52,850</td></tr><tr><td>6/7/2023</td><td>81,400</td></tr><tr><td>7/7/2023</td><td>57,030</td></tr><tr><td>11/7/2023</td><td>57,650</td></tr><tr><td>12/7/2023</td><td>55,480</td></tr><tr><td>15/7/2023</td><td>62,800</td></tr><tr><td>18/7/2023</td><td>29,980</td></tr><tr><td>21/7/2023</td><td>91,490</td></tr><tr><td>22/7/2023</td><td>69,420</td></tr><tr><td>25/7/2023</td><td>69,060</td></tr><tr><td>Average</td><td>62,716</td></tr><tr><td>tCOD/m³</td><td>0.0627</td></tr></table>		DATE	COD Removed	5/7/2023	52,850	6/7/2023	81,400	7/7/2023	57,030	11/7/2023	57,650	12/7/2023	55,480	15/7/2023	62,800	18/7/2023	29,980	21/7/2023	91,490	22/7/2023	69,420	25/7/2023	69,060	Average	62,716	tCOD/m ³	0.0627
DATE	COD Removed																											
5/7/2023	52,850																											
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25/7/2023	69,060																											
Average	62,716																											
tCOD/m ³	0.0627																											
Justification of choice of data or description of measurement methods and procedures applied	Laboratory analysis																											
Purpose of data	Calculation of baseline and project emissions																											
Comments	Difference between the inflow COD and the outflow COD																											

AMS-I.D, version 18.0

Data / Parameter	EF _{grid,CM,y}
Data unit	tCO ₂ /MWh
Description	Grid Emission Factor for Peninsula Malaysia
Source of data	Ministry of Energy, Science, Technology, Environment and Climate Change
Value applied	0.585 During actual monitoring, any updated GEFs officially published by the Government will be assessed. For each monitoring period, a direct comparison will be made between the latest available official GEF and the 2017 GEF. The lower of the two values will be applied in the ex-post calculation of emission reductions
Justification of choice of data or description of measurement methods and procedures applied	Calculated by official sources according to the “Tool to calculate the emission factor for an electricity system”
Purpose of data	Calculation of baseline emissions
Comments	Year 2017

Data / Parameter	EG _{PJ,facility,y}
Data unit	MWh/year
Description	Quantity of net electricity generation supplied by the project to the grid in year y
Source of data	Measurement
Value applied	7884 MWh
Justification of choice of data or description of measurement methods and procedures applied	Measured continuously by electricity meter
Purpose of data	Calculation of baseline emissions
Comments	-

Tool 05: Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation, version 3.0

Data / Parameter	EC _{PJ,j,y}
Data unit	MWh/yr
Description	Quantity of electricity consumed by the project electricity consumption source j in year y
Source of data	Direct measurement
Description of measurement methods and procedures to be applied	Use electricity meters installed at the electricity consumption sources
Frequency of monitoring/recording	Continuous measurement and at least monthly recording
Value applied	0
Monitoring equipment	Transformer power meter
QA/QC procedures to be applied	In cases where electricity meters are regulated (e.g. the electricity is supplied by the electric grid), the electricity meter will be subject to regular maintenance and testing in accordance with the stipulation of the meter supplier and/or as per the requirements set by the grid operators or national requirements. The calibration of meters, including the frequency of calibration, should be done in accordance with national standards or requirements set by the meter supplier or requirements set by the grid operators. The accuracy class of the meters should be in accordance with the stipulation of the meter supplier and/or as per the requirements set by the grid operators or national requirements. In cases where electricity meters are not regulated (e.g. the electricity is supplied by captive power plants), the electricity meter will be subject to regular maintenance and testing in accordance with the stipulation of the meter supplier or national requirements. The calibration of meters, including the frequency of calibration, should be done in accordance with national standards or requirements set by the meter supplier. The accuracy class of the meters should be in accordance with the stipulation of the meter supplier or national requirements. If these standards are not available, and meter supplier does not specify, calibrate the meters every 3 years and use the meters with at least 0.5

	accuracy class (e.g. a meter with 0.2 accuracy class is more accurate and thus it is accepted).
Purpose of data	Calculation of project emissions.
Calculation method	-
Comments	-

5.2 Data and Parameters Monitored

Grouped Project

AMS-III.H, version 19.0.

Data / Parameter	$Q_{ww,i,y}$
Data unit	m ³ /month
Description	Volume of wastewater treated in the project wastewater system in year y
Source of data	On-site measurement
Description of measurement methods and procedures to be applied	Measured continuously using flow meter
Frequency of monitoring/recording	Monitored continuously (at least hourly measurements are undertaken, if less, confidence/precision level of 90/10 shall be attained). Parameter aggregated monthly and annually for calculations.
Value applied	Value to be monitored ex-post
Monitoring equipment	Flow meter
QA/QC procedures to be applied	Flow meters will undergo maintenance/calibration subject to appropriate industry standards. If local/national standards or the manufacturer's specifications are not available, international standards may be used. In the case of temporary situation where the flow meter malfunctions leading to no readings captured, the historical record at least 10 days before or after the affected period will be used to calculate the lower bound of 95% confidence interval. The lower bound of the interval boundaries calculated will be applied to the affected period as a conservative approach
Purpose of data	Calculation of baseline and project emissions

Calculation method	-
Comments	-

Data / Parameter	COD _{ww,untreated,y} or COD _{inflow,y}
Data unit	tCOD/m ³
Description	Chemical oxygen demand of the wastewater entering the biogas treatment system in year y (tonnes/m ³)
Source of data	On-site measurement
Description of measurement methods and procedures to be applied	Measured the COD according to national or international standards. COD is measured through international sampling
Frequency of monitoring/recording	Periodically, at least monthly
Value applied	Value to be monitored ex-post
Monitoring equipment	Laboratory analysis
QA/QC procedures to be applied	Sampling and analysis will be carried out adhering to internationally recognized procedures. Samples and measurements shall ensure a 90/10 confidence level. COD samples will be tested by external laboratories at least quarterly for cross checking purposes.
Purpose of data	Calculation of baseline and project emissions
Calculation method	
Comments	-

Data / Parameter	COD _{ww,treated,y} or COD _{outflow,y}
Data unit	tCOD/m ³
Description	Chemical Oxygen Demand of the wastewater at the outlet (outflow) of the anaerobic digester (tonnes/m ³)

Source of data	On-site measurement
Description of measurement methods and procedures to be applied	Measured the COD according to national or international standards. COD is measured through international sampling
Frequency of monitoring/recording	Periodically, at least monthly
Value applied	Value to be monitored ex-post
Monitoring equipment	Laboratory analysis
QA/QC procedures to be applied	Sampling and analysis will be carried out adhering to internationally recognized procedures. Samples and measurements shall ensure a 90/10 confidence level. COD samples will be tested by external laboratories at least quarterly for cross checking purposes.
Purpose of data	Calculation of baseline and project emissions
Calculation method	-
Comments	-

Data / Parameter	COD _{ww,discharged,y}
Data unit	tCOD/m ³
Description	Chemical oxygen demand of the treated wastewater discharged into sea, river or lake (tonnes/m ³)
Source of data	On-site measurement
Description of measurement methods and procedures to be applied	Measured the COD according to national or international standards. COD is measured through international sampling
Frequency of monitoring/recording	Periodically, at least monthly
Value applied	Value to be monitored ex-post
Monitoring equipment	Laboratory analysis

QA/QC procedures to be applied	Sampling and analysis will be carried out adhering to internationally recognized procedures. Samples and measurements shall ensure a 90/10 confidence level. COD samples will be tested by external laboratories at least quarterly for cross checking purposes.
Purpose of data	Calculation of baseline and project emissions
Calculation method	-
Comments	-

Data / Parameter	$S_{final,PJ,y}$
Data unit	t
Description	Method of disposal of the sludge
Source of data	Record of site and date of sludge disposal
Description of measurement methods and procedures to be applied	If the methane emissions from anaerobic decay of the final sludge are to be neglected because the sludge is controlled combusted, disposed of in a landfill with methane recovery, or used for soil application, then the end-use of the final sludge will be monitored during the crediting period. If the baseline emissions include the anaerobic decay of final sludge generated by the baseline treatment systems in a landfill without methane recovery, the baseline disposal site shall be clearly defined, and verified by the DOE.
Frequency of monitoring/recording	At each incidence of sludge disposal.
Value applied	-
Monitoring equipment	-
QA/QC procedures to be applied	Plant manager's signature is required on the record.
Purpose of data	Monitoring of sludge disposal.
Calculation method	-
Comments	-

Data / Parameter	$BG_{burnt,y}$
Data unit	m^3
Description	Biogas volume in year y
Source of data	Onsite measurement for $BG_{fuelled,y}$ and $BG_{flared,y}$
Description of measurement methods and procedures to be applied	In all cases, the amount of biogas recovered, fuelled, flared or otherwise utilized (e.g. injected into a natural gas distribution grid or distributed via a dedicated piped network) shall be monitored ex post, using continuous flow meters. If the biogas streams flared and fuelled (or utilized) are monitored separately, the two fractions can be added together to determine the total biogas recovered, without the need to monitor the recovered biogas before the separation. The methane content measurement shall be carried out close to a location in the system where a biogas flow measurement takes place.
Frequency of monitoring/recording	Monitored continuously (at least hourly measurements are undertaken, if less, confidence/precision level of 90/10 shall be attained)
Value applied	Value is not used for ex ante estimates. Value to be monitored ex-post at $BG_{fuelled,y}$ and $BG_{flared,y}$
Monitoring equipment	Gas flow meter(s)
QA/QC procedures to be applied	Flowmeter to be maintained according to manufacturer's specifications. In the case of temporary situation where the flow meter malfunctions leading to no readings captured, the historical record at least 10 days before or after the affected period will be used to calculate the lower bound of 95% confidence interval. The lower bound of the interval boundaries calculated will be applied to the affected period as a conservative approach
Purpose of data	Calculation of project emissions
Calculation method	Add the values for $BG_{flared,y}$ and $BG_{fuelled,y}$ (if applicable)
Comments	-

Data / Parameter	BG _{fuelled,y}
Data unit	m ³
Description	Amount of biogas fuelled in year y
Source of data	Gas flowmeter
Description of measurement methods and procedures to be applied	In all cases, the amount of biogas fuelled shall be monitored ex post, using continuous flow meters. The methane content measurement shall be carried out close to a location in the system where a biogas flow measurement takes place
Frequency of monitoring/recording	Monitored continuously (at least hourly measurements are undertaken, if less, confidence/precision level of 90/10 shall be attained)
Value applied	Value to be monitored ex-post
Monitoring equipment	Flowmeter
QA/QC procedures to be applied	Flowmeter to be maintained according to manufacturer's specifications. In the case of temporary situation where the flow meter malfunctions leading to no readings captured, the historical record at least 10 days before or after the affected period will be used to calculate the lower bound of 95% confidence interval. The lower bound of the interval boundaries calculated will be applied to the affected period as a conservative approach
Purpose of data	Calculation of project emissions
Calculation method	-
Comments	-

Data / Parameter	BG _{flared,y}
Data unit	m ³
Description	Amount of biogas flared in year y
Source of data	Gas flowmeter
Description of measurement methods and procedures to be applied	In all cases, the amount of biogas flared shall be monitored ex post, using continuous flow meters. The methane content measurement shall be carried out close to a location in the system where a biogas flow measurement takes place

Frequency of monitoring/recording	Monitored continuously (at least hourly measurements are undertaken, if less, confidence/precision level of 90/10 shall be attained)
Value applied	Value to be monitored ex-post
Monitoring equipment	Flowmeter
QA/QC procedures to be applied	Flowmeter to be maintained according to manufacturer's specifications. In the case of temporary situation where the flow meter malfunctions leading to no readings captured, the historical record at least 10 days before or after the affected period will be used to calculate the lower bound of 95% confidence interval. The lower bound of the interval boundaries calculated will be applied to the affected period as a conservative approach
Purpose of data	Calculation of project emissions
Calculation method	-
Comments	-

Data / Parameter	$W_{CH_4,y}$
Data unit	%
Description	Methane content in biogas in the year y
Source of data	On site measurement
Description of measurement methods and procedures to be applied	The fraction of methane in the gas should be measured with a continuous analyser or, alternatively, with periodical measurements at a 90/10 confidence/precision level. It shall be measured using equipment that can directly measure methane content in the biogas - the estimation of methane content of biogas based on measurement of other constituents of biogas such as CO₂ is not permitted. The methane content measurement shall be carried out close to a location in the system where a biogas flow measurement takes place
Frequency of monitoring/recording	Monitored continuously (at least hourly measurements are undertaken, if less, confidence/precision level of 90/10 shall be attained)
Value applied	Value to be monitored ex-post
Monitoring equipment	Methane analyser

QA/QC procedures to be applied	Methane analyser to be maintained according to manufacturer's specifications. In the case of temporary situation where the methane analyser malfunctions leading to no readings captured, below approaches will be applied in the calculation: 1. To measure manually with portable gas analyser. The values measured by using the portable analyser will be recalculated based on 95% confidence interval to obtain the lower bound of the value. The lower bound of the 95% Confidence Interval will be applied as conservative approach. 2. To use the historical record at least 10 days before or after to calculate the lower bound of 95% confidence interval. The lower bound of the interval boundaries calculated will be applied to the affected period as a conservative approach
Purpose of data	Calculation of project emissions
Calculation method	-
Comments	-

Data / Parameter	T
Data unit	°C
Description	Temperature of the biogas
Source of data	On site measurement
Description of measurement methods and procedures to be applied	The temperature of the gas is required to determine the density of the methane combusted. If the biogas flow meter employed measures flow, pressure and temperature and displays or outputs the normalised flow of biogas, then there is no need for separate monitoring of pressure and temperature of the biogas
Frequency of monitoring/recording	Shall be measured at the same time when methane content in biogas ($w_{CH_4,y}$) is measured
Value applied	Value to be monitored ex-post
Monitoring equipment	Temperature transmitter
QA/QC procedures to be applied	Temperature transmitter to be maintained according to manufacturer's specifications.

	In the case of temporary situation where the temperature transmitter malfunctions leading to no readings captured, the historical record at least 10 days before or after the affected period will be used to calculate the lower bound of 95% confidence interval. The lower bound of the interval boundaries calculated will be applied to the affected period as a conservative approach
Purpose of data	Calculation of project emissions
Calculation method	-
Comments	-

Data / Parameter	P
Data unit	Pa
Description	Pressure of the biogas
Source of data	On site measurement
Description of measurement methods and procedures to be applied	The pressure of the gas is required to determine the density of the methane combusted. If the biogas flow meter employed measures flow, pressure and temperature and displays or outputs the normalised flow of biogas, then there is no need for separate monitoring of pressure and temperature of the biogas
Frequency of monitoring/recording	Shall be measured at the same time when methane content in biogas ($w_{CH_4,y}$) is measured
Value applied	Value to be monitored ex-post
Monitoring equipment	Pressure transmitter
QA/QC procedures to be applied	Pressure transmitter to be maintained according to manufacturer's specifications. In the case of temporary situation where the pressure transmitter malfunctions leading to no readings captured, the historical record at least 10 days before or after the affected period will be used to calculate the lower bound of 95% confidence interval. The lower bound of the interval boundaries calculated will be applied to the affected period as a conservative approach
Purpose of data	Calculation of project emissions
Calculation method	-
Comments	-

Data / Parameter	Flame _m
Data unit	Flame on or Flame off
Description	Flame detection of flare in the minute m
Source of data	On site measurement
Description of measurement methods and procedures to be applied	Measure using a fixed installation optical flame detector
Frequency of monitoring/recording	Once per minute. Detection of flame recorded as a minute that the flame was on, otherwise recorded as a minute that the flame was off
Value applied	Value to be monitored ex-post
Monitoring equipment	Optical flame detector: ultra violet detector or infra-red or both
QA/QC procedures to be applied	The flame detection will be monitored and cross checked with the amount of biogas sent to flare. If there is data for biogas sent to flare means the flame is on. Equipment will be maintained and calibrated in accordance with manufacturer's recommendations.
Purpose of data	Calculation of project emissions
Calculation method	-
Comments	For all PAIs where biogas is fueled, utilized and excess flared.

Data / Parameter	T _{EG,m}
Data unit	°C
Description	Temperature in the exhaust gas of the enclosed flare in minute m
Source of data	Measurements by project participants
Description of measurement methods	Measure the temperature of the exhaust gas stream in the flare by a Type N thermocouple. A temperature above 500 °C indicates

and procedures to be applied	that a significant amount of gases are still being burnt and that the flare is operating.
Frequency of monitoring/recording	Continuously.
Value applied	Value to be monitored ex-post
Monitoring equipment	Temperature transmitter
QA/QC procedures to be applied	Flow meters are to be periodically calibrated according to the manufacturer's recommendation or industry standard.
Purpose of data	Calculation of project emissions
Calculation method	-
Comments	-

Data / Parameter	$EC_{PJ,j,y}$
Data unit	MWh/year
Description	Electricity consumption in the project scenario in the year y
Source of data	On site measurement
Description of measurement methods and procedures to be applied	Measurement by electricity meters on the electricity entering the Biogas Plant (from the Mill and/or directly from the national electricity grid).
Frequency of monitoring/recording	Continuous
Value applied	Value to be monitored ex-post
Monitoring equipment	Electricity meter
QA/QC procedures to be applied	Maintenance and calibration of the meters according to the manufacturer's recommendations, industry standard or on the national grid operator's requirement and standard practice.

	The meter data can be cross-checked against the electricity sales invoices presented to the Biogas Plant from the electricity providers. In the case of temporary situation where the electricity meter malfunctions leading to no readings captured, the power consumption will use the historical data at least 12 months before or after the affected period, The upper bound of the interval boundaries calculated by using 95% confidence interval will be applied to the affected period as a conservative approach
Purpose of data	Calculation of project emissions
Calculation method	-
Comments	-

AMS-I.D, version 18.0

Data / Parameter	EG _{PJ,facility,y}
Data unit	MWh/year
Description	Quantity of net electricity generation supplied by the project to the grid in year y
Source of data	On site measurement
Description of measurement methods and procedures to be applied	Measured continuously by electricity meter
Frequency of monitoring/recording	Continuous, at least monthly recording
Value applied	Value to be monitored ex-post
Monitoring equipment	Electricity meter
QA/QC procedures to be applied	Maintenance and calibration of the meters according to the manufacturer's recommendations, industry standard or on the national grid operator's requirement and standard practice.

	The meter data can be cross-checked against the electricity sales invoices presented to the Biogas Plant from the electricity providers. In the case of temporary situation where the electricity meter malfunctions leading to no readings captured, the power consumption will use the historical data at least 12 months before or after the affected period, The upper bound of the interval boundaries calculated by using 95% confidence interval will be applied to the affected period as a conservative approach
Purpose of data	Calculation of baseline emissions
Calculation method	-
Comments	-

Data / Parameter	EG _{PJ,add,y}
Data unit	MWh
Description	Quantity of net electricity generation supplied to the grid in year y by the project plant/unit that has been added under the project activity
Source of data	On site measurement
Description of measurement methods and procedures to be applied	Measured continuously by electricity meter
Frequency of monitoring/recording	Continuous, at least monthly recording
Value applied	Value to be monitored ex-post
Monitoring equipment	Electricity meter
QA/QC procedures to be applied	Maintenance and calibration of the meters according to the manufacturer's recommendations, industry standard or on the national grid operator's requirement and standard practice. The meter data can be cross-checked against the electricity sales invoices presented to the Biogas Plant from the electricity providers.

	In the case of temporary situation where the electricity meter malfunctions leading to no readings captured, the power consumption will use the historical data at least 12 months before or after the affected period, The upper bound of the interval boundaries calculated by using 95% confidence interval will be applied to the affected period as a conservative approach
Purpose of data	Calculation of baseline emissions
Calculation method	-
Comments	-

Tool 06: Project emissions from flaring, version 04.0

Data / Parameter	$F_{CH_4,EG,t}$
Data unit	kg
Description	Mass flow of methane in the exhaust gas of the flare on a dry basis at reference conditions in the time period t
Source of data	Measurements undertaken by a third party accredited entity
Description of measurement methods and procedures to be applied	Measure the mass flow of methane in the exhaust gas according to an appropriate national or international standard e.g. UKs Technical Guidance LFTGN05. The time period t over which the mass flow is measured must be at least one hour. The average flow rate to the flare during the time period t must be greater than the average flow rate observed for the previous six months
Frequency of monitoring/recording	Biannual
Value applied	Value to be monitored ex-post
Monitoring equipment	Flow meter
QA/QC procedures to be applied	According to the standard applied
Purpose of data	Calculation of project emissions.

Calculation method	
Comments	Monitoring of this parameter is required in the case of enclosed flares and if the project participants decide to determine flare efficiency through biannual measurement

Data / Parameter	$T_{EG,m}$
Data unit	°C
Description	Temperature in the exhaust gas of the enclosed flare in the minute <i>m</i>
Source of data	Project participants
Description of measurement methods and procedures to be applied	Measure the temperature of the exhaust gas in the flare by an appropriate temperature measurement equipment, Measurements outside the operational temperature specified by the manufacturer may indicate that the flare is not functioning correctly and may require maintenance. Flare manufacturers must provide suitable monitoring ports for the monitoring of the temperature of the flare. These would normally be expected to be in the middle third of the flare. Where more than one temperature port is fitted to the flare, the flare manufacturer must provide written instructions detailing the conditions under which each location shall be used and the port most suitable for monitoring the operation of the flare according to manufacturer's operating specifications for temperature.
Frequency of monitoring/recording	Once per minute
Value applied	Temperature measurement equipment should be replaced or calibrated in accordance with their maintenance schedule
Monitoring equipment	Gas analyser
QA/QC procedures to be applied	Temperature measurement equipment should be replaced or calibrated in accordance with their maintenance schedule.
Purpose of data	Calculation of project emissions.
Calculation method	-
Comments	Unexpected changes such as a sudden increase/drop in temperature can occur for different reasons. These events should

	be noted in the site records along with any corrective action that was implemented to correct the issue. Monitoring of this parameter is applicable in case of enclosed flares. Measurements are required to determine if manufacturer's flare operating specifications for operating temperature are met.
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Data / Parameter	$V_{i,RG,m}$
Data unit	-
Description	Volumetric fraction of component i in the residual gas on a dry basis in the minute m where $i = CH_4, CO, CO_2, O_2, H_2, H_2S, NH_4, N_2$
Source of data	Measurements by project participants using a continuous gas analyser (values are recorded with the same frequency as the flow).
Description of measurement methods and procedures to be applied	Measurement may be made on either dry or wet basis. If the value is made on a wet basis, then it shall be converted to dry basis for reporting
Frequency of monitoring/recording	Continuously. Values are to be averaged on a minute basis
Value applied	Value to be monitored ex-post
Monitoring equipment	Gas analyser
QA/QC procedures to be applied	Analysers must be periodically calibrated according to the manufacturer's recommendation. A zero check and a typical value check should be performed by comparison with a standard certified gas. In the case of temporary situation where the gas analyser malfunctions leading to no readings captured, below approaches will be applied in the calculation: 1. To measure manually with portable gas analyser. The values measured by using the portable analyser will be recalculated based on 95% confidence interval to obtain the lower bound of the value. The lower bound of the 95% Confidence Interval will be applied as conservative approach. 2. To use the historical record at least 10 days before or after to calculate the lower bound of 95% confidence interval. The lower bound of the interval boundaries

	calculated will be applied to the affected period as a conservative approach
Purpose of data	Calculation of project emissions.
Calculation method	
Comments	As a simplified approach, project participants may only measure the content CH₄, CO and CO₂ of the residual gas and consider the remaining part as N₂. Monitoring of this parameter is only applicable in case of enclosed flares and continuous monitoring of the flare efficiency. The methane content measurement shall be carried out close to a location in the system where a biogas flow measurement takes place.

Data / Parameter	V _{RG,m}
Data unit	m ³
Description	Volumetric flow of the residual gas on a dry basis at reference conditions in the minute m
Source of data	Measurements by project participants using a flow meter
Description of measurement methods and procedures to be applied	Instruments with recordable electronic signal (analogical or digital)
Frequency of monitoring/recording	Continuously. Values to be averaged on a minute basis
Value applied	Value to be monitored ex-post
Monitoring equipment	Flow meter
QA/QC procedures to be applied	Flow meters are to be periodically calibrated according to the manufacturer's recommendation In the case of temporary situation where the flow meter malfunctions leading to no readings captured, the historical record at least 10 days before or after the affected period will be used to calculate the lower bound of 95% confidence interval. The lower bound of the interval boundaries calculated will be applied to the affected period as a conservative approach

Purpose of data	Calculation of project emissions.
Calculation method	
Comments	Monitoring of this parameter is applicable in case of enclosed flares and continuous monitoring of the flare efficiency and if project participant selects to calculate VRG_m instead of monitoring directly. Monitoring of this parameter may also be necessary for confirming that the manufacturer's operating specifications for flow rate/heat flux are met. In this case the flow rate should be measured in a m^3/h basis

Data / Parameter	$M_{RG,m}$
Data unit	kg
Description	Mass flow of the residual gas on a dry basis at reference conditions in the minute m
Source of data	-
Description of measurement methods and procedures to be applied	Instruments with recordable electronic signal (analogical or digital)
Frequency of monitoring/recording	Continuous. Values to be averaged on a minute basis
Value applied	Value to be monitored ex-post
Monitoring equipment	Flow meter
QA/QC procedures to be applied	Periodic calibration against a primary device provided by an independent accredited laboratory is mandatory. Calibration and frequency of calibration is according to manufacturer's specifications. In the case of temporary situation where the flow meter malfunctions leading to no readings captured, the historical record at least 10 days before or after the affected period will be used to calculate the lower bound of 95% confidence interval. The lower bound of the interval boundaries calculated will be applied to the affected period as a conservative approach

Purpose of data	Calculation of project emissions.
Calculation method	
Comments	Monitoring of this parameter is applicable in case of enclosed flares and continuous monitoring of the flare efficiency and if project participant selects to monitor $M_{RG,m}$ directly, instead of calculating. Monitoring of this parameter may also be necessary for confirming that the manufacturer's specifications for flow rate/heat flux are met. In this case the flow rate should be measured in a kg/h basis

Data / Parameter	$V_{O_2,EG,m}$
Data unit	-
Description	Volumetric fraction of O_2 in the exhaust gas on a dry basis at reference conditions in the minute m
Source of data	Measurements by project participants using a continuous gas analyser
Description of measurement methods and procedures to be applied	Extractive sampling analysers with water and particulates removal devices or in situ analysers for wet basis determination. The point of measurement (sampling point) shall be in the upper section of the flare (80% of total flare height). Sampling shall be conducted with appropriate sampling probes adequate to high temperatures level (e.g. inconel probes)
Frequency of monitoring/recording	Continuously. Values to be averaged on a minute basis
Value applied	Value to be monitored ex-post
Monitoring equipment	Gas analyser
QA/QC procedures to be applied	Analysers must be periodically calibrated according to the manufacturer's recommendation. A zero check and a typical value check should be performed by comparison with a standard gas. In the case of temporary situation where the gas analyser malfunctions leading to no readings captured, below approaches will be applied in the calculation: 1. To measure manually with portable gas analyser. The values measured by using the portable analyser will be recalculated based on 95% confidence interval to obtain

	the lower bound of the value. The lower bound of the 95% Confidence Interval will be applied as conservative approach. 2. To use the historical record at least 10 days before or after to calculate the lower bound of 95% confidence interval. The lower bound of the interval boundaries calculated will be applied to the affected period as a conservative approach
Purpose of data	Calculation of project emissions.
Calculation method	
Comments	Monitoring of this parameter is only applicable in case of enclosed flares and continuous monitoring of the flare efficiency

Data / Parameter	$f_{CH_4,EG,m}$
Data unit	mg/m ³
Description	Concentration of methane in the exhaust gas of the flare on a dry basis at reference conditions in the minute m
Source of data	Measurements by project participants using a continuous gas analyser
Description of measurement methods and procedures to be applied	Extractive sampling analysers with water and particulates removal devices or in situ analyser for wet basis determination. The point of measurement (sampling point) shall be in the upper section of the flare in order that the sampling is of the gas after consumption has taken place (80% of total flare height). Sampling shall be conducted with appropriate sampling probes adequate to high temperatures level (e.g. inconel probes)
Frequency of monitoring/recording	Continuously. Values to be averaged on a minute basis
Value applied	Value to be monitored ex-post
Monitoring equipment	Gas analyser
QA/QC procedures to be applied	Analysers must be periodically calibrated according to the manufacturer's recommendation. A zero check and a typical value check should be performed by comparison with a standard gas.

	In the case of temporary situation where the gas analyser malfunctions leading to no readings captured, below approaches will be applied in the calculation: 1. To measure manually with portable gas analyser. The values measured by using the portable analyser will be recalculated based on 95% confidence interval to obtain the lower bound of the value. The lower bound of the 95% Confidence Interval will be applied as conservative approach. 2. To use the historical record at least 10 days before or after to calculate the lower bound of 95% confidence interval. The lower bound of the interval boundaries calculated will be applied to the affected period as a conservative approach
Purpose of data	Calculation of project emissions.
Calculation method	
Comments	Monitoring of this parameter is only applicable in case of enclosed flares and continuous monitoring of the flare efficiency. Measurement instruments may read ppmv or % values. To convert from ppmv to mg/m ³ simply multiply by 0.716. 1% equals 10 000 ppmv

Data / Parameter	Flame _m
Data unit	Flame on or Flame off
Description	Flame detection of flare in the minute m
Source of data	Project participants
Description of measurement methods and procedures to be applied	Measure using a fixed installation optical flame detector: Ultra Violet detector or Infra-Red or both
Frequency of monitoring/recording	Once per minute. Detectionn of flame recorded das a minute that the flame was on, otherwise recoded as a minute that the flame was off.
Value applied	Value to be monitored ex-post
Monitoring equipment	Gas analyser

QA/QC procedures to be applied	Equipment shall be maintained and calibrated in accordance with manufacturer's recommendations.
Purpose of data	Calculation of project emissions.
Calculation method	-
Comments	Applicable to all flares

Data / Parameter	Maintenance _y
Data unit	Calendar dates
Description	Maintenance events completed in year y
Source of data	Specific PAI
Description of measurement methods and procedures to be applied	Record the date that maintenance events were completed in year y. Records of maintenance logs must include all aspects of the maintenance including the details of the person(s) undertaking the work, parts replaced, or needing to be replaced, source of replacement parts, serial numbers and calibration certificates
Frequency of monitoring/recording	Annual
Value applied	Value to be monitored ex-post
Monitoring equipment	Gas analyser
QA/QC procedures to be applied	Records must be kept in a maintenance log for two years beyond the life of the flare
Purpose of data	Calculation of project emissions.
Calculation method	
Comments	Monitoring of this parameter is required for the case of enclosed flares and the project participant selects Option B to determine flare efficiency. These dates are required so that they can be compared to the maintenance schedule to check that maintenance events were completed within the minimum time between maintenance events specified by the manufacturer ($SPEC_{flare}$)

Tool 05: Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation, version 3.0

Data / Parameter	$EF_{grid,CM,y}$
Data unit	tCO ₂ /MWh
Description	Combined margin emission factor for the grid in year y
Source of data	Calculate the combined margin emission factor, using the procedures in the latest approved version of the “Tool to calculate the emission factor for an electricity system”
Description of measurement methods and procedures to be applied	As per the “Tool to calculate the emission factor for an electricity system”
Frequency of monitoring/recording	As per the “Tool to calculate the emission factor for an electricity system”
Value applied	
Monitoring equipment	-
QA/QC procedures to be applied	As per the “Tool to calculate the emission factor for an electricity system”
Purpose of data	Calculation of project emissions.
Calculation method	-
Comments	Option A1 is chosen. During actual monitoring, any updated GEFs officially published by the Government will be assessed. For each monitoring period, a direct comparison will be made between the latest available official GEF and the 2017 GEF. The lower of the two values will be applied in the ex-post calculation of emission reductions,

Data / Parameter	$TDL_{j,y}$ and $TDL_{k,y}$ and $TDL_{l,y}$
Data unit	-
Description	Average technical transmission and distribution losses for providing electricity to source j, k or l in year y

Source of data	1. Use annual average value based on the most recent data available within the host country; 2. Use as default values of 20% for: (a) project or leakage electricity consumption sources; (b) baseline electricity consumption sources if the electricity consumption by all project and leakage electricity consumption sources to which scenario A or scenario C (cases C.I or C.III) applies is larger than the electricity consumption of all baseline electricity consumption sources to which scenario A or scenario C (cases C.I or C.III) applies; 3. Use as default values of 3% for: (a) baseline electricity consumption sources; (b) project and leakage electricity consumption sources if the electricity consumption by all project and leakage electricity consumption sources to which scenario A or scenario C (cases C.I or C.III) applies is smaller than the electricity consumption of all baseline electricity consumption sources to which scenario A or scenario C (cases C.I or C.III) applies
Description of measurement methods and procedures to be applied	The distribution losses can either be calculated by the project participants or be based on references from utilities, network operators or other official documentation
Frequency of monitoring/recording	Annually. In the absence of data from the relevant year, most recent figures should be used, but not older than 5 years
Value applied	Value to be monitored ex-post
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Calculation of project emissions.
Calculation method	-
Comments	-

Data / Parameter	$FC_{n,i,t}$
Data unit	Mass or () volume unit per year (in m ³ , ton or l)
Description	Quantity of fossil fuel type i fired in the captive power plant n in the time period t
Source of data	Annual data during the crediting period: Onsite measurements Historical data: Historical records / onsite measurements
Description of measurement methods and procedures to be applied	1. In cases where fuel is supplied from small daily tanks, rulers can be used to determine mass or volume of the fuel consumed, with the following conditions: The ruler gauge must be part of the daily tank and calibrated at least once a year and have a book of control for recording the measurements (on a daily basis or per shift); 2. Accessories such as transducers, sonar and piezoelectronic devices are accepted if they are properly calibrated with the ruler gauge and receiving a maintenance per supplier specifications; 3. In case of daily tanks with pre-heaters for heavy oil, the calibration will be made with the system at typical operational conditions
Frequency of monitoring/recording	Continuously, aggregated at least annually
Value applied	Value to be monitored ex-post
Monitoring equipment	Mass or Volumetric meter
QA/QC procedures to be applied	The consistency of metered fuel consumption quantities should be cross-checked with an annual energy balance that is based on purchased quantities and stock changes.
Purpose of data	Calculation of project emissions.
Calculation method	-
Comments	-

Data / Parameter	$EG_{n,t}$
Data unit	MWh
Description	Quantity of electricity generated in captive power plant n in the time period t
Source of data	Onsite measurements
Description of measurement methods and procedures to be applied	Use electricity meters
Frequency of monitoring/recording	Continuously, aggregated at least annually
Value applied	Value to be monitored ex-post
Monitoring equipment	Electricity Meter
QA/QC procedures to be applied	Cross check measurement results with records for sold electricity where relevant
Purpose of data	Calculation of baseline emissions.
Calculation method	-
Comments	-

Data / Parameter	$NCV_{i,t}$
Data unit	GJ / mass or volume unit
Description	Average net calorific value of fossil fuel type i used in the period t
Source of data	(a) To be provided by fuel supplier (b) Measurements by the project proponents (c) Regional or national default values (d) IPCC default values at the upper or lower limit – whatever is more conservative – of the uncertainty at a 95% confidence

	interval as provided in Table 1.2 of Chapter 1 of Vol. 2 (Energy) of the 2006 IPCC Guidelines on National GHG Inventories
Description of measurement methods and procedures to be applied	Measurements taken will be in line with national or international fuel standards
Frequency of monitoring/recording	(a & b) The NCV should be obtained for each fuel delivery, from which weighted average values for the period t should be calculated (c) Review appropriateness of the values annually (d) Any future revision of the IPCC Guidelines should be taken into account
Value applied	Value to be monitored ex-post
Monitoring equipment	-
QA/QC procedures to be applied	Verify if the values are within the uncertainty range of the IPCC default values as provided in Table 1.2, Vol. 2 of the 2006 IPCC Guidelines. If the values fall out this range collect additional information from the testing laboratory to justify the outcome or conduct additional measurements. The laboratories should have ISO17025 accreditation or justify that they can comply with similar quality standards
Purpose of data	Calculation of project emissions.
Calculation method	-
Comments	

Data / Parameter	EF _{CO2,i,t}
Data unit	t CO ₂ / GJ
Description	CO ₂ emission factor of fossil fuel type i used in the period t
Source of data	(a) To be provided by fuel supplier (b) Measurements by the project proponents (c) Regional or national default values

	(d) IPCC default values at the upper or lower limit – whatever is more conservative – of the uncertainty at a 95% confidence interval as provided in Table 1.4 of Chapter 1 of Vol. 2 (Energy) of the 2006 IPCC Guidelines on National GHG Inventories
Description of measurement methods and procedures to be applied	Measurements taken will be in line with national or international fuel standards
Frequency of monitoring/recording	(a & b) The CO ₂ emission factor should be obtained for each fuel delivery, from which weighted average values for the period t should be calculated (c) Review appropriateness of the values annually (d) Any future revision of the IPCC Guidelines should be taken into account
Value applied	Value to be monitored ex-post
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Calculation of project emissions.
Calculation method	-
Comments	

Data / Parameter	EC _{PJ,j,y}
Data unit	MWh/yr
Description	Quantity of electricity consumed by the project electricity consumption source j in year y Net increase in electricity consumption of source l in year y as a result of leakage
Source of data	Direct measurement or calculated based on measurements from more than one electricity meters

Description of measurement methods and procedures to be applied	Use electricity meters installed at the electricity consumption sources
Frequency of monitoring/recording	Continuous measurement and at least monthly recording
Value applied	Value to be monitored ex-post
Monitoring equipment	Transformer power meter
QA/QC procedures to be applied	In cases where electricity meters are regulated (e.g. the electricity is supplied by the electric grid), the electricity meter will be subject to regular maintenance and testing in accordance with the stipulation of the meter supplier and/or as per the requirements set by the grid operators or national requirements. The calibration of meters, including the frequency of calibration, should be done in accordance with national standards or requirements set by the meter supplier or requirements set by the grid operators. The accuracy class of the meters should be in accordance with the stipulation of the meter supplier and/or as per the requirements set by the grid operators or national requirements. In cases where electricity meters are not regulated (e.g. the electricity is supplied by captive power plants), the electricity meter will be subject to regular maintenance and testing in accordance with the stipulation of the meter supplier or national requirements. The calibration of meters, including the frequency of calibration, should be done in accordance with national standards or requirements set by the meter supplier. The accuracy class of the meters should be in accordance with the stipulation of the meter supplier or national requirements. If these standards are not available, and meter supplier does not specify, calibrate the meters every 3 years and use the meters with at least 0.5 accuracy class (e.g. a meter with 0.2 accuracy class is more accurate and thus it is accepted).
Purpose of data	Calculation of project emissions.
Calculation method	-
Comments	-

Data / Parameter	$EC_{BL,k,y}$
Data unit	MWh/yr
Description	Quantity of electricity that would be consumed by the baseline electricity consumption source k in year y
Source of data	Direct measurement or calculated based on measurements from more than one electricity meters
Description of measurement methods and procedures to be applied	Use electricity meters installed at the electricity consumption sources
Frequency of monitoring/recording	Continuous measurement and at least monthly recording
Value applied	Value to be monitored ex-post
Monitoring equipment	Transformer power meter
QA/QC procedures to be applied	In cases where electricity meters are regulated (e.g. the electricity is supplied by the electric grid), the electricity meter will be subject to regular maintenance and testing in accordance with the stipulation of the meter supplier and/or as per the requirements set by the grid operators or national requirements. The calibration of meters, including the frequency of calibration, should be done in accordance with national standards or requirements set by the meter supplier or requirements set by the grid operators. The accuracy class of the meters should be in accordance with the stipulation of the meter supplier and/or as per the requirements set by the grid operators or national requirements. In cases where electricity meters are not regulated (e.g. the electricity is supplied by captive power plants), the electricity meter will be subject to regular maintenance and testing in accordance with the stipulation of the meter supplier or national requirements. The calibration of meters, including the frequency of calibration, should be done in accordance with national standards or requirements set by the meter supplier. The accuracy class of the meters should be in accordance with the stipulation of the meter supplier or national requirements. If these standards are not available, and meter supplier does not specify, calibrate the meters every 3 years and use the meters with at least 0.5 accuracy class (e.g. a meter with 0.2 accuracy class is more accurate and thus it is accepted).

Purpose of data	Calculation of baseline emissions.
Calculation method	-
Comments	-

Data / Parameter	EG _{PJ,grid,y} or EG _{PJ,facility,I,y}
Data unit	MWh/yr
Description	Quantity of electricity generated and supplied by the project power plant to the grid in year y Quantity of electricity generated and supplied by the project power plant to the consumers/electricity consuming facility i in year y
Source of data	Direct measurement or calculated based on measurements from more than one electricity meters
Description of measurement methods and procedures to be applied	Use electricity meters installed at the grid interface for electricity export to grid and for supply to captive consumers use electricity meters installed at the entrance of the electricity consuming facility. In case of grid and net electricity generation: This parameter should be either monitored using bi-directional energy meter or calculated as difference between (a) the quantity of electricity supplied by the project plant/unit to the grid; and (b) the quantity of electricity the project plant/unit from the grid. If it is calculated, then the following parameters shall be measured: (a) The quantity of electricity supplied by the project plant/unit to the grid; and (b) The quantity of electricity delivered to the project plant/unit from the grid
Frequency of monitoring/recording	Continuous measurement and at least monthly recording
Value applied	Value to be monitored ex-post
Monitoring equipment	Electricity meter
QA/QC procedures to be applied	In cases where electricity meters are regulated (e.g. the electricity is supplied by the electric grid), the electricity meter will be subject to regular maintenance and testing in accordance with the

	stipulation of the meter supplier and/or as per the requirements set by the grid operators or national requirements. The calibration of meters, including the frequency of calibration, should be done in accordance with national standards or requirements set by the meter supplier or requirements set by the grid operators. The accuracy class of the meters should be in accordance with the stipulation of the meter supplier and/or as per the requirements set by the grid operators or national requirements. In cases where electricity meters are not regulated (e.g. the electricity is supplied by captive power plants), the electricity meter will be subject to regular maintenance and testing in accordance with the stipulation of the meter supplier or national requirements. The calibration of meters, including the frequency of calibration, should be done in accordance with national standards or requirements set by the meter supplier. The accuracy class of the meters should be in accordance with the stipulation of the meter supplier or national requirements. If these standards are not available, and meter supplier does not specify, calibrate the meters every 3 years and use the meters with at least 0.5 accuracy class (e.g. a meter with 0.2 accuracy class is more accurate and thus it is accepted).
Purpose of data	Calculation of baseline emissions.
Calculation method	-
Comments	-

Tool 03: Tool to calculate project or leakage CO₂ emissions from fossil fuel combustion, version 3.0

Data / Parameter	FC _{i,j,y}
Data unit	Mass or volume unit per year (e.g. ton/yr or m ³ /yr)
Description	Quantity of fuel type i combusted in process j during the year y
Source of data	Onsite measurements
Description of measurement methods and procedures to be applied	(a) In cases where fuel is supplied from small daily tanks, rulers can be used to determine mass or volume of the fuel consumed, with the following conditions: The ruler gauge must be part of the daily tank and calibrated at least once a year and have a book of control for recording the measurements (on a daily basis or per shift);

	(b) Accessories such as transducers, sonar and piezoelectronic devices are accepted if they are properly calibrated with the ruler gauge and receiving a reasonable maintenance; (c) In case of daily tanks with pre-heaters for heavy oil, the calibration will be made with the system at typical operational conditions
Frequency of monitoring/recording	Continuous
Value applied	Value to be monitored ex-post
Monitoring equipment	Mass or Volumetric Meter
QA/QC procedures to be applied	The consistency of metered fuel consumption quantities should be cross-checked by an annual energy balance that is based on purchased quantities and stock changes. Where the purchased fuel invoices can be identified specifically for the CDM project, the metered fuel consumption quantities should also be cross-checked with available purchase invoices from the financial records
Purpose of data	Calculation of baseline and project emissions.
Calculation method	
Comments	

Data / Parameter	WC,i,y
Data unit	tC/mass unit of the fuel
Description	Weighted average mass fraction of carbon in fuel type i in year y
Source of data	(a) Values provided by the fuel supplier in invoices (b) Measurements by the project participants
Description of measurement methods and procedures to be applied	Measurements should be undertaken in line with national or international fuel standards

Frequency of monitoring/recording	The mass fraction of carbon should be obtained for each fuel delivery, from which weighted average annual values should be calculated
Value applied	Value to be monitored ex-post
Monitoring equipment	-
QA/QC procedures to be applied	Verify if the values are within the uncertainty range of the product of the IPCC default values as provided in Table 1.2 and Table 1.3, Vol. 2 of the 2006 IPCC Guidelines. If the values fall below this range collect additional information from the testing laboratory to justify the outcome or conduct additional measurements. The laboratories in (b) should have ISO17025 accreditation or justify that they can comply with similar quality standards
Purpose of data	Calculation of baseline and project emissions.
Calculation method	-
Comments	-

Data / Parameter	$\rho_{i,y}$
Data unit	Mass unit/volume unit
Description	Weighted average density of fuel type i in year y
Source of data	(a) Values provided by the fuel supplier in invoices (b) Measurements by the project participants (c) Regional or national default values
Description of measurement methods and procedures to be applied	Measurements should be undertaken in line with national or international fuel standards
Frequency of monitoring/recording	The density of the fuel should be obtained for each fuel delivery, from which weighted average annual values should be calculated
Value applied	Value to be monitored ex-post
Monitoring equipment	-

QA/QC procedures to be applied	-
Purpose of data	Calculation of baseline and project emissions.
Calculation method	-
Comments	-

Data / Parameter	NCV _{i,y}
Data unit	GJ per mass or volume unit (e.g. GJ/m ³ , GJ/ton)
Description	Weighted average net calorific value of fuel type i in year y
Source of data	(a) Values provided by the fuel supplier in invoices (b) Measurements by the project participants (c) Regional or national default values (d) IPCC default values at the upper limit of the uncertainty at a 95% confidence interval as provided in Table 1.2 of Chapter 1 of Vol. 2 (Energy) of the 2006 IPCC Guidelines on National GHG Inventories
Description of measurement methods and procedures to be applied	Measurements should be undertaken in line with national or international fuel standards
Frequency of monitoring/recording	(a & b) The NCV should be obtained for each fuel delivery, from which weighted average annual values should be calculated. (c) Review appropriateness of the values annually (d) Any future revision of the IPCC Guidelines should be taken into account
Value applied	Value to be monitored ex-post
Monitoring equipment	-
QA/QC procedures to be applied	Verify if the values are within the uncertainty range of the IPCC default values as provided in Table 1.2, Vol. 2 of the 2006 IPCC Guidelines. If the values fall below this range collect additional information from the testing laboratory to justify the outcome or conduct additional measurements. The laboratories should have

	ISO17025 accreditation or justify that they can comply with similar quality standards
Purpose of data	Calculation of baseline and project emissions.
Calculation method	-
Comments	-

Data / Parameter	EF _{CO₂, (c), y}
Data unit	tCO ₂ /GJ
Description	Weighted average CO ₂ emission factor of fuel type (c) in year y
Source of data	(a) Values provided by the fuel supplier in invoices (b) Measurements by the project participants (c) Regional or national default values (d) IPCC default values at the upper limit of the uncertainty at a 95% confidence interval as provided in Table 1.4 of Chapter 1 of Vol. 2 (Energy) of the 2006 IPCC Guidelines on National GHG Inventories
Description of measurement methods and procedures to be applied	Measurements should be undertaken in line with national or international fuel standards
Frequency of monitoring/recording	(a & b) The CO₂ emission factor should be obtained for each fuel delivery, from which weighted average annual values should be calculated. (c) Review appropriateness of the values annually (d) Any future revision of the IPCC Guidelines should be taken into account
Value applied	Value to be monitored ex-post
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Calculation of baseline and project emissions.

Calculation method	
Comments	

PAI1

AMS-III.H, version 19.0

Data / Parameter	$Q_{ww,i,y}$
Data unit	m ³ /month
Description	Volume of wastewater treated in the project wastewater system in year y
Source of data	On-site measurement
Description of measurement methods and procedures to be applied	Measured continuously using flow meter
Frequency of monitoring/recording	Monitored continuously (at least hourly measurements are undertaken, if less, confidence/precision level of 90/10 shall be attained). Parameter aggregated monthly and annually for calculations.
Value applied	Value to be monitored ex-post
Monitoring equipment	Flow meter
QA/QC procedures to be applied	Flow meters will undergo maintenance/calibration subject to appropriate industry standards. If local/national standards or the manufacturer's specifications are not available, international standards may be used. In the case of temporary situation where the flow meter malfunctions leading to no readings captured, the historical record at least 10 days before or after the affected period will be used to calculate the lower bound of 95% confidence interval. The lower bound of the interval boundaries calculated will be applied to the affected period as a conservative approach.
Purpose of data	Calculation of baseline and project emissions
Calculation method	-

Comments	-
Data / Parameter	$COD_{ww,untreated,y}$ or $COD_{inflow,y}$
Data unit	tCOD/m ³
Description	Chemical oxygen demand of the wastewater entering the biogas treatment system in year y (tonnes/m ³)
Source of data	On-site measurement
Description of measurement methods and procedures to be applied	Measured the COD according to national or international standards. COD is measured through international sampling
Frequency of monitoring/recording	Periodically, at least monthly
Value applied	Value to be monitored ex-post
Monitoring equipment	Laboratory analysis
QA/QC procedures to be applied	Sampling and analysis will be carried out adhering to internationally recognized procedures. Samples and measurements shall ensure a 90/10 confidence level. COD samples will be tested by external laboratories at least quarterly for cross checking purposes.
Purpose of data	Calculation of baseline and project emissions
Calculation method	-
Comments	-

Data / Parameter	$COD_{ww,treated,y}$ or $COD_{outflow,y}$
Data unit	tCOD/m ³
Description	Chemical oxygen demand of the wastewater after the treatment system affected by the project activity.

Source of data	On site measurement
Description of measurement methods and procedures to be applied	Measure the COD according to national or international standards. COD is measured through representative sampling
Frequency of monitoring/recording	Periodically, at least monthly.
Value applied	Value to be monitored ex-post
Monitoring equipment	Laboratory analysis
QA/QC procedures to be applied	Sampling and analysis will be carried out adhering to internationally recognized procedures. Samples and measurements shall ensure a 90/10 confidence level. COD samples will be tested by external laboratories at least quarterly for cross checking purposes.
Purpose of data	Calculation of baseline and project emissions
Calculation method	-
Comments	-

Data / Parameter	COD _{ww,discharged,y}
Data unit	tCOD/m ³
Description	Chemical oxygen demand of the treated wastewater discharged into sea, river or lake (tonnes/m ³)
Source of data	On site measurement
Description of measurement methods and procedures to be applied	Measure the COD according to national or international standards. COD is measured through representative sampling.
Frequency of monitoring/recording	Periodically, at least monthly.
Value applied	Value to be monitored ex-post

Monitoring equipment	Laboratory analysis
QA/QC procedures to be applied	Sampling and analysis will be carried out adhering to internationally recognized procedures. Samples and measurements shall ensure a 90/10 confidence level. COD samples will be tested by external laboratories at least quarterly for cross checking purposes.
Purpose of data	Calculation of baseline and project emissions
Calculation method	-
Comments	-

Data / Parameter	BG _{burnt,y}
Data unit	m ³
Description	Biogas volume in year y
Source of data	Onsite measurement for BG _{fuelled,y} and BG _{flared,y}
Description of measurement methods and procedures to be applied	In all cases, the amount of biogas recovered, fuelled, flared or otherwise utilized (e.g. injected into a natural gas distribution grid or distributed via a dedicated piped network) shall be monitored ex post, using continuous flow meters. If the biogas streams flared and fuelled (or utilized) are monitored separately, the two fractions can be added together to determine the total biogas recovered, without the need to monitor the recovered biogas before the separation. The methane content measurement shall be carried out close to a location in the system where a biogas flow measurement takes place.
Frequency of monitoring/recording	Monitored continuously (at least hourly measurements are undertaken, if less, confidence/precision level of 90/10 shall be attained)
Value applied	Value is not used for ex ante estimates. Value to be monitored ex-post at BG _{fuelled,y} and BG _{flared,y}
Monitoring equipment	Gas flow meter(s)
QA/QC procedures to be applied	Flowmeter to be maintained according to manufacturer's specifications. In the case of temporary situation where the flow meter malfunctions leading to no readings captured, the historical

	record at least 10 days before or after the affected period will be used to calculate the lower bound of 95% confidence interval. The lower bound of the interval boundaries calculated will be applied to the affected period as a conservative approach
Purpose of data	Calculation of project emissions
Calculation method	Add the values for $BG_{\text{flared},y}$ and $BG_{\text{fuelled},y}$
Comments	-

Data / Parameter	$BG_{\text{fuelled},y}$
Data unit	m ³
Description	Amount of biogas fuelled in year y
Source of data	Gas flowmeter
Description of measurement methods and procedures to be applied	In all cases, the amount of biogas fuelled shall be monitored ex post, using continuous flow meters. The methane content measurement shall be carried out close to a location in the system where a biogas flow measurement takes place
Frequency of monitoring/recording	Monitored continuously (at least hourly measurements are undertaken, if less, confidence/precision level of 90/10 shall be attained)
Value applied	Value to be monitored ex-post
Monitoring equipment	Flowmeter
QA/QC procedures to be applied	Flowmeter to be maintained according to manufacturer's specifications. In the case of temporary situation where the flow meter malfunctions leading to no readings captured, the historical record at least 10 days before or after the affected period will be used to calculate the lower bound of 95% confidence interval. The lower bound of the interval boundaries calculated will be applied to the affected period as a conservative approach
Purpose of data	Calculation of project emissions
Calculation method	-

Comments	-
Data / Parameter	BG _{flared,y}
Data unit	m ³
Description	Amount of biogas flared in year y
Source of data	Gas flowmeter
Description of measurement methods and procedures to be applied	In all cases, the amount of biogas flared shall be monitored ex post, using continuous flow meters. The methane content measurement shall be carried out close to a location in the system where a biogas flow measurement takes place
Frequency of monitoring/recording	Monitored continuously (at least hourly measurements are undertaken, if less, confidence/precision level of 90/10 shall be attained)
Value applied	Value to be monitored ex-post
Monitoring equipment	Flowmeter
QA/QC procedures to be applied	Flowmeter to be maintained according to manufacturer's specifications. In the case of temporary situation where the flow meter malfunctions leading to no readings captured, the historical record at least 10 days before or after the affected period will be used to calculate the lower bound of 95% confidence interval. The lower bound of the interval boundaries calculated will be applied to the affected period as a conservative approach
Purpose of data	Calculation of project emissions
Calculation method	-
Comments	-

Data / Parameter	$W_{CH_4,y}$
Data unit	%
Description	Methane content in biogas in the year y
Source of data	On site measurement
Description of measurement methods and procedures to be applied	The fraction of methane in the gas should be measured with a continuous analyser or, alternatively, with periodical measurements at a 90/10 confidence/precision level. It shall be measured using equipment that can directly measure methane content in the biogas - the estimation of methane content of biogas based on measurement of other constituents of biogas such as CO ₂ is not permitted. The methane content measurement shall be carried out close to a location in the system where a biogas flow measurement takes place
Frequency of monitoring/recording	Monitored continuously (at least hourly measurements are undertaken, if less, confidence/precision level of 90/10 shall be attained)
Value applied	Value to be monitored ex-post
Monitoring equipment	Methane analyser
QA/QC procedures to be applied	Methane analyser to be maintained according to manufacturer's specifications. In the case of temporary situation where the methane analyser malfunctions leading to no readings captured, below approaches will be applied in the calculation: 1. To measure manually with portable gas analyser. The values measured by using the portable analyser will be recalculated based on 95% confidence interval to obtain the lower bound of the value. The lower bound of the 95% Confidence Interval will be applied as conservative approach. 2. To use the historical record at least 10 days before or after to calculate the lower bound of 95% confidence interval. The lower bound of the interval boundaries calculated will be applied to the affected period as a conservative approach
Purpose of data	Calculation of project emissions
Calculation method	-

Comments	-
Data / Parameter	T
Data unit	°C
Description	Temperature of the biogas
Source of data	On site measurement
Description of measurement methods and procedures to be applied	The temperature of the gas is required to determine the density of the methane combusted. If the biogas flow meter employed measures flow, pressure and temperature and displays or outputs the normalised flow of biogas, then there is no need for separate monitoring of pressure and temperature of the biogas
Frequency of monitoring/recording	Shall be measured at the same time when methane content in biogas ($w_{CH_4,y}$) is measured
Value applied	Value to be monitored ex-post
Monitoring equipment	Temperature transmitter
QA/QC procedures to be applied	Temperature transmitter to be maintained according to manufacturer's specifications. In the case of temporary situation where the temperature transmitter malfunctions leading to no readings captured, the historical record at least 10 days before or after the affected period will be used to calculate the lower bound of 95% confidence interval. The lower bound of the interval boundaries calculated will be applied to the affected period as a conservative approach
Purpose of data	Calculation of project emissions
Calculation method	-
Comments	-
Data / Parameter	P
Data unit	Pa

Description	Pressure of the biogas
LI Source of data	On site measurement
Description of measurement methods and procedures to be applied	The pressure of the gas is required to determine the density of the methane combusted. If the biogas flow meter employed measures flow, pressure and temperature and displays or outputs the normalised flow of biogas, then there is no need for separate monitoring of pressure and temperature of the biogas
Frequency of monitoring/recording	Shall be measured at the same time when methane content in biogas ($w_{CH_4,y}$) is measured
Value applied	Value to be monitored ex-post
Monitoring equipment	Pressure transmitter
QA/QC procedures to be applied	Pressure transmitter to be maintained according to manufacturer's specifications. In the case of temporary situation where the pressure transmitter malfunctions leading to no readings captured, the historical record at least 10 days before or after the affected period will be used to calculate the lower bound of 95% confidence interval. The lower bound of the interval boundaries calculated will be applied to the affected period as a conservative approach
Purpose of data	Calculation of project emissions
Calculation method	-
Comments	-

Data / Parameter	Flame _m
Data unit	Flame on or Flame off
Description	Flame detection of flare in the minute m
Source of data	On site measurement
Description of measurement methods and procedures to be applied	Measure using a fixed installation optical flame detector

Frequency of monitoring/recording	Once per minute. Detection of flame recorded as a minute that the flame was on, otherwise recorded as a minute that the flame was off
Value applied	Value to be monitored ex-post
Monitoring equipment	Optical flame detector: ultra violet detector or infra-red or both
QA/QC procedures to be applied	The flame detection will be monitored and cross checked with the amount of biogas sent to flare. If there is data for biogas sent to flare means the flame is on. Equipment will be maintained and calibrated in accordance with manufacturer's recommendations.
Purpose of data	Calculation of project emissions
Calculation method	-
Comments	For all PAIs where biogas is fueled, utilized and excess flared.

Data / Parameter	EC _{PJ,j,y}
Data unit	MWh/year
Description	Electricity consumption in the project scenario in the year y
Source of data	On site measurement
Description of measurement methods and procedures to be applied	Measurement by electricity meters on the electricity entering the Biogas Plant (from the Mill and/or directly from the national electricity grid).
Frequency of monitoring/recording	Continuous, ongoing
Value applied	Value to be monitored ex-post
Monitoring equipment	Electricity meter
QA/QC procedures to be applied	Maintenance and calibration of the meters according to the manufacturer's recommendations, industry standard or on the national grid operator's requirement and standard practice. The meter data can be cross-checked against the electricity sales

	invoices presented to the Biogas Plant from the electricity providers. For PAI1, the electricity will be supplied from mill. The electricity consumption will be recorded from the electricity meter owned by the mil. In the case of temporary situation where the electricity meter malfunctions leading to no readings captured, the power consumption will use the historical data at least 12 months before or after the affected period, The upper bound of the interval boundaries calculated by using 95% confidence interval will be applied to the affected period as a conservative approach
Purpose of data	Calculation of project emissions
Calculation method	-
Comments	-

Data / Parameter	$EG_{PJ, facility, y}$
Data unit	MWh/year
Description	Quantity of net electricity generation supplied by the project to the grid in year y
Source of data	On site measurement
Description of measurement methods and procedures to be applied	Measured continuously by electricity meter
Frequency of monitoring/recording	Continuous,
Value applied	Value to be monitored ex-post
Monitoring equipment	Electricity meter
QA/QC procedures to be applied	Maintenance and calibration of the meters according to the manufacturer's recommendations, industry standard or on the national grid operator's requirement and standard practice. The meter data can be cross-checked against the electricity sales invoices presented to the Biogas Plant from the electricity

	providers. In the case of temporary situation where the electricity meter malfunctions leading to no readings captured, the power consumption will use the historical data at least 12 months before or after the affected period, The upper bound of the interval boundaries calculated by using 95% confidence interval will be applied to the affected period as a conservative approach
Purpose of data	Calculation of baseline emissions
Calculation method	-
Comments	-

5.3 Monitoring Plan

Grouped Project

i) Organizational structure and responsibilities

The project owner organises a specific team in assigning responsibility for data collection, supervision, and witness the whole process of data measuring and recording. The organizational structure, responsibilities and competencies of the personnel that will carry out the monitoring and management of the project activities are as follows:

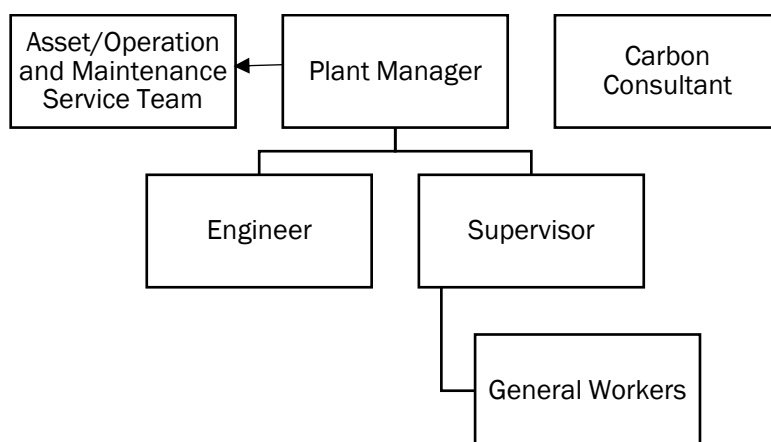


Figure 7: Grouped PAI VCS Organization Chart

The followings are the position and overall responsibility of the operation and monitoring team:

Table 9: Description of roles for PAI VCS team

Personnel	Description of Roles in Monitoring Plan
Asset/Operation and	Overall managing the carbon project

Personnel	Description of Roles in Monitoring Plan
Maintenance Service Team	
Plant Manager	- Report and seek approval from the asset/operation and maintenance service team on VCS- related matters - Take full responsibility in the overall monitoring of the project - Conduct internal meetings on VCS matters - Oversees and coordinate the VCS monitoring plan data gathering - Monitor & ensure that data QA/QC is carried out by appointing (internal) independent verifiers to do cross-checking on irregular basis - Ensure all documentations were prepared accordingly based on VCS-PD requirement. - Coordinate with VCS consultant for matters (data/documents/records etc) that require rectification from plant - Point of contact for the mill, to respond to grievance from external parties.
Engineer	- Assists the VCS Manager in performing the VCS monitoring works - Assisting in the daily and monthly monitoring records for all the VCS related equipment. - Prepare monthly summary record of the operations of plants. - Ensure maintenance and calibration is carried out for proper running of machinery and equipment
Supervisors	- Report to the Engineer on any VCS monitoring issues - Conduct daily checking of data that have been identified for collection. - Ensuring on the functionality of the data recording devices and calibrated as planned (including performing QA/QC) - Responsible with the security of locked Programmable Logic Controller (PLC) control room.
General Workers	- Perform regular operational and maintenance tasks. - Conduct the daily recordings in daily monitoring log sheets and request verification from the supervisors on the log sheets. - Report any fault to supervisor-in-charge when necessary
VCS Consultant	- Provides advice on all VCS - related matters - Prepare monitoring reports for verification audit. - Liaises with verifier on verification processes - Conducts data audits on VCS monitoring

j) Data and parameters to be monitored

AMS-III.H, version 19.0

The parameters to be monitored in the monitoring plan includes:

Data to be monitored	Description
$Q_{ww,i,y}$	Volume of wastewater treated in the project wastewater system in year y (m^3 /month)
$COD_{ww,untreated,PJ,y}$ or $COD_{inflow,PJ,y}$	Chemical oxygen demand of the wastewater entering the biogas treatment system in year y (tonnes/ m^3)
$COD_{ww,treated,y}$ or $COD_{ww,outflow,y}$	Chemical oxygen demand of the wastewater after the treatment system affected by the project activity (tonnes/ m^3)
$COD_{ww,discharge,y}$	Chemical oxygen demand of the treated wastewater discharged into sea, river or lake (tonnes/ m^3)
$S_{final,PJ,y}$	Method of disposal of the sludge (t)
$BG_{burnt,y}$	Biogas volume in year y (m^3)
$BG_{fuelled,y}$	Amount of biogas fuelled in year y (m^3)
$BG_{flared,y}$	Amount of biogas flared in year y (m^3)
$W_{CH4,y}$	Methane content in biogas in the year y (%)
T	Temperature of the biogas at the flow measurement site ($^{\circ}C$)
P	Pressure of the biogas at the flow measurement site (Pa)
Flame _m	Flame detection of flare in the minute m (flame on and off)
$T_{EG,m}$	Temperature in the exhaust gas of the enclosed flare in minute m ($^{\circ}C$)
$EC_{PJ,j,y}$	Electricity consumption in the project scenario in the year y (MWh/year)

AMS-I.D. version 18.0

The parameters to be monitored in the monitoring plan includes:

Data to be monitored	Description
EC _{PJ, facility,j,y}	Quantity of net electricity generation supplied by the project to the grid in year y (MWh)
EG _{PJ,add,y}	Quantity of net electricity generation supplied to the grid in year y by the project plant/unit that has been added under the project activity

TOOL06: Project emissions from flaring

The parameters to be monitored in the monitoring plan includes:

Data to be monitored	Description
F _{CH₄,EG,t}	Mass flow of methane in the exhaust gas of the flare on a dry basis at reference conditions in the time period t (kg)
T _{EG,m}	Temperature in the exhaust gas of the enclosed flare in the minute m (°C)
V _{i,RG,m}	Volumetric fraction of component i in the residual gas on a dry basis in the minute m where i = CH ₄ , CO, CO ₂ , O ₂ , H ₂ , H ₂ S, NH ₄ , N ₂
V _{RG,m}	Volumetric flow of the residual gas on a dry basis at reference conditions in the minute m (m ³)
M _{RG,m}	Mass flow of the residual gas on a dry basis at reference conditions in the minute m (kg)
V _{O₂,EG,m}	Volumetric fraction of O ₂ in the exhaust gas on a dry basis at reference conditions in the minute m
f _{CH₄,EG,m}	Concentration of methane in the exhaust gas of the flare on a dry basis at reference conditions in the minute m (mg/m ³)
Flame _m	Flame detection of flare in the minute m (on or off)
Maintenance _y	Maintenance events completed in year y (calendar dates)

TOOL05: Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation

The parameters to be monitored in the monitoring plan includes:

Data to be monitored	Description
EF _{grid,CM,y}	Grid Emission Factor for Peninsula Malaysia, Sabah, and Sarawak (tCO ₂ /MWh)

Data to be monitored	Description
$TDL_{j,y}$ and $TDL_{k,y}$ and $TDL_{l,y}$	Average technical transmission and distribution losses for providing electricity to source j, k or l in year y
$FC_{n,i,t}$	Quantity of fossil fuel type i fired in the captive power plant n in the time period t (m^3 , ton or l)
$EG_{n,t}$	Quantity of electricity generated in captive power plant n in the time period t (MWh)
$NCV_{i,t}$	Average net calorific value of fossil fuel type i used in the period t (GJ / mass or volume unit)
$EF_{CO_2,i,t}$	CO ₂ emission factor of fossil fuel type i used in the period t (t CO ₂ / GJ)
$EC_{PJ,j,y}$	Quantity of electricity consumed by the project electricity consumption source j in year y
$EC_{BL,k,y}$	Quantity of electricity that would be consumed by the baseline electricity consumption source k in year y (MWh/yr)
$EG_{PJ,grid,y}$ or $EG_{PJ,facility,l,y}$	Quantity of electricity generated and supplied by the project power plant to the grid in year y Quantity of electricity generated and supplied by the project power plant to the consumers/electricity consuming facility i in year y (MWh/yr)

Tool 03: Tool to calculate project or leakage CO₂ emissions from fossil fuel combustion:

Data to be monitored	Description
$FC_{i,j,y}$	Quantity of fuel type i combusted in process j during the year y (ton/yr or m^3 /yr)
$WC_{i,y}$	Weighted average mass fraction of carbon in fuel type i in year y (tC/mass unit of the fuel)
$\rho_{i,y}$	Weighted average density of fuel type i in year y (Mass unit/volume unit)
$NCV_{i,y}$	Weighted average net calorific value of fuel type i in year y (GJ/ m^3 , GJ/ton)
$EF_{CO_2,i,y}$	CO ₂ emission factor of fossil fuel type i used in the period t (t CO ₂ / GJ)

c) Training of monitoring personnel

All plant staff will be trained by Green Lagoon Technology Sdn Bhd, prior to the commissioning of the plants. The involved personnel shall receive adequate and regular training on the project performances and the monitoring plan involving methane capture up to utilization of the biogas.

Training of the running of day-to-day operations is under the responsibility of the individual PAIs. This include collection of data and ensuring data integrity is not tolerated.

A VCS monitoring Manual will be drafted and brief training will be provided by the appointed VCS consultant on the monitoring parameters.

d) Procedures for handling non-conformances with the validated monitoring plan

In case of a temporary situations such as the installed flow meter(s)/gas analyser malfunctioned or gave unrepresentative results, the data shall be measured according to the method listed below:

1. Manually with a back-up flow meter/gas analyser. The number of samples needed for the calculation will be calculated by using raosoft calculator, assuming the 95% confidence interval, 10% margin of error. The lower bound of the 95% confidence interval will be applied in the calculation sheet as a conservative approach
2. Alternatively, if a back-up meter is not available, the historical records and future data (at least one (1) month before and after the malfunction period) will be used to estimate the quantity. The 95% confidence interval will be calculated from the data obtained. The lower bound of the 95% confidence interval will be applied in the calculation sheet as a conservative approach

In relation to malfunctions of measurement of COD, the following will be observed. In case of temporary situation such as the COD sampling device/meter malfunctioned, the amount of COD shall be based on the most recent testing result

e) Sampling approaches and precision level

For AMS-III.H, related PAIs, all data that are collected in accordance with the VCS-PD monitoring frequency and data collection, are included and considered in the emission reduction calculation.

In relation to the measurement of COD: According to raosoft, the total number of months in a year is 12 months and the total number of samples needed to be collected is 11 samples to meet the 90% confidence interval.

f) Quality Assurance and Quality Control & maintenance procedures

Maintenance of the equipment shall be conducted regularly to ensure a good and proper functioning of the equipment.

Calibration of monitoring equipment and instruments will be done periodically according to the standard and procedures to maintain the accuracy of the measuring instrument

Emergency/Contingency preparedness for monitoring system is presented as below:

1. Data collection

If the person-in-charge for daily raw data logging is unavailable, there will be another back-up person to take over the job.

2. Data storage, transfer and backup

The monitored parameters that are collected and will be stored in the following ways:

- (a) Hard copy - Log sheets/ logbooks will be kept in proper manner, and available for audit verification purpose,

- (b) Soft copy - Raw data from log sheets/ logbooks will be stored as soft copy; together with monthly CER calculation, the entire data of the project activity will be back-up to an external hard drive or another computer every quarterly.

Immediate action shall be taken immediately if any matters/ uncertainties affecting the data quality.

In case of leakage occurrences in any parts of the biogas plant system, the team shall take immediate action to put the plant operation back on track. The plant operation manual shall include the procedure on safety and always made available on site to ensure the operators are fully aware of emergency procedures.

PAI1

The monitoring plan for this project includes all monitoring requirements following the procedures set by AMS-III.H (version 19.0) and AMS-I.D (version 18.0). The overall monitoring plan can be illustrated in the figure below:

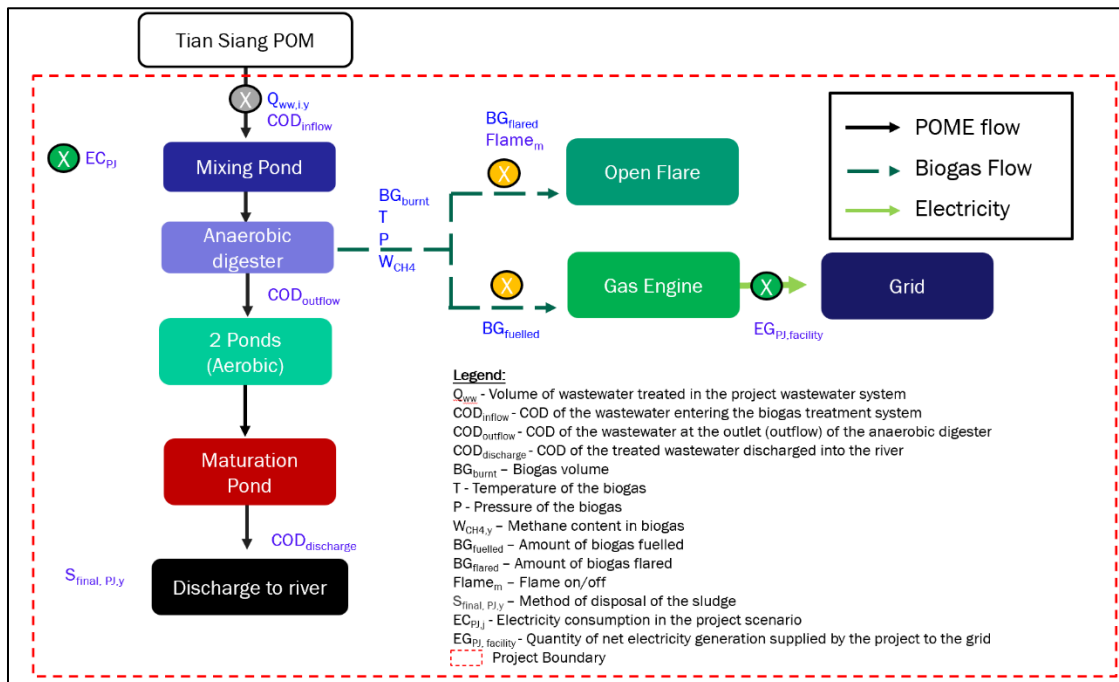


Figure 8: Monitoring Plan for PAI1

Implementation of the Monitoring Plan

The biogas plant operator will adopt the instructions presented in the monitoring plan and ensure all activities related to the biogas plant operation and monitoring are implemented. The main responsibilities of monitoring include:

- Data reading and handling: maintaining an adequate system for collecting, recording and storing data in accordance with the protocols determined by the monitoring plan as well as checking data quality, collection and record-keeping procedures on a regular basis. This also

includes making sure that all relevant electronic data is backed-up electronically for at least 2 years after the end of the crediting period in case of an unanticipated loss of data.

- In the case of erroneous data recorded for the monitored parameter occurred during short period, data for the affected period shall be derived based on the approach based on 95% confidence interval principles (source: “IPCC Good Practice Guidance and Uncertainty Management in National Greenhouse Gas Inventories”, page 6.6). A total of 10 readings before and after the constant period will be used to calculate the lower bound of 95% confidence interval with reference to the above-mentioned guideline. The lower bound of the interval boundaries calculated will be applied to the period for the constant data as a conservative approach. Alternately, if the 10 readings before and after the erroneous period is unavailable, the record from the daily manual log-sheet will be used to calculate the lower bound of 95% confidence interval. The lower bound of the interval boundaries calculated will be applied to the period for the constant data as a conservative approach.
- Reporting: preparing periodic summaries and reports that include emission reductions generated and observations regarding monitoring plan procedures;
- Training: assuring assigned personnel received adequate and regular training on project performances and the monitoring plan. Any new staff taking over the role of monitoring and reporting will be fully trained as well; and
- Quality control and quality assurance: complying with quality control and quality assurance procedures to facilitate periodical audits and verification. Any observation of matters affecting data quality will be reported to the management immediately.

Responsibility of Monitoring Plan

A dedicated VCS monitoring team will be assigned to be responsible for undertaking the monitoring as well as the quality control and assurance tasks. The following are the position and responsibility of the VCS monitoring team:

Table 9: Description of roles for PAI1 VCS team

Personnel	Description of Roles in Monitoring Plan
Plant Manager	• Report and seek approval from the higher management on VCS-related matters • Take full responsibility in the overall monitoring of the project • Conduct internal meetings on VCS matters • Oversees and coordinate the VCS monitoring plan data gathering • Monitor & ensure that data QA/QC is carried out by appointing (internal) independent verifiers to do cross-checking on irregular basis • Ensure all documentations were prepared accordingly based on VCS-PD requirement. • Coordinate with VCS consultant for matters (data/documents/records etc) that require rectification from plant

Personnel	Description of Roles in Monitoring Plan
Engineer	- Assists the VCS Manager in performing the VCS monitoring works - Assisting in the daily and monthly monitoring records for all the VCS related equipment. - Prepare monthly summary record of the operations of plants. - Ensure maintenance and calibration is carried out for proper running of machinery and equipment
Supervisors	- Report to the Engineer on any VCS monitoring issues - Conduct daily checking of data that have been identified for collection. - Ensuring on the functionality of the data recording devices and calibrated as planned (including performing QA/QC) - Responsible with the security of locked Programmable Logic Controller (PLC) control room.
General Workers	- Perform regular operational and maintenance tasks. - Conduct the daily recordings in daily monitoring log sheets and request verification from the supervisors on the log sheets. - Report any fault to supervisor-in-charge when necessary
VCS Consultant	- Provides advice on all VCS - related matters - Prepare monitoring reports for verification audit. - Liaises with verifier on verification processes - Conducts data audits on VCS monitoring

Quality Control, and Quality Assurance

Quality control and quality assurance procedures will be carried out on all parameters monitored. The practices to be undertaken in the context of the proposed project are as follows:

1. Monitoring records:

- Daily readings of all meters will be filled out in paper worksheets and filed consequently. All data collected will also be entered in electronic worksheets and stored in a computer (immediately) and on discs (periodically).
- Periodic controls of the monitoring records will be carried out to check any deviation from the monitoring plan.

2. Equipment calibration and maintenance:

- Flow meters, gas analyzers and other sensors will be subject to regular maintenance and testing according to the technical specifications by the manufacturers to ensure accuracy and good performance.
- Calibration of equipment will be conducted periodically according to technical specifications.

3. Corrective actions:

- Actions to correct deviations from the monitoring plan will be implemented as these deviations are observed either by the operator or during internal audits.
- Apart from periodic meetings, additional technical meetings will be held in order to define any necessary corrective actions to be undertaken.

4. Site audits:

- Regular site audits will be conducted to ensure that monitoring and operational procedures are being observed in accordance with the monitoring plan.

5. Training:

- A training plan will be set out for the operator. This will ensure that monitoring staff involved in the project are properly trained so that they are able to undertake the tasks required by this monitoring plan. Appropriate staff training must be provided before the project starts operating and generating emission reductions.

6. Storage of documents:

- List of monitoring equipment (e.g. flow meters, gas analyzers, thermometers) including their model numbers, names, manufacturers, specifications, user requirements, etc.
- Calibration lists and reports, including equipment or parts calibrated, date, method and procedures of calibration, precision following these processes, personnel, devices needed, etc.
- Maintenance lists and reports, including equipment or parts maintained, date, method and procedure of maintenance, performance following these procedures, personnel, devices needed, etc.
- Operation manual of the proposed project.
- Registration of minutes during meetings.
- Non-conformance reports.
- Monthly and annual worksheets.
- Training plan.
- Internal audit reports, including personnel, time, findings, corrective actions, follow-up inspections, etc.
- Annual monitoring review

APPENDIX 1: COMMERCIALLY SENSITIVE INFORMATION

Use the table below to describe the commercially sensitive information included in the project description to be excluded in the public version.

Section	Information	Justification